One Reform Divided the Elite, the Other Replaced It

Electoral Reform and Parliamentary Speech in the Belgian *Belle Époque*, 1880–1912

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Abstract

Did electoral reform change how Belgium's governing parties spoke to one another? We study two reforms: plural male suffrage, first applied in 1894, and proportional representation, introduced in 1900. The first changed who could vote; the second changed how votes translated into seats. Using 665,658 plenary interventions from both parliamentary houses, we measure Catholic–Liberal rhetorical polarization with the leave-out partisanship estimator of Gentzkow et al. (2019). We compare the Chamber with the Senate. Both faced the new electorate, but restrictive rules for Senate candidates limited the entry of socialists and reduced senators' exposure to electoral competition. After 1894, Catholic–Liberal distinctiveness rose by 0.0095 more in the Chamber than in the Senate, and by 0.0094 within the same politicians. The break occurs at the reform and is the largest among fourteen feasible cutoff dates. The increase lasts until 1900, is driven mainly by differences in how the parties spoke *about the same subjects*, and is strongest on labour and social policy. Catholic incumbents changed their language; among Liberals, the more distinctive speakers were chiefly the newly elected members. The 1900 reform presents a different identification problem. The election renewed both houses, and the Senate became more distinctive while the Chamber's score fell. We therefore cannot isolate the reform's effect on rhetorical polarization. We can, however, document substantial turnover: roughly half of the Senate's speaking members and three in ten Chamber members were new, while Catholic deputies returned to the coal districts from which majority rule had largely excluded them. The evidence distinguishes a reform that changed the speech of sitting politicians from one that clearly changed parliamentary representation, without claiming to identify the second reform's effect on speech.

Keywords: elite polarization, electoral reform, electoral institutions, plural suffrage, proportional representation, bicameralism, parliamentary speech, text-as-data.

JEL: D72, N43, P16, Z13.

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1 Introduction

Why would an electoral reform make established parties speak more differently? A larger electorate may force incumbents to respond to new voters. A different electoral rule may instead change which politicians win seats. Both changes could affect polarization, but they need not do so in the same way. We study them separately in Belgium, where plural male suffrage was first used in 1894 and proportional representation in 1900.

Polarization matters because disagreement between governing parties can make legislative compromise more difficult (Jones, 2001; McCarty et al., 2006). It may also reinforce divisions among voters (Iyengar et al., 2019; Boxell et al., 2024). We use polarization in its basic sense: greater distance between groups (DiMaggio et al., 1996; Esteban and Ray, 1994). A growing literature measures that distance from politicians' words. For the United States, Gentzkow et al. (2019) estimate how much a speaker's party can be inferred from the phrases used in congressional debates. Their approach corrects the small-sample bias of simpler text measures. Related estimates are available for Norway (Fiva et al., 2025), Finland (Simola et al., 2025), the United Kingdom (Peterson and Spirling, 2018), and Ireland (Lauderdale and Herzog, 2016), building on earlier work in economics (Jensen et al., 2012; Card et al., 2022). Speech is especially useful when party discipline makes roll-call votes less informative about disagreement within parliament (Poole and Rosenthal, 1985).

We know much less about what makes parliamentary language become more partisan. Existing explanations stress social identities, economic shocks, crises, and long-run political realignment (Iyengar et al., 2019; Autor et al., 2013, 2020; Funke et al., 2016; McCarty et al., 2006). Research on electoral institutions often asks why elites choose particular rules (Boix, 1999; Emmenegger and Walter, 2019; Renwick, 2010; Ahmed, 2013; Cox, 1997; Iversen and Soskice, 2006). Fiva and Folke (2016) study how proportional representation affects electoral outcomes in Norway. Research on franchise extension has focused mainly on its origins and policy effects (Acemoglu and Robinson, 2000; Lizzeri and Persico, 2004; Meltzer and Richard, 1981; Aidt and Jensen, 2014; Lott and Kenny, 1999). Studies of legislative behaviour after enfranchisement (Berlinski and Dewan, 2011; Aidt and Franck, 2015) and work exploiting historical parliamentary reforms (Cirone and Van Coppenolle, 2019) show why these institutions are useful for studying political responses. Our question is different: *did changing the electoral rules alter the rhetorical distance between the parties already in power?*

The Belgian case provides two consecutive reforms with different immediate effects. Plural suffrage, adopted in 1893 after a general strike, extended the vote to almost all adult men, although some received additional votes based on property or education. The first election under the new franchise, in 1894, brought the socialists into the Chamber. In 1899, Belgium adopted proportional representation, first used in 1900. That reform changed the allocation of seats without another enlargement of the electorate. The two elections therefore allow us to compare a change in electoral demand with a change in the rules of political representation.

We measure the language of Catholic and Liberal politicians from 1880 to 1912. Our corpus contains 665,658 interventions in the plenary debates of both houses. Following Gentzkow et al. (2019), we estimate how reliably a phrase reveals the speaker's party, correcting for the speaker's contribution to the word frequencies used to score that phrase. The resulting measure captures *rhetorical distinctiveness*: whether Catholics and Liberals use different language. It does not directly measure ideological positions or whether the legislature passes fewer laws.

A simple comparison before and after each election would be misleading. The reforms followed strikes, social conflict, and the rise of organized labour (Deneckere, 1997; Polasky, 1992). Those events could have changed political speech even without institutional reform. We instead compare the Chamber with the Senate. Both houses faced the same national events and, after 1893, the same enlarged electorate. But the Senate retained restrictive eligibility rules, a higher minimum age, and some indirectly elected seats. These

features made it harder for the new electorate to replace sitting senators with socialist candidates. The design therefore uses the *difference in exposure* to electoral competition across the two houses, not a claim that the Senate was unaffected by the reforms. We discuss the identifying assumption and its limits in Section 5.

We find a clear change after plural suffrage. Between 1894 and 1899, Catholic–Liberal distinctiveness in the Chamber rose by 0.0095 relative to the Senate (Table 3, Panel A).[1] The estimate is virtually unchanged (0.0094; Table 3, column 3) when we follow the same politicians over time. The event study shows no statistically significant pre-reform leads, and the 1894 cutoff produces the largest break among fourteen feasible dates. The result is reasonably stable across alternative measures, vocabularies, and time windows, although some yardstick choices reduce its size (Section A.2). The effect ends with the 1900 election.

The change came mainly from politicians who were already in office. Catholic incumbents used more distinctive language after the franchise extension. Liberal incumbents did not show a comparable change; the more distinctive Liberals were mostly those first elected under the new franchise. Moreover, most of the increase occurred *within topics*, especially labour and social policy. The parties were not simply discussing different subjects. They increasingly used different language when discussing the same ones. This distinction matters because our companion paper studies how enfranchisement changed the *attention* politicians paid to labour issues (Graziani and Vanschoonbeek, 2026). Attention and partisan distinctiveness are related, but not identical, outcomes (Budge and Farlie, 1983; Petrocik, 1996; Downs, 1957; Dolezal et al., 2014).

For proportional representation, the evidence is less conclusive. The Chamber–Senate difference turns negative after 1900, but the Senate's own score rises while the Chamber's falls. The 1900 dissolution renewed the Senate under the new rule, so we cannot treat it as an unaffected comparison group. The negative coefficient is therefore not a causal estimate of the reform's effect on polarization. The change in *who held seats*, however, is clear. The election brought substantial turnover in both houses. In the coal districts, where Liberals and then socialists had taken nearly all the seats under the majoritarian system, Catholics returned under proportional representation. The corpus contains no Catholic intervention from the core coal districts between the reforms, compared with 18,312 in 1900–1912. This tells us about representation, not about how an unchanged set of politicians would have spoken under different electoral rules.

These results add to historical accounts of Belgium's increasingly entrenched religious and class divisions (Lipset and Rokkan, 1967; Lorwin, 1966; Huyse, 1970; Strikwerda, 1997; Kossmann, 1978). They also suggest two distinctions for the study of polarization. First, changes in who can vote and changes in how seats are allocated can affect different aspects of parliamentary politics. In Belgium, the franchise reform changed the language of sitting politicians, whereas the visible effect of proportional representation was on membership. Second, a rise in between-party differences need not mean stronger party discipline. The within-party spread does not significantly contract after 1894 (-0.0073, s.e. 0.0058; Table 65), although the estimate is too imprecise to rule out meaningful compression (Canen et al., 2026); see Section 7.

The rest of the paper introduces the historical setting and data, explains our text measure and identification strategy, and presents the results and their robustness. The appendices document the measurement choices, additional tests, and the limits of the comparisons.

2 Historical setting

Four institutional features matter for the design; Appendix A provides the historical background. Under the censitary franchise about 2% of the population voted (Belgische Senaat, 2019), and the two governing parties—Catholics and Liberals, divided since the school war of 1879–1884 on the confessional axis—alternated in a closed arena. The 1893 revision, conceded under general strike, created plural manhood

[1] Table 3 reports three decimals. The four-decimal coefficients and standard errors quoted in the text for its columns (1) and (3) are in Table 59.

suffrage: every man over twenty-five voted, property and education adding up to two further votes, a tenfold expansion of the electorate first applied in October 1894, which also seated twenty-eight socialists in the Chamber. The 1899 law replaced majoritarian seat allocation with proportional representation (d'Hondt), first applied at the general election of 27 May 1900, which followed a dissolution of both chambers and so renewed the Chamber and every directly elected Senate seat at once. Throughout, the Senate was returned by the same electorate as the Chamber but under stricter candidacy rules—a high eligibility census that kept the socialist leadership off its ballots, so that socialists entered it first through the provincial councils—which is the asymmetry §5.2 builds on.

3 Data

We draw on a linked, politician-level dataset for the Belgian Parliament, 1880–1912, that follows the same individuals across chambers, parties, and reform periods (Graziani and Vanschoonbeek, 2026). It combines biographical and career records for all members who served in either house with the plenary proceedings of both chambers, the Chamber's from the Belgian Chamber Debates dataset (Graziani, 2026). The career file covers 876 unique politicians who sat in the legislatures 1880–1912—509 only in the Chamber, 282 only in the Senate, and 85 in both—each assigned a stable identifier that links party affiliation (recorded by legislature), chamber, arrondissement, and time in office, with names standardized and disambiguated across historical spelling variants. In the speech data, a member's party label is set to his career mode where that mode covers at least 90% of his labelled speeches; below that threshold the labels stay as recorded unless the correction registry fixes them, and documented party switchers are exempt (Appendix B.3). In the Chamber the speaker of each intervention is the member or minister to whom the dataset links it: the name printed in the speaker label is matched to the closest name among the members who sat in the Chamber on the day of the sitting, and otherwise among the ministers in office, allowing for misread characters, particles and compound surnames (Graziani, 2026). Over our window 92.8% of the Chamber's interventions are matched exactly, 4.5% on a near match and 2.7% to a minister (Table 58). In the Senate the speaker is linked by rule rather than taken from the fuzzy match of the digitization: he must be a member who sat in that house on that day, or a minister in office, whose surname matches the speaker field (the digitization's link is also kept for 253 interventions delivered outside the member's recorded mandate), with ministerial and other roles resolved by date. Interventions from the chair, and those whose speaker cannot be resolved, are excluded in both houses (Appendix B.3).

The speech corpus comprises all plenary debates in both chambers over the legislatures 1880–1912, from June 1880 to May 1914: 665,658 interventions in total (594,396 in the Chamber and 71,262 in the Senate). Both houses' proceedings are OCR-derived, but through different pipelines. The Chamber's come from the Belgian Chamber Debates dataset (Graziani, 2026), which takes the OCR text of the *Annales parlementaires* served by *plenum.be*, tags every speaker label with rules that tolerate the OCR's misreadings, splits each sitting into typed blocks—speeches, collective interjections, the clerks' text, documents read into the record, roll calls—and keeps the text of every block as printed. We take the speech blocks of identified members and ministers, in French or Dutch, without the turns of the chair, and join the consecutive blocks of one speaker in a sitting into a single intervention, so that an interjection from the floor does not split a speech: 669,369 blocks make the 594,396 interventions. The reactions of the floor and the stage directions that the record prints inside a speech—*(Rires à gauche.)*, *(Très bien! à droite.)*, *(Interruption.)*—are not the speaker's words, and we take them out of the text of both houses before any phrase is counted (§4.1). The Senate proceedings were recovered from scanned archives through a dedicated OCR and post-correction pipeline of our own, then parsed into a speaker-level corpus by layout recognition that exploits recurrent typographical conventions to mark speakers, roles, and intervention boundaries.

The PLENUM project offers a second structuring of the same OCR text of the Chamber, and the choice

between the two matters for the design. The two cover the same sittings (3,831 of the dataset's 3,835) and before 1894 they agree closely: the median ratio of the dataset's interventions to PLENUM's in a sitting is about one. But PLENUM's interventions hold 71% of the dataset's words in the same sittings before 1894 and 49% from 1894, and in 1894 and from 1900 the dataset finds between 1.3 and 3.5 times as many interventions per sitting, in part because PLENUM merges several speakers into one record (Table 58). Text that goes missing at a rate that changes at the reform, in the treated house only, is the kind of measurement asymmetry a difference-in-exposure cannot net out; the dataset removes it. Because *both* corpora are OCR-derived rather than born-digital, differential text quality between the two houses is a potential source of measurement asymmetry that could, in principle, contaminate the Chamber–Senate contrast; we treat it as a first-order robustness concern and address it directly in Section A.8, showing the plural-suffrage estimate is not an artefact of OCR noise. The unit of observation is an intervention in a plenary session—in the Chamber a speaker's turn, in the Senate a single uninterrupted intervention—linked to politician, party, chamber, date, legislature, and arrondissement; in the Chamber the arrondissement is the one the member represents in that legislature. The chamber field records *where the intervention was delivered*, which for almost every member coincides with the house he sat in but for one group does not: ministers defend their bills before both houses, so a minister returned for the Chamber also speaks in the Senate. One hundred and twenty politicians in the estimation sample have speech in both houses on this definition: the government of the period—Beernaert, Malou, Frère-Orban, de Smet de Naeyer, de Burlet, de Broqueville and their colleagues—and forty-seven members whose speech passes from one house to the other with a clean break in time. In the career file, eighty-five *sat* in both over the window. The two counts measure different things and we keep them distinct: membership in the career file, venue in the speech corpus. The treatment indicator follows membership rather than venue: each member is assigned to the house in which he delivers most of his interventions, so that the indicator is constant within politician, and 21,429 interventions, about one in twenty of the estimation sample, are delivered outside the member's house. Section A.2 reports two alternatives—the venue of each speech, and the house in which the career file places the speaker on the day he spoke—and the plural-suffrage estimate stays between $+0.0085$ and $+0.0115$ under either (Table 14). For the main Catholic–Liberal analysis we work from a lemmatized rendering of each intervention, tokenized into phrase features as described in Section 4. For the Senate it is the corpus's own lemmatized field; for the Chamber, whose text the dataset keeps as printed, we build it with the recipe of the PLENUM corpus—lower case, the stopwords that corpus removes, Snowball stems for French and for Dutch—and on a held-out sample of interventions it reproduces 98% of PLENUM's tokens and 91% of its candidate phrases (Table 58). The Socialists, who enter only from 1894 and sit overwhelmingly in the Chamber (in the Senate they arrive first through the provincial councils), are analysed separately as a third party in the dyadic and proximity measures.

Label and dating audit. Because the identifying cells can be thin, every party label and every date the design leans on has been audited against the parliamentary record, with a correction registry in Appendix B.3; the audit's material corrections (a father–son merge, two anachronistic identities, the mining-window recount, the linking of speakers to members and the exclusion of the chair) are incorporated throughout. In the Chamber most of them are now made upstream, by the dataset's own links and party records; the registry still runs over both houses.

4 Measuring elite polarization

We measure *partisanship* by asking how accurately a randomly chosen phrase from a speech identifies the speaker's party. If the two parties spoke identically the answer is chance, $\frac{1}{2}$; the more distinctive their vocabularies, the closer to 1. The measure has four components: the phrase vocabulary (§4.1), the leave-

out estimator (§4.2), its per-speech decomposition that serves as the regression outcome (§4.3), and two safeguards—a penalized estimator (§A.4) and permutation bands (§A.5). We also define a measure of proximity to socialist language (§A.3); §6.7 explains why the membership data do not support reading it as a mechanism.

4.1 Phrases and the vocabulary

Let a *phrase* be a two-word (bigram) feature formed from the lemmatized speech stream. Bigrams can reveal partisan differences that single words miss, for example "*class_ouvri*" and "*patron_ouvri*" rather than isolated words. Following Gentzkow et al. (2019) we form bigrams *before* removing function words, so that genuine collocations spanning a stopword are preserved, and prune noisy bigrams *afterward*. A bigram is dropped if

- **both** of its tokens are generic function words (set $\mathscr{S}$, "stopwords"); or
- **either** token is a parliamentary/procedural term ($\mathscr{P}$: *ministr*, *séanc*, *scrutin*, ...), a lemmatization fragment ($\mathscr{F}$: stranded suffixes *ment*, *tion*, foreign tokens), a party self-label ($\mathscr{L}$: *cathol*, *libéral*), or a politician surname (set $\mathscr{N}$).

We exclude surnames because a member's own name predicts his party mechanically, and leaving it in would inflate measured partisanship. The set $\mathscr{N}$ holds the surnames of the members and ministers the speeches are linked to—1,543 name tokens, with the forms in which the OCR systematically prints them in the speaker labels—and nothing else. Harvesting it from the whole speaker field of the digitization, which also carries titles, functions and fragments of the agenda, would remove with the names common words such as *suffrag*, *enseign*, *religion*, *libert*, *impôt* and *agricol*, and every phrase built on them—and, with the words for the sides of the house (*gauch*, *droit*, *banc*, *centr*), most of the reactions of the floor that the record prints inside a speech (*Très bien! à droite*, *Rires à gauche*). Those reactions are not the speaker's words, and we remove them from the text itself before any phrase is formed: every parenthesized stage direction that records a reaction or an event of the sitting—applause, laughter, approval, protest, exclamations, noise, interruptions, the president's bell—is deleted from the speech in both houses, 142,037 segments in the Chamber and the reaction tokens of 12,119 interventions in the Senate (Appendix B.9). They are party-diagnostic by construction, because a member's own benches applaud him and the opposite benches protest, so that left in the text they would pass the record's stage directions off as partisan language; with them in, the plural-suffrage estimate is $+0.0096$ and the Panel B contrast -0.0093, against $+0.0094$ and -0.0099 without them, all with politician fixed effects and $p < 0.01$ (Table 59). Two features of the Belgian corpus make naive stopword lists fail, and we correct for both. First, the text is lemmatized and truncated, so reference forms must match the corpus forms (*vandervelde→vanderveld*; *honorable→honor*); we therefore harvest $\mathscr{S}$ and $\mathscr{P}$ *empirically* from the corpus's own high-frequency tokens, and expand $\mathscr{N}$ with truncated surname variants $\{n, n_{[:-1]}, n_{[:-2]}\}$. Second, lemmatization fuses the party self-label *ouvrier* with the content noun *ouvrière*; we keep *ouvri*, because it carries the substantive labour signal, while still dropping *cathol* and *libéral*.

After pruning we apply a GST-style hard filter on the global vocabulary. A phrase is retained only if it (i) occurs at least $\tau_c = 35$ times in the corpus, (ii) is used by at least $\tau_s = 6$ distinct speakers, (iii) reaches a count of at least 15 in *some* legislature, so that the phrase is not an artefact of a single period, and (iv) is built from tokens of at least two characters. These thresholds are deliberately looser in absolute terms than those of Gentzkow et al. (2019), who retain phrases spoken at least ten times in some session, in at least ten distinct speaker-sessions, and at least a hundred times across the whole *Congressional Record*: their corpus is orders of magnitude larger than ours, so the same absolute cut-offs would leave a vocabulary too thin to estimate on. Relative to corpus size the two filters are comparable, and ours is if anything the more

conservative on the dimension that matters for finite-sample bias, since we also require a within-legislature count. The same fixed vocabulary is used for every estimate below, so that all comparisons live in one phrase space.

The filter retains 36,297 phrases, which carry about a quarter of the bigram tokens of Catholic and Liberal speech; its coverage, its stability across the reform, and its consequences for the estimates are documented in Appendix B.2.[2]

4.2 Leave-out partisanship

Index speakers by i and phrases by j. Let $X_{ij} \geq 0$ be the number of times speaker i uses phrase j, and let $y_i \in \{0,1\}$ record party, $y_i = 1$ for Catholic and $y_i = 0$ for Liberal. Write the speaker's token total $m_i = \sum_j X_{ij}$, the party phrase totals

$$C_j^1 = \sum_{i:\, y_i=1} X_{ij}, \qquad C_j^0 = \sum_{i:\, y_i=0} X_{ij},$$

and the party token totals $M^1 = \sum_{i:y_i=1} m_i$, $M^0 = \sum_{i:y_i=0} m_i$.

The empirical phrase frequencies C_j^1/M^1 and C_j^0/M^0 define, for any token of phrase j, the posterior probability that its speaker is Catholic,

$$\rho_j \;=\; \frac{C_j^1/M^1}{C_j^1/M^1 + C_j^0/M^0}.$$

Plugging the same data into both the frequencies and the guess overfits: with J large and speakers finite, a phrase used once by one Catholic looks perfectly Catholic. Gentzkow et al. (2019) remove this bias by *leaving each speaker out* of the frequencies used to classify her own tokens. Concretely, for a token of phrase j spoken by Catholic speaker i (who uses it X_{ij} times), the leave-out posterior that the token points to i's *own* party is

$$\rho_{ij}^{\text{self}} = \frac{q_{ij}^R}{q_{ij}^R + q_{ij}^D}, \qquad q_{ij}^R = \frac{C_j^1 - X_{ij}}{M^1 - m_i}, \quad q_{ij}^D = \frac{C_j^0}{M^0}, \tag{1}$$

and symmetrically for a Liberal speaker,

$$\rho_{ij}^{\text{self}} = \frac{q_{ij}^D}{q_{ij}^R + q_{ij}^D}, \qquad q_{ij}^D = \frac{C_j^0 - X_{ij}}{M^0 - m_i}, \quad q_{ij}^R = \frac{C_j^1}{M^1}. \tag{2}$$

The subtraction of X_{ij} from the numerator and m_i from the denominator is the leave-out correction: speaker i's own counts never enter the distribution used to score her. Averaging (1)–(2) *within* each party, token-weighted, gives the party-specific partisanship

$$\pi^R = \frac{1}{M^1} \sum_{i:\, y_i=1} \sum_j X_{ij}\, \rho_{ij}^{\text{self}}, \qquad \pi^D = \frac{1}{M^0} \sum_{i:\, y_i=0} \sum_j X_{ij}\, \rho_{ij}^{\text{self}}, \tag{3}$$

and weighting the two parties equally gives the partisanship of the cell,

$$\boxed{\Pi = \tfrac{1}{2}\left(\pi^R + \pi^D\right)} \tag{4}$$

with $\Pi = \frac{1}{2}$ under no information and $\Pi \to 1$ as the two vocabularies separate. Equal party weighting prevents the more talkative party from dominating; the token weighting inside each party lets frequent, discriminating phrases count for more. We compute (4) pooled, to anchor the regression outcome defined next; the descriptive trajectory by legislature and chamber uses its token-weighted counterpart (§4.3).

[2]Thresholds match `FilterParams`: `min_total_count=35`, `min_speakers=6`, `min_count_in_one_period=15`, `min_token_len=2`—the medium rung. A looser 25/5 and a stricter 50/8 vocabulary are reported as a robustness ladder in Appendix B.9; the differences are discussed where they matter (notably the socialist-proximity mechanism, §6.7).

4.3 Per-speech distinctiveness: the regression outcome

Equation (4) is a cell-level scalar; causal inference needs a unit-level outcome. We decompose Π to the speech. For a speech s by speaker i, let $x_{ij}^{(s)}$ be the in-vocabulary phrase counts *in that speech*. Define

$$s_{\text{distinct}}^{(i,s)} = \frac{\sum_j x_{ij}^{(s)} \rho_{ij}^{\text{self}}}{\sum_j x_{ij}^{(s)}}, \qquad w^{(i,s)} = \sum_j x_{ij}^{(s)}, \tag{5}$$

i.e. the token-weighted average, over the phrases the speech contains, of the *same* leave-out quantities ρ_{ij}^{self} from (1)–(2), with weight $w^{(i,s)}$ equal to the speech's in-vocabulary token count. By construction the w-weighted mean of s_{distinct} over the speeches of one party in a cell reproduces that party's partisanship in the cell, evaluated on the fixed yardstick of §5.4:

$$\frac{\sum_{(i,s)\in\text{cell},\, y_i=1} w^{(i,s)} s_{\text{distinct}}^{(i,s)}}{\sum_{(i,s)\in\text{cell},\, y_i=1} w^{(i,s)}} = \pi_{\text{cell}}^{R},$$

symmetrically for π_{cell}^{D}, so that $\Pi_{\text{cell}} = \frac{1}{2}\left(\pi_{\text{cell}}^{R} + \pi_{\text{cell}}^{D}\right)$. The w-weighted mean over *all* the speeches of a cell weights the two parties by their token shares rather than equally: it equals $\Pi_{\text{cell}} + (\omega_{\text{cell}} - \frac{1}{2})(\pi_{\text{cell}}^{R} - \pi_{\text{cell}}^{D})$, where ω_{cell} is the Catholic share of the cell's in-vocabulary tokens, and so coincides with Π_{cell} only when the two parties contribute equal token totals or are equally partisan. In the descriptive figures and tables the "Π" of a cell is a token-weighted mean of this kind: of s_{distinct} itself in Figure 1 and Tables 4, 19 and 29, and of own-party leave-out posteriors scored on yardsticks of their own, stated in their captions and notes, in Figure 4 and Tables 20 and 22. Thus s_{distinct} is a genuine micro-decomposition of the partisanship index, not a separate construct: within each party, regressions on s_{distinct} weighted by w aggregate tokens exactly as π^{R} and π^{D} do, and the party fixed effect removes the level difference between the parties. We use this outcome in the difference-in-differences design of Section 5.

The descriptive and regression measures differ in one important way because it governs how the descriptive and the causal figures relate. The regression outcome s_{distinct} is built on a single, globally estimated yardstick $\hat{\rho}$ that is held fixed across chambers and periods, so that it differences out of the bicameral contrast (§5.4). A cell-specific index, which re-estimates $\hat{\rho}$ inside each chamber×legislature cell, would be a different object: we use cell-specific posteriors only for the descriptive phrase lists (§4.6) and for the leave-cell-out ruler of §A.2, and the series in the figures, including Figure 4, are scored on a yardstick held fixed across legislatures. The two can move differently in a given cell: a thin cell can compress a cell-specific index even as the fixed-yardstick gap between chambers widens.

4.4 The second de-biasing route

A penalized Poisson–LASSO co-estimator attacks the same small-sample bias by shrinkage instead of leave-out; construction, and the demonstration that the two de-biasers agree where the design needs them to, in Appendix A. In the text it appears as a robustness column.

4.5 Permutation bands, diagnosticity, and the phrase lists

Three pieces of apparatus complete the measure and are stated fully in Appendix A. Descriptive figures carry a null band: the 5th–95th percentile of Π under random permutation of party across members, group sizes held fixed, on a yardstick held fixed across legislatures. A per-phrase diagnosticity statistic ranks vocabulary by how sharply it separates the parties and feeds the contribution decompositions used in §6.4. And the most party-diagnostic phrases are what a reader of the period would predict—the school war and the confessional

network on the Catholic side, laicization and public instruction on the Liberal. The choice model carries no covariates, by design: the yardstick must stay common across chambers and periods for the cancellation of §5.4 to hold, so composition is handled by the permutation band and by fixed effects downstream, not inside the measure.

4.6 Which phrases carry the signal: diagnosticity and its period decomposition

The index Π combines thousands of phrases. To see which expressions drive it, we calculate each phrase's contribution and report the most informative phrases by period. The construction follows Gentzkow et al. (2019) and Simola et al. (2025) with one simplification we flag rather than absorb: their diagnosticity statistic is the change in the expected posterior when a phrase is removed from the vocabulary, while $\zeta_{j,p}$ in (10) is the plainer max-posterior form. We restate the construction because we apply it period by period and, later, topic by topic.

The contribution decomposition. The two quantities combine explicitly. Because Π_p—here the token-weighted index of window p, scored with the window's own leave-out posteriors rather than the fixed yardstick of §5.4—is the token-weighted average of those posteriors, each phrase contributes to it in proportion to how often it is spoken *and* how diagnostic it is: writing $w_{j,p}$ for phrase j's share of window-p tokens,

$$\Pi_p \;=\; \sum_j w_{j,p}\,\zeta^{\mathrm{LO}}_{j,p}, \qquad \sum_j w_{j,p} = 1, \tag{6}$$

with $\zeta^{\mathrm{LO}}_{j,p}$ the average, over the window-p tokens of phrase j, of the leave-out posterior of the speaker's own party, computed within the window as in (1)–(2). This own-party form, rather than the max-posterior form of (10), is what makes (6) an identity; for a phrase spoken by one party only, it is the leave-out counterpart of (10). The product $w_{j,p}\zeta^{\mathrm{LO}}_{j,p}$ is the phrase's share of measured partisanship, and it is what moves the index when a party adopts or abandons a vocabulary. Comparing the ranked lists across windows therefore shows not only *which* phrases separate the parties at a point in time but which entered, left, or changed sides across a reform—the phrase-level face of the aggregate movement.

4.7 Proximity to the socialist lexicon

An additional measure, s_{prox}, measures each speech's proximity to the socialist lexicon on the post-1894 three-party pool (construction in Appendix A). The paper does not build on it: §6.7 shows that the membership data do not support an accommodation-and-retreat reading of its coefficients, which are reported in Appendix B.8 beside the member counts.

Diagnosticity and the phrase lists. A per-phrase statistic, ζ, ranks vocabulary by how sharply it separates the parties and feeds the period and topic decompositions used below; the most party-diagnostic phrases are what a reader of the period would predict—the school war and the confessional network on the Catholic side, laicization and public instruction on the Liberal—and the full apparatus, the phrase lists by period, and the discussion of what the index does and does not register are in Appendix A.

5 Identification

5.1 Design

The index might change because of war, a budget crisis, or a long-run shift in political language. To distinguish those changes from the effects of electoral reform, we compare Belgium's two parliamentary houses.

Let $\mathrm{DK}_i = 1$ if member i sits in the directly elected Chamber and 0 in the less exposed Senate—his house being the one in which he delivers most of his interventions (§3), and let Post_t switch on after a reform. The baseline difference-in-differences is

$$s^{(i,s)}_{\text{distinct}} = \beta\left(\mathrm{DK}_i \times \mathrm{Post}_t\right) + \gamma \mathrm{Post}_t + \lambda_p + \mu_a + \theta_{\text{party}} + \varepsilon_{i,s}, \qquad [\mathrm{aw} = w^{(i,s)}], \tag{7}$$

estimated with politician (λ_p), arrondissement (μ_a), and party (θ) fixed effects and a common post-reform indicator Post_t, weighted by $w^{(i,s)}$—the speech's in-vocabulary token count, so the regression aggregates speeches with exactly the weights the phrase-count estimator uses—with standard errors clustered on arrondissement. Specifications without politician fixed effects include DK_i itself. Because the politician fixed effect absorbs the level DK_i and the post-reform indicator absorbs the time shift common to both houses, β is identified from the *interaction*: the change in Chamber distinctiveness, after the reform, relative to the contemporaneous change in the Senate. The main specification models the common time effect parsimoniously with this single indicator; saturating it to full legislature fixed effects δ_ℓ, as every column of Table 9 does, leaves β essentially unchanged ($+0.0093$ against $+0.0094$). The Senate differences out anything common to both houses—the secular trend, national shocks—and any exposure the Senate shared moves β toward zero, so on the side of the design the estimate is conservative for the members actually exposed to the reform; the side of the measurement is taken up in §A.2.

The distinction between exposure and insulation is important. Throughout the paper we speak of a difference in *exposure*, not of an "insulated" control. The Senate of the *Belle Époque* was not hermetically sealed off from mass politics—no chamber of a national parliament is—and, as §5.2 details, the 1900 rule change formally reached it too. What identification requires is only that the Senate's exposure to each reform be *strictly smaller* than the Chamber's. Any residual Senate exposure moves the control in the same direction as the treated house and attenuates β toward zero; it cannot manufacture a positive estimate. This is the precise sense in which the design is conservative. Whether the measure is as well is a separate question: the yardstick and the vocabulary are estimated on the treated period too, and §A.2 shows that this can push the estimate up.

5.2 Why the Senate is a valid control

Our comparison requires the Senate to have been less exposed than the Chamber. The institutional rules and representation data support this claim for the 1894 reform, although the two houses were not identical. The Belle-Époque Senate was built to stand at a distance from mass politics by three compounding filters: *wealth*— eligibility required a direct-tax *cens* of 1,200 francs, or ownership of property with a cadastral income of at least 12,000 francs, where voting for the Chamber after 1893 required no *cens* at all, leaving a pool of eligible men in the low thousands out of six million (Belgische Senaat, 2019; Luykx and Platel, 1985); *age and indirection*—a minimum age of forty, with part of the seats filled through the provincial councils; and *selection*, a membership drawn from the propertied, professional and aristocratic classes (Witte et al., 2009).

The 1893 reform also changed the Senate electorate, an important limit to the comparison. The reform did not spare the Senate's electorate: after 1893 both houses were returned by the same plural manhood suffrage, and the Senate's own record describes one electorate for both chambers (Belgische Senaat, 2019). The one asymmetry the shared franchise retains is the voting age—thirty against twenty-five (Belgische Senaat, 2019), with the excluded 25–30 cohort disproportionately working-class; real, but small next to the candidacy filter below, and it runs against us: to the extent the Senate's electorate excludes the youngest and most working-class cohort, the control retains marginally more insulation and the contrast loses, not gains, a sliver of its identifying variation. A Senate treated as unexposed to the enfranchisement would be a control the sources do not support, and we do not claim one.

The main remaining difference was in who could stand for election. The 1893 revision loosened it—it lowered the senatorial *cens* from 2,116 to 1,200 francs and created the provincial senators, chosen by the provincial councils without any *cens* (Belgische Senaat, 2019)—but the directly elected seats stayed closed to all but a few thousand men over forty, and the 1899 law left eligibility as it was. Whatever exposure the loosening gave the Senate works against the contrast, not for it. The filter's role is to convert a *common* shock into different chambers. The same enlarged electorate that returned twenty-eight socialists to the Chamber in 1894 returned almost none to the Senate, because the men the new electors wanted to send could not stand for the upper house. The comparison therefore relies less on who could vote than on which candidates those voters could elect. That difference affected how vulnerable incumbents were to electoral competition. Stated as an incentive rather than a composition: a deputy after 1894 holds a seat the new elector can credibly give to someone else, so ignoring the social question carries an electoral price; a senator faces the enlarged electorate, but no candidate that electorate wants can stand against him, so the price does not exist there. The channel is demand-side in origin—the electorate is what changed—with its bite governed by the candidacy rules the reform left in place; it operates whether or not a single socialist ever takes the Senate floor, and it is distinct from, and additional to, the compositional fact that the Chamber also acquired a new interlocutor (§6.2).

Candidacy law is not the only fixed feature distinguishing the houses: so are the minimum age, the mandate length, and partial renewal, and with two chambers none is separable from the others. Stated as a limit, this would make the treatment a bundle; stated as it operates, it is closer to an insurance: each of the four lowers the Senate's electoral accountability—a seat the new elector cannot contest, a member who cannot be young, a mandate that outlasts any single electorate's anger, a renewal that dilutes it—so all four moderate a common shock in the *same* direction, and the design needs only that the upper house was less answerable to the vote, which each of them implies. That the candidacy filter is the binding one is an interpretation, supported by the socialist-entry counts of Table 30 and by the mining triple-difference, which finds no concentration of the response where the new electors were densest (Appendix B.6); we hold it as an interpretation, not an identified fact.

This is what separates *institution-driven* from *electorate-driven* polarization. A common shock—a downturn, the agrarian import crisis, the rise of organized labour—hits both houses, loads on the common time path (and, in the saturated specification, the legislature effects δ_ℓ), and differences out of β. Only a force that moved the fully exposed Chamber differently from the less exposed Senate, inside the reform window, survives the contrast; and because the reforms are the salient thing distinguishing the two houses' incentives at exactly those dates, β isolates the reform channel.

This design works better for franchise reform than for changes in the seat formula. The candidacy filter differentiates exposure strongly for a reform of the *electorate*—the enlarged electorate can express its new preferences in one chamber only—and far more weakly for a reform of the *seat formula*, which reallocates seats among candidates who already exist in both houses. The design is therefore systematically more sensitive to demand-side reforms; for the second it is, in addition, exposed to the Senate's own renewal under the new rule in 1900 (§6.6).

We also test this interpretation with two placebo comparisons (§7): the identical estimator on a *territorial* partition of the same speakers finds no break at 1894, and the confessional domain never moves on the fixed yardstick (§4.6). A generic rise in corpus separability, or a confound loading on geography or religion rather than on the reform, would move at least one of them; neither moves.

Both houses touched by 1900, and the gap closes afterwards. Proportional representation reached both houses at once, and the sequence is documented rather than assumed. The proportional rule was adopted in December 1899 (Belgische Senaat, 2019; Velaers, 2019); both chambers were dissolved on 7 May 1900; and the general election of 27 May 1900 was the first held under it. Dissolution is what makes the Senate's expo-

sure immediate: under the electoral code senators sat eight-year terms renewed by halves every four years, but a dissolution triggered a *renouvellement intégral* (Code électoral, art. 248), so all seventy-six directly elected senatorial seats went to the polls under the new rule at its first application—against thirty-seven of seventy-six at the partial renewal of 1898 and forty-three of eighty-three in 1904. Only the provincial senators, chosen indirectly by the provincial councils, stood outside it. There is accordingly no staggered arrival of the rule to lean on, and Panel B's exposure difference has the same source as Panel A's—candidacy law, which the 1899 reform did not alter.

That difference is what the design measures, and it is worth putting a number on it before anything else. The socialist share of interventions is 30.0% in the Chamber against 8.4% in the Senate over 1894–99, and 32.5% against 11.8% after 1900 (Table 1). Before the reform it is zero in both houses. The same electorate, in other words, produced one chamber in which three interventions in ten were socialist and one in which under one in ten was, and it did so immediately and then held a gap of some twenty percentage points.

The gap in speech is what holds, not the seat count, and the two should not be run together. Directly elected socialist senators do rise across the window—zero through 1898, ten by 1912 (Table 30), as candidacy ceases to bind absolutely—so on seats the control is progressively contaminated. On *speech* it does not: the socialist share of Senate interventions goes from 8.4% to 11.8% while the Chamber's goes from 30.0% to 32.5% (Table 1), so the gap is as wide at the end of the window as at the start. The senators are far more numerous and speak proportionally little, which a long mandate, a high age floor and a revising chamber would each predict. Since the outcome is measured on speech, the exposure that identifies the contrast is the one that is stable, and Panel B needs no attenuation-over-time argument. What limits Panel B is stated in §6.6: the reform reached both houses, and the candidacy filter that separates them is the same filter that separates them in Panel A.

Table 1: Socialist share of interventions, by chamber and reform window, and the share of the two governing parties. The same electorate returned both houses after 1893; candidacy rules that stayed restrictive kept the new party largely out of the upper house, and the gap this opened does not close over the window.

	Chamber	Senate
pre-1894	0.0%	0.0%
1894–1899	30.0%	8.4%
post-1900	32.5%	11.8%
Catholic and Liberal share		
pre-1894	98.3%	100.0%
1894–1899	66.0%	91.6%
post-1900	65.5%	88.2%

Share of plenary interventions delivered by members labelled Socialist (upper rows) and by members labelled Catholic or Liberal (lower rows), legislatures 1880–1912, with the speaker links of Appendix B.3 and without interventions from the chair. The pre-1894 row is zero by construction after the party-entry rule of Appendix B.3: a party cannot be spoken before it sits.

Senate placebo at 1894. We can check the comparison directly: if the Chamber–Senate gap reflects differential exposure to the reforms, then re-running the reform "treatment" on the Senate alone should produce nothing. For plural suffrage it does. Estimating the distinctiveness equation on the Senate sample, Post-1894 is -0.0022 (n.s.) against $+0.0071$ ($p < 0.01$) in the Chamber (Table 2), and the wild cluster bootstrap of the Senate placebo does not come close to rejecting the null ($p = 0.88$). If anything the Senate drifts *down* after 1894, so the bicameral gap is not riding a shared rise; and the two within-house changes recompose the bicameral estimate almost exactly ($+0.0071 - (-0.0022) = 0.0093$, against the Panel A contrast of $+0.0094$). For proportional representation the placebo cannot settle the question. The Chamber's own Post-1900 co-

efficient is -0.0060 ($p < 0.01$) and the Senate's $+0.0039$ (n.s.; $p = 0.59$ on the bootstrap; Table 2): the treated house falls back, but about two fifths of the negative Panel B contrast is the control rising, and the 1900 election renewed every directly elected Senate seat under the new rule, so that part is not a common shock the contrast can net out (§6.6).

Table 2: Senate placebo. Reform indicators estimated on the less exposed Senate alone (a valid control shows no break), against the Chamber coefficients. The Senate shows no break at 1894; at 1900, when every directly elected Senate seat was renewed under the new rule, it rises, though not significantly, while the Chamber falls back.

	Senate (placebo)	Chamber
Post-1894	-0.0022 (0.0034)	$+0.0071^{***}$ (0.0020)
Post-1900	$+0.0039$ (0.0028)	-0.0060^{***} (0.0011)
Senate placebo, WCB p, 1894 / 1900	0.881 / 0.590	

Standard errors clustered at arrondissement; WCB = wild cluster bootstrap (Webb weights). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What underwrites parallel trends. Both houses sit in one building, debate the same bills under the same procedure, and face the same national shocks; the pre-reform leads are jointly indistinguishable from zero (§6.3) and the full statement of the assumption is in Appendix A.

Composition of the two houses. Senators are, by recruitment, older, wealthier, and more professionalized than representatives, and one might worry the two houses would respond differently to *any* shock for reasons of composition rather than exposure. The comparison, however, is within-politician: the politician fixed effect in (7) absorbs every time-invariant member characteristic—age at entry, profession, wealth—so a compositional gap between the houses is a level that differences out of β; only a *differential change* in the behaviour of the same members survives the contrast. The Senate placebo above then tests the residual worry directly: whatever its composition, the control house shows no break at plural suffrage.

5.3 Mining intensity as exogenous treatment intensity

To probe *where* a reform bit hardest we add a third difference in the local intensity of the mass electorate, proxied by the coal-mining share of a district's workforce, mine_a, from the 1896 industrial census. Mining intensity is attractive precisely because it is plausibly exogenous to elite rhetorical choices: the location of the Walloon coal belt (the *sillon* Haine–Sambre–Meuse) is set by Carboniferous geology, fixed centuries before our window and unrelated to any politician's vocabulary. Coal districts were the heartland of the new working-class electorate and of socialist mobilization, so mine_a measures the local dose of the reform shock without being chosen by the actors whose behaviour we study. We use it both as a continuous standardized measure (mine_cont) and as an indicator for the top of the distribution (TopMining), and report robustness across the heavy-industry variants.

A caution we honour throughout: an exogenous *cross-section* is not a clean *difference-in-time*. A district's mining share can correlate with a *standing* rhetorical gap—coal deputies may always have spoken differently—which is a level, not a reform break. We test for this directly with event studies and placebo dates, and (Section 6) we find that the Catholic–Liberal mining triple is negative at plural suffrage: the widening is smaller where the new electorate was densest, on the continuous measure and on the TopMining indicator alike. Because the indicator rests on four arrondissements, its leads drift the same way before 1894, and placebo cutoffs on both measures return coefficients of the same sign, we report it in the appendix as a gradient consistent with the contestability account of §6.2 that may partly predate the reform, not as

a second identified effect; in the main analysis the coal districts appear as a count of who represents them (§6.7).

We estimate (7) separately for plural suffrage (Panel A, Post = Post_{1894}, on legislatures up to 1899) and for proportional representation (Panel B, Post = Post_{1900}, on legislatures from 1894). To probe geographic heterogeneity we add a third difference in mining intensity mine_a. Written out in full, the triple-difference is

$$\begin{aligned} s^{(i,s)}_{\text{distinct}} = \beta_3 \left(\text{DK}_i \times \text{Post}_t \times \text{mine}_a\right) + \beta_1 \left(\text{DK}_i \times \text{Post}_t\right) + \beta_2 \left(\text{Post}_t \times \text{mine}_a\right) \\ + \gamma \text{Post}_t + \lambda_p + \mu_a + \theta_{\text{party}} + \varepsilon_{i,s}, \qquad [\text{aw} = w^{(i,s)}], \end{aligned} \tag{8}$$

with standard errors again clustered on arrondissement. The remaining lower-order terms are not omitted but *absorbed*: the chamber level DK_i and the chamber–geology interaction $\text{DK}_i \times \text{mine}_a$ are constant within a politician, because DK_i records the member's house rather than the venue of each speech (§3), and vanish into λ_p, while the mining level mine_a is constant within an arrondissement and vanishes into μ_a. In the specifications without politician fixed effects both enter explicitly. Section A.2 re-estimates on the venue of each speech, where DK does vary within the ministers who speak in both houses, and the plural-suffrage coefficient barely moves. Each surviving coefficient has a reading: β_1 is the chamber-wide reform response of (7); β_2 is the post-reform drift of mining-district speech common to both houses (national coal-belt politics, absorbed as a control); and the triple β_3 asks whether the reform bit *harder* in the exposed chamber precisely where the mass electorate was most concentrated, over and above both.

5.4 The generated regressand

The outcome s_{distinct} is itself estimated, through the leave-out posteriors ρ^{self}_{ij}. We treat the resulting generated-regressand problem deliberately. The posteriors are estimated *once*, leave-speaker-out, on the pooled sample, and then held fixed across the whole panel. Because the same fixed $\hat{\rho}$ enters every arm of the difference-in-differences—Chamber and Senate, pre and post—any first-order error in $\hat{\rho}$ is common across arms and differences out of the contrast that identifies β. The clustered standard errors in (7)–(8) therefore understate, rather than overstate, precision, and serve as a conservative benchmark. For inference that accounts for the estimation of $\hat{\rho}$ directly, we report a block bootstrap that resamples arrondissement clusters and re-estimates both $\hat{\rho}$ and β on each draw, propagating measurement uncertainty into the reform effect.

What a fixed yardstick does. The single pooled yardstick $\hat{\rho}$, fixed across chambers and periods so that it differences out of the bicameral contrast, discards all partisan signal carried by vocabulary that changes sides across the reform, which attenuates; so does the socialist arrival thinning the governing pool, while the leave-out sample-size dependence, which weakens in the Chamber and strengthens in the Senate across 1894, inflates. Two checks bracket the estimate. A ruler that the treated cell cannot calibrate returns four fifths of it, and a vocabulary selected on pre-1894 speech alone between less than half and nine tenths, depending on the ruler and the fixed effects, while a split-sample ruler that keeps the yardstick common returns a quarter more (§A.2; full statement in Appendix A). We therefore claim neither that the estimate is a lower bound nor that it is an upper one.

6 Results

We first show how Catholic–Liberal language changed after the franchise extension, then ask whether that change came from incumbents or new members (§6.2). We next examine timing and topic composition. For the 1900 reform, we separate what the Chamber–Senate estimates show from what they cannot identify

(§6.6). We then document turnover in the coal districts (§6.7) and compare the governing parties with the socialists (§6.8). Additional geographic tests are reported in Appendix B.6.

6.1 The result in one raw picture

Figure 1 plots the token-weighted average of per-speech distinctiveness, s_{distinct}, by legislature and house. This is the same outcome used in the regressions, scored on a fixed vocabulary and weighted by each speech's in-vocabulary token count (§4.3). The raw series combines changes among sitting members with changes in membership; it cannot separate the two.

Three patterns are visible. First, Chamber partisanship is close to 0.500 in 1880–1882 and reaches about 0.503–0.505 in 1884–1892 (Figure 1; Table 57). It rises to 0.510 in 1894, peaks at 0.513 in 1896, and remains at 0.511 in 1898. The two parties contribute different amounts of text: the Catholic share of their combined Chamber tokens rises from 0.64 in 1892 to 0.83 in 1894, following the Liberals' electoral losses. When we weight Catholics and Liberals equally, the increase is smaller: the Chamber rises from 0.505 in 1892 to 0.506 in each of 1894, 1896, and 1898 (Table 57). Party and politician fixed effects address differences in average levels between speakers and parties.

Second, the Chamber measure falls to 0.504 in 1900 and ranges from 0.503 to 0.508 through 1912 (Table 57). Some of the later variation comes from language. Dutch interventions increase after the 1898 language-equality law, and Dutch was spoken disproportionately by Catholic Flemish members. Restricting the analysis to French speeches slightly lowers the post-1900 Chamber level (§A.2).

Third, the Senate does not show a comparable increase in 1894. Its score is 0.496 in 1894, 0.498 in 1896, and 0.501 in 1898 (Table 57). It then rises after 1900, when the new electoral system also applies to the renewed Senate and directly elected socialists begin entering it (Table 30). That increase helps explain the negative Chamber–Senate difference after 1900 (§6.6).

The Senate has relatively few speeches per legislature, so individual points are imprecise. We therefore use pooled regressions for inference. To distinguish changes within politicians from entry and exit, we also report the within-politician series in Figure 3 and the incumbent–entrant decomposition in Section 6.2.

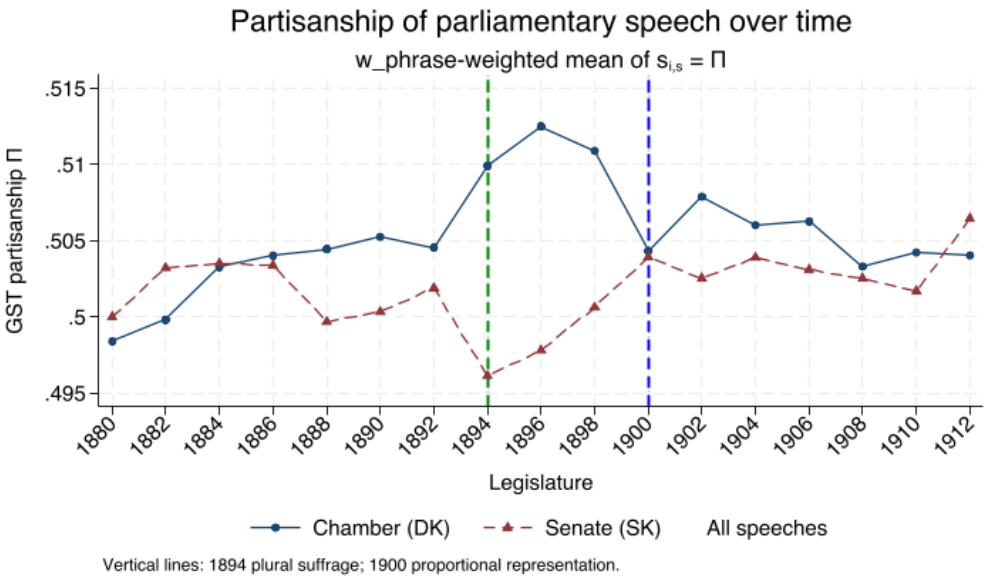


Figure 1: **Partisanship of parliamentary speech by legislature and chamber, 1880–1912.** Each point is the $w^{(i,s)}$-weighted legislature mean of the per-speech leave-out outcome s_{distinct} on the single globally fixed yardstick $\hat{\rho}$, pooled over both governing parties, which are therefore weighted by their token shares rather than equally as in (4) (§4.3); all speeches, each member counted in the house in which he delivers most of his interventions. Vertical lines mark plural suffrage (1894) and proportional representation (1900). The Senate's per-legislature cells are thin, so its series is noisy; identification rests on the pooled difference-in-exposure of §5, not on the visual gap. Produced by `polarization_did.do` (Section 3b).

6.2 The demand-side reform divided the Chamber—through adaptation

Plural suffrage increased Catholic–Liberal distinctiveness in the Chamber relative to the Senate. The estimated Chamber×Post-1894 effect is $+0.0095$ ($p < 0.01$) with arrondissement and party fixed effects, and it is $+0.0094$ ($p < 0.01$) after adding politician fixed effects (Table 3, Panel A, columns 1–3). The change is therefore not explained simply by replacing one set of speakers with another: the same politicians also changed how they spoke.

Table 24 separates incumbents and entrants. The effect among incumbents is $+0.0103$ ($p < 0.01$); the additional entrant effect is -0.0043 and not statistically significant. The party-specific estimates (Table 25) reveal an important difference. Catholic incumbents become more distinctive ($+0.007$, $p < 0.01$), while the Catholic entrant premium is not significant ($+0.006$). Liberal incumbents do not show a comparable change (-0.003, n.s.; Table 25). Instead, newly elected Liberals are more distinctive than incumbents ($+0.014$, $p < 0.10$; Table 25). The Catholic majority appears to have adjusted its language while remaining in office. For the Liberals, reduced to twenty members in a Chamber of 152 in 1894, the change is more closely associated with the new members.

Timing supports the interpretation that the franchise extension mattered. The event study shows no statistically significant pre-reform leads and a break in 1894 (§6.3). Among fourteen possible cutoff dates, 1894 produces the largest estimated break (Table 6). Across two- and three-legislature windows, a donut specification, and a window that includes 1900, estimates stay between $+0.007$ and $+0.009$ (all $p < 0.05$; Table 9). The penalized Poisson–LASSO estimator gives $+0.0094$ ($p < 0.01$; Table 17). A cluster bootstrap that re-estimates the text measure gives a mean effect of 0.014 and a 95% interval of $[0.006, 0.021]$ (Table 18). Excluding the treated cell from the scoring yardstick, or selecting the vocabulary only from

pre-1894 speeches, reduces the estimated effect (§A.2).

The regression standard errors require an important qualification. Treatment varies across two houses, not independently across the forty-one arrondissements used for clustering. The conventional standard errors and wild cluster bootstrap ($p = 0.021$ with politician fixed effects and 0.001 without them; §7.1, Table 10) rely on a stronger independence assumption than the design itself provides. We therefore report two exact checks. First, 1894 ranks first among fourteen possible cutoff dates ($p = 1/14 = 0.071$, the smallest possible value for this test; Table 6). Second, among 133 politicians speaking on both sides of the reform, a studentized chamber-label permutation gives $p = 0.032$ when speeches are weighted by volume and $p = 0.36$ with equal member weights (§7.1; Tables 66 and 67). The estimates are robust across specifications, but the evidence is stronger for frequent speakers than for the average member.

How large is the increase of 0.0094 (Table 3, column 3)? It amounts to about 0.40 of a speech-level standard deviation of s_{distinct} (SD = 0.023) and a similar fraction of the politician-level standard deviation (SD = 0.023; Table 57). On this scale it is a meaningful, though not extreme, change. The outcome is noisy at the speech level ($R^2 \approx 0.06$–0.15; Table 3).

The index also has a useful reference point: 0.500 means that phrases do not reveal party affiliation. Before 1894, the Chamber was only 0.0030 above that benchmark (Table 57). Adding the estimated effect would put it about 0.012 above chance, roughly four times the initial excess. The observed 1894–99 excess is 0.0110 (Table 57), closer to three and two thirds times the pre-reform value. These ratios depend heavily on a small baseline; changing it to 0.0020 or 0.0040 would move the first multiplier to about five and a half or three and a third. We therefore emphasize the standardized coefficient and use the ratios only to explain the index's scale.

Calibration against legislative outcomes. We also test whether the change in language is associated with legislative output: Appendix A assembles the 2,917 bills deposited over the window and asks whether more distinguishable legislatures adopt less of what is put before them. They do not: pooled, the association does not even have the gridlock sign, and disaggregated by bill type and by the author's party no coefficient is distinguishable from zero. The index measures how the parties *spoke*, and we can attach no legislative-output consequence to it.

Table 3: Catholic–Liberal distinctiveness: bicameral difference-in-differences and mining triple-difference (mining share, continuous standardized)

	DiD			DDD		
	(1)	(2)	(3)	(4)	(5)	(6)
	FE	FE + Turn.	Pol. FE	FE	FE + Turn.	Pol. FE
Panel A: pre-1894 vs. 1894–1899						
Post-1894	-0.003	-0.002	-0.002	0.000	0.001	0.001
	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)
Chamber × Post-1894	0.010***	0.010***	0.009***			
	(0.002)	(0.002)	(0.003)			
Chamber × Post-1894 × Mining				-0.006***	-0.005***	-0.003***
				(0.001)	(0.001)	(0.001)
Incumbent at 1894		0.005**			0.004	
		(0.002)			(0.002)	
Entrant after 1894		0.000			0.000	
Observations	206146	206146	206116	206146	206146	206116
R^2	0.059	0.061	0.146	0.068	0.069	0.148
Panel B: 1894–1899 vs. 1900+						
Post-1900	0.006**	0.005*	0.004	0.004*	0.003	-0.000
	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)
Chamber × Post-1900	-0.011***	-0.011***	-0.010***			
	(0.002)	(0.002)	(0.003)			
Chamber × Post-1900 × Mining				0.004**	0.004*	-0.000
				(0.002)	(0.002)	(0.001)
Incumbent at 1900		-0.001			-0.001	
		(0.001)			(0.001)	
Entrant after 1900		0.000			0.000	
Observations	245296	245296	245268	245296	245296	245268
R^2	0.053	0.053	0.117	0.055	0.055	0.118
Party FE	Yes	Yes	Yes	Yes	Yes	Yes
Politician FE	No	No	Yes	No	No	Yes
Arrondissement FE	Yes	Yes	Yes	Yes	Yes	Yes

Outcome is the per-speech leave-out distinctiveness s_{distinct}, weighted by the in-vocabulary token count $w^{(i,s)}$; standard errors clustered at arrondissement in parentheses. Produced by `polarization_did.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.3 The timing of the break: an event study

The difference-in-differences coefficient averages across the post-reform legislatures. Figure 2 instead estimates the within-politician Chamber×legislature interaction for each election, using 1892 as the reference. The coefficient table is in Appendix B.1 (Table 7).

The six pre-reform coefficients, from 1880 through 1890, range from -0.0024 to $+0.0004$ (Figure 2; Table 7, column 1). None is individually significant, and they are jointly indistinguishable from zero ($F(6,40) = 0.79$, $p = 0.58$). The first positive jump appears in 1894: $+0.0093$ ($p < 0.05$), followed by

+0.0097 in 1896 ($p < 0.01$) and +0.0051 in 1898 ($p < 0.05$; Table 7). The effect thus lasts through the three legislatures between the reforms.

After 1900, the interactions return to the 1892 level or slightly below it. They range from −0.0031 to +0.0009 through 1910; two of them are individually significant and negative (−0.0026 in 1900, $p < 0.10$, and −0.0031 in 1904, $p < 0.05$), and the 1912 coefficient is −0.0069 ($p < 0.01$; Table 7, column 2). Since the Senate itself changes after 1900 (§6.6), these later interactions cannot be interpreted as the Chamber's response alone.

The pre-reform trajectory is also nearly flat under a linear specification: +0.0003 per legislature (s.e. 0.0005; Table 7). This matters because strikes and socialist mobilization intensified in 1891–1893, before the new franchise was used. The main increase in partisanship appears with the 1894 election, not before it. On French-language speech alone, the three post-1894 coefficients remain positive (+0.009, +0.011, +0.006; Table 61), but the pre-reform coefficients trend upward and jointly reject zero. We therefore treat the French-only results as supportive of the 1894 break, not as an equally clean parallel-trends check (§A.2).

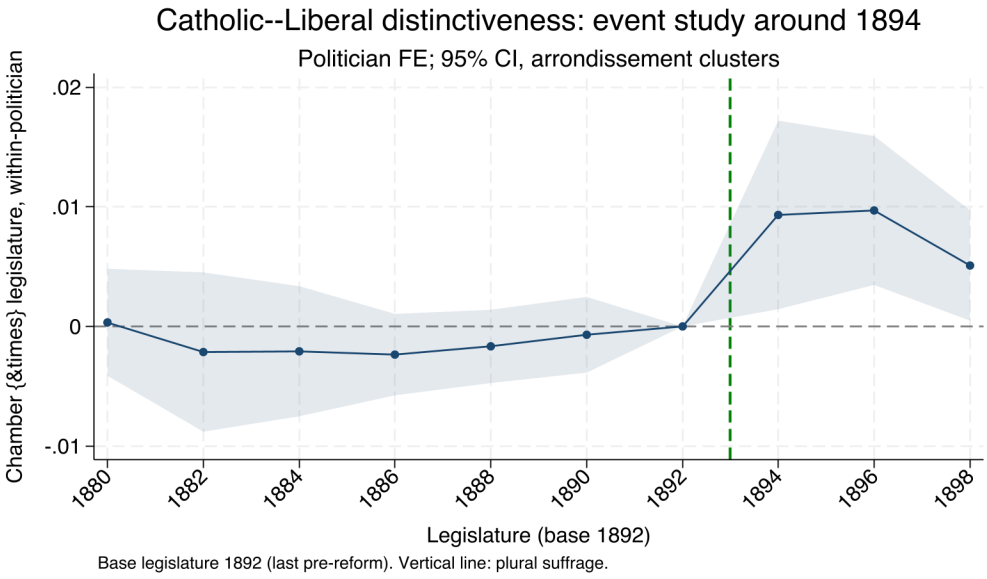


Figure 2: **Catholic–Liberal distinctiveness: event study around plural suffrage, 1880–1898.** Points are within-politician Chamber × legislature interactions with 1892 as the omitted base; bands are 95% confidence intervals from standard errors clustered at arrondissement. The vertical line marks plural suffrage (1894). Leads are jointly indistinguishable from zero ($F(6,40) = 0.79$, $p = 0.58$); the break is at the reform. Produced by `event_study_main.do`.

6.4 Within-topic divergence versus agenda reallocation

The partisanship index includes phrases that are especially common in labour and social-policy debates. If the Chamber discussed more labour issues after 1894, the index could rise even without any change in how Catholics and Liberals spoke *within* each issue. This is a serious possibility: our companion study finds that incumbents devoted more attention to labour policy after enfranchisement (Graziani and Vanschoonbeek, 2026). Politician fixed effects alone cannot distinguish a change in language from a change in topics discussed.

We therefore separate the reform effect into changes *within* policy topics and changes in the *share* of attention given to each topic. Speeches are assigned to topics with a CamemBERT classifier trained on

hand-labelled interventions; details are in Appendix A.

The within-topic component accounts for 78.6% of the increase using the share decomposition and 94.1% using the regression decomposition (Table 19). The parties therefore became more distinctive mainly when discussing the same subjects, rather than by switching to different subjects. This is why the polarization result is not simply a restatement of the labour-salience result in Graziani and Vanschoonbeek (2026).

6.5 Where the parties diverged: polarization by topic

By topic, the plural-suffrage widening loads on the distributive domains—labour above all—while the confessional domain never moves on the fixed yardstick (by-topic table in Appendix A). The specificity is identification evidence and is used in §5.2.

6.6 The supply-side reform changed who sat; the design cannot say how they spoke

For the 1900 reform, the Chamber–Senate difference is negative: -0.011 ($p < 0.01$) in the fixed-effects specification and -0.010 ($p < 0.01$) with politician fixed effects (Table 3). The estimate is similar across vocabulary choices (Table 52) and in both halves of the post-1900 window. It is not, however, a causal estimate of what the reform did to Chamber speech.

Estimated separately, the Chamber's Post-1900 coefficient is -0.0060 ($p < 0.01$), while the Senate's is $+0.0039$ (not significant; wild cluster bootstrap $p = 0.59$; Table 2). About two fifths of the negative Chamber–Senate difference comes from the Senate's increase. That matters because the 1900 election followed the dissolution of both houses. Every directly elected Senate seat was contested under proportional representation (§5.2). The comparison house was therefore also treated. Neither its increase nor the Chamber's before–after decline can isolate the reform's causal effect on polarization.

We can still describe how membership changed. An accounting decomposition (Table 4) separates changes among continuing members from the contributions of entrants and departures. In 1894, 0.0070 of the 0.0116 bicameral change came from continuing speakers, about three fifths of the total. In 1900, by contrast, the Chamber's own change is -0.0060 and the Senate's is $+0.0054$ (Table 4). New senators account for $+0.0046$ of the Senate's increase (Table 4). Members who were not in the Senate before 1900 delivered 82% of its post-1900 in-vocabulary tokens, compared with 47% in the Chamber (Table 4). Overall, entry accounts for -0.0076 of the -0.0114 bicameral difference (Table 4). These are accounting contributions, not separate causal effects.

Within the Chamber, the incumbents fall back at 1900 relative to their Senate counterparts (-0.014, $p < 0.01$; Table 24), the Catholics more than the Liberals (-0.0095, $p < 0.05$, against -0.0030, n.s.; Table 25). Entrants exist only after the reform, so their coefficient is a level gap at debut rather than a change ($+0.007$, $p < 0.01$); measured against the incumbents of their own house they debut no more distinctive and do not drift with tenure (premium at debut $+0.0003$, $p = 0.84$; tenure slope $+0.0003$, $p = 0.25$; Table 44). The debut-cohort series agrees: the Chamber cohort first returned in 1900 debuts at 0.503, below the cohorts of 1894 and 1896 (0.507 and 0.514), and pooling the cohorts of 1900–1912 against those of 1894–1898 gives a Chamber–Senate difference of -0.0094 (s.e. 0.0064; Table 68).

The clearest change in 1900 is in parliamentary membership. The 1900 election renewed 29.4% of the Chamber's members, 2.3 standard deviations above the mean of the non-reform elections—a renewal the window exceeds only once, at plural suffrage (30.3%, $z = 2.4$; Table 5); on in-vocabulary tokens it is 20.7% ($z = 2.5$), against 28.7% in 1894. In the Senate, which went to the polls in full, 52% of the members who spoke in the 1900 legislature were new (note to Table 5). What sets the 1900 renewal apart is therefore not its size in the Chamber but where it fell: in the Senate, and in the coal districts. The coal districts show the recomposition most sharply (§6.7), and the companion study finds the same channel for proportional representation on labour salience: geographically concentrated recomposition rather than

convergence, consistent with elections selecting representatives rather than inducing incumbents to converge (Lee et al., 2004; Graziani and Vanschoonbeek, 2026).

We therefore state the second result within the limits of the evidence. The rule change rotated the personnel of both houses; whether it also moved the distance between the governing parties in the Chamber the bicameral contrast cannot say, because at this date it compares two treated houses. A design for proportional representation would need a control that the 1900 election did not renew.

Table 4: Where each chamber's change in partisanship comes from. An exact identity, not a regression: the change in Π across each reform splits into what members present on both sides did (adaptation), what entrants brought relative to the pre-reform level (entry), and what leavers took with them (exit). The bicameral difference inherits the split.

	Plural suffrage (1894)			Proportional representation (1900)		
	Chamber	Senate	Diff.	Chamber	Senate	Diff.
Total change in Π	0.0080	−0.0036	0.0116	−0.0060	0.0054	−0.0114
Adaptation (stayers)	0.0056	−0.0014	0.0070	−0.0028	0.0008	−0.0035
Entry (vs. pre level)	0.0020	−0.0007	0.0027	−0.0030	0.0046	−0.0076
Exit	0.0004	−0.0015	0.0019	−0.0002	0.0000	−0.0003
Adaptation share of diff.		60.7%			31.1%	
Entrant token share w_E	0.310	0.171		0.472	0.817	

Token-weighted cell means on the globally fixed yardstick $\hat{\rho}$; the three components sum to the total change to machine precision, which the code asserts. In 1900 the bicameral difference is driven by the Senate's entrants, and since the Senate was itself renewed under the new rule it is not an estimate of the reform's effect on the Chamber (§6.6). Produced by `recomposition_accounting.do`.

6.7 What the coal districts actually show

The coal districts offer direct evidence of turnover: Catholic members are all but absent from the mining belt while majority rule fills its seats—Liberal before 1894, largely socialist from 1894, with at most two Catholic deputies per legislature before 1894 and none at all in the six years between the reforms—and once proportional representation gives the local minority seats, as many as twelve Catholic deputies per legislature speak for it, 18,312 times over 1900–1912. The same counts rule out reading the districts through a lexical accommodation-and-retreat mechanism: the window holds no Catholic mining deputy at all, so a within-group coefficient for it is not identified. Appendix A.1 reports the full account: the counts by window, the socialist-proximity coefficients beside them, and the contestability interpretation, whose non-monotone dose prediction the negative mining triple of §5.3 is consistent with but cannot by itself establish.

6.8 Pairwise polarization, 1894–1912

Socialist speech is more distinctive than that of either established party in the Chamber. Scored on the paper's vocabulary with posteriors estimated on post-1894 speech, Catholic–Socialist partisanship sits at 0.5157–0.5211 across the post-1894 windows and Liberal–Socialist at 0.5037 in 1894–99 and 0.5136 after 1900, against 0.5015–0.5051 for Catholic–Liberal (Table 22). On a vocabulary built on the three-party pool itself, pooled over 1894–1912, both socialist dyads lie well above their permutation bands (0.519 and 0.516, against upper bounds of 0.504 and 0.503), and the Catholic–Liberal comparison also exceeds its band, though by less (0.508 against 0.503; Figure 6 and Table 22). The reform-timed movement the design identifies is nonetheless the governing parties' (full tables and by-legislature trajectory in Appendix A).

Table 5: Renewal of the Chamber at each election. Entrants are members whose first sitting is that legislature; z is computed against the mean and standard deviation of the non-reform elections.

Election	Entrant share		z	
	tokens	heads	tokens	heads
1892	0.193	0.219	2.19	1.19
1894	**0.287**	**0.303**	**4.07**	**2.44**
1896	0.079	0.087	−0.10	−0.80
1898	0.091	0.145	0.14	0.08
1900	0.207	0.294	2.48	2.31
1902	0.015	0.043	−1.39	−1.44
1904	0.081	0.122	−0.05	−0.27
1906	0.052	0.068	−0.65	−1.08
1908	0.035	0.122	−0.99	−0.26
1910	0.031	0.066	−1.06	−1.10
1912	0.113	0.207	0.58	1.01

Chamber only; 1880–1890 omitted from the display. The token measure is the one the outcome is weighted by. In the Senate, renewed in full in 1900, entrants are 52.4% of the members who spoke in the 1900 legislature and 41.7% of its in-vocabulary tokens. Produced by `turnover_base_rate.do`.

7 Robustness

The plural-suffrage estimate is similar across several time windows and estimators. With two- or three-legislature bandwidths, a donut that omits the adjacent legislatures, and a window extending to 1900, it remains between +0.007 and +0.009 (all $p < 0.05$; Table 9). The penalized Poisson–LASSO estimator gives +0.0094 ($p < 0.01$). Re-estimating the phrase scores in a cluster bootstrap gives a mean of 0.014 with a 95% interval of $[0.006, 0.021]$ (Table 18). A yardstick that excludes the scored cell and a vocabulary selected before 1894 give smaller effects; a split-sample yardstick gives a larger one (§A.2).

The mining results are less straightforward. The continuous Chamber×Post-1894×Mining effect is negative (−0.006 without and −0.003 with politician fixed effects, both $p < 0.01$; Table 3, columns 4 and 6). The TopMining indicator also gives a negative average (−0.004, $p < 0.05$; Table 31), including in the three legislatures after 1894. But it covers only four arrondissements, and its pre-reform coefficients tend downward. Placebo cutoff dates show a similar tendency. We therefore report mining as a geographic gradient, not as another causal reform estimate (Appendix B.6).

Finally, we rebuild the vocabulary after successive cleaning steps. The 1894 Chamber–Senate increase survives removing surnames, parliamentary boilerplate, lemmatization fragments, and party names. Estimates that shrink markedly when these features are excluded receive less interpretive weight.

Specificity: the reforms widen parties, not places. One concern is that speeches became easier to classify after 1894 for reasons unrelated to party competition. If so, differences between other groups of speakers should rise as well. We test this directly by re-running the entire leave-out estimator with the partition redefined: instead of splitting speakers by party, we split them by whether their arrondissement is a coal-mining district, and measure how distinctive mining members are from the rest *within their own party*, on the same fixed pooled yardstick, the same vocabulary, and the same bicameral difference-in-differences (Appendix B.7). If the post-1894 widening were generic, this territorial axis would widen too. Within party it does not. On plural suffrage the within-party interaction is negative and insignificant (Table 40: −0.007, and −0.003 within politician), and the event study (Table 41) shows no break at 1894 (+0.002 in 1894,

-0.003 and -0.002 in 1896 and 1898, none significant). Pooled across the two parties the mining members do become more distinguishable at plural suffrage ($+0.006$, $p < 0.10$; $+0.008$, $p < 0.05$, within politician), but between the reforms the Chamber's coal-district benches held no Catholic (Appendix A.1), so the pooled partition there is partly the party partition, and the within-party columns are the ones that answer the worry. The territorial axis does carry its own structure, and it runs the other way: the leads are small and insignificant ($+0.006$ and $+0.004$ in 1880 and 1882; Table 41), and after 1900 the mining members of each party become less distinguishable from the rest of it (-0.011 to -0.015 from 1902; Table 41; -0.012, $p < 0.01$, within party and within politician at proportional representation; Table 40). That this axis does not widen—within politician, exactly where the partisan axis breaks—is the specificity the argument needs: plural suffrage widened the distance between *parties*, not the distance between *places*.

Does the yardstick score itself? A five-fold split-sample ruler, which removes each speaker's own text from the posteriors that score him while keeping the yardstick common across the arms, returns more than the estimate (127% of the baseline, $p = 0.013$ on the wild cluster bootstrap; Table 13). A leave-cell-out ruler, which removes the whole chamber×legislature cell so that the treated cell takes no part in its own measurement, returns 78% ($p = 0.035$; Table 13), and a vocabulary selected on pre-1894 speech alone between three quarters and nine tenths on the split-sample ruler and between a little under a half and two thirds on the leave-cell-out one (Table 64). Part of the estimate may therefore come from the treated cells calibrating their own ruler, but not most of it; the construction, the full results and the two readings they admit are in Appendix A.

Language is not driving the result. The corpus is overwhelmingly francophone and the Dutch share stays between 0.4% and 0.8% of interventions through 1896, stepping up only in 1898 (Table 12), so no linguistic recomposition coincides with the break. Re-estimated on French text alone, with the vocabulary rebuilt on that text, the effect is 114% of the baseline (Table 60), and the restriction is close to neutral at the reform; on that sample the pre-reform leads drift upward, which we report with the estimate (Appendix A).

Venue or membership? The ministers. The chamber indicator records the member's house, the one in which he delivers most of his interventions, while ministers speak in both houses (§3). Re-estimating on the venue of each speech, with the interventions delivered outside the member's house dropped, or with the house in which the career file places him on the day he spoke, moves the coefficient by at most 0.0021, under a quarter of the effect (Table 14). Dropping the members who speak in both houses altogether is a different matter, and we report it as such: they carry seven in ten Panel A speeches, and without them the effect is still nearly nine tenths of the reference ($+0.0083$, $p < 0.05$; Table 14). The full audit is in Appendix A (Table 14).

Divergence, or enforcement? The second moment. Canen et al. (2026) show that polarization measured on roll calls confounds two objects—the ideological distance between legislators and the pressure a party leadership exerts on its own rank-and-file—and estimate that pressure accounts for more than two thirds of measured U.S. polarization by 2018. Their explicit target is the class of roll-call models that give party organizations no formal role, but the objection carries to any measure of the same kind, and the leave-out speech index used here is one. The objection is specific: the movement documented above might not be Catholics and Liberals diverging, but their parties beginning to discipline how they speak.

The two accounts agree on the first moment and disagree on the second, in opposite directions. Enforcement works by compressing the rank-and-file onto a line, so it widens the distance *between* parties while narrowing the spread *within* them—which is also why Canen et al. (2026), following Aldrich (1995) and

Rohde (1991), find party strength co-moving with within-party homogeneity. Electoral adaptation predicts no such compression: members answer to districts that differ, so the within-party spread stays flat or widens.

We estimate the second moment on the same design. Collapsing to politician×legislature means, taking each member's absolute deviation from his own party's contemporaneous centroid in his own house—computed leaving that member out, as the index itself is—and running the bicameral contrast with politician, party and legislature fixed effects gives -0.0073 (s.e. 0.0058) on 1,627 politician-cells (Table 65, column 1). The point estimate is a narrowing of about four fifths of the between-party effect, and it is not significant; but the interval $[-0.019, +0.004]$ is too wide to rule out a compression as large as the effect itself, so on this moment the test does not decide between the two accounts. The median absolute deviation, which tracks the body of each party rather than its tails, does not move ($+0.0002$, s.e. 0.0017; Table 65, column 2). Across legislatures the two moments do not co-move in the direction enforcement implies: the correlation between between-party distance and within-party spread is essentially zero in the Chamber (-0.03) and positive in the Senate ($+0.15$) over seventeen legislatures (Table 65), where enforcement implies a clearly negative one.

Two things make this a stronger answer than a time series can give. The Catholic and Liberal organizations were *unitary across both houses*—one leadership, one line—so an explanation based on party enforcement predicts compression in the Senate as well, and the bicameral contrast differences that channel out; what it leaves is the electoral incentive, which the reform changed in one house only. And the claim is the absence of a significant narrowing rather than a break, so it does not rest on a pre-trend. The cell-level measures—the dispersion of politician means within each party×house×legislature, raw and adjusted—point the same way as the politician-level one and are no more precise (raw: -0.008, s.e. 0.008; Table 65, column 5), so the second moment leaves the enforcement account weakened rather than excluded.

Weighting by speech length. The outcome is token-weighted because within each party the weighted mean reproduces the GST aggregate and weighting is precision weighting for a generated regressand; the cost is concentration—a Kish effective size of 28 at the politician level. The politician-level diagnostics of §7.1 address that concern; the full discussion, with the unweighted replication, is in Appendix A.

7.1 Few clusters: what survives the wild cluster bootstrap

The whole analysis clusters on arrondissement, of which Belgium has only 44 in our window—and of which 41 carry identifying variation in the estimation sample, the remaining three contributing no speech that survives the constituency crosswalk. With so few clusters, analytic cluster-robust standard errors can be anti-conservative (Cameron et al., 2008; Bertrand et al., 2004)—and serially correlated speech within a district compounds the problem—so we re-test every headline coefficient with the wild cluster bootstrap, using Webb weights and 9,999 replications. Table 10 reports the result. The plural-suffrage difference-in-exposure survives it at the five per cent level on the most demanding, politician-fixed-effects specification ($p = 0.021$) and at $p = 0.001$ with fixed effects only, and the Senate placebo stays far from rejection ($p = 0.88$), so the control does not react to 1894 even under the most demanding inference. The socialist-proximity coefficients do not clear the same bar, and they are not identified in the form a mechanism would need: for Catholic speech from the four TopMining arrondissements the 1894–99 coefficient is not identified with politician fixed effects, and the post-1900 one, which within a member rests on a single deputy whose recorded district changes after 1900, is small and imprecise (-0.002, s.e. 0.004; $p = 0.68$ on the bootstrap; 0.28 and 0.84 for the two legs on the continuous measure, which spreads the variation over all the arrondissements without rescuing it; Table 10). We read this as decisive rather than as a matter of degree. The causal claim of the paper is the plural-suffrage effect on Catholic–Liberal distinctiveness. The socialist-proximity coefficients are not a second claim at all: §6.7 and Appendix A.1 show that the window on which the first would be estimated contains no Catholic deputy for the four arrondissements, and, since the second is estimated from

1894 on and no Catholic deputy for the four sat between the reforms, within a member it moves only for that one deputy. Neither compares the same members across a reform, and the bootstrap failure is what one should expect of them.

Two groups, not forty-one clusters. The bootstrap above resamples forty-one arrondissements, but assignment does not vary within a chamber: every Chamber member is exposed and every Senate member is not, so the identifying variation is a single 2×2 contrast and the arrondissements are not independent draws on treatment. Cluster-robust inference of any flavour, bootstrapped or not, cannot manufacture the independence the design does not have; this is the problem Conley and Taber (2011) and Ferman and Pinto (2019) address, and with a single treated group neither of their procedures has resolution here (Conley–Taber's feasible form admits a smallest p of $1/9$ with the eight control differences the design supplies), and the placebo-based alternative of Hagemann (2019) needs more clusters than the design supplies; randomization inference with few treated clusters (MacKinnon and Webb, 2020) points the same way. We therefore add an exact test that makes no cluster-count assumption, in the approximate-symmetry framework of Canay et al. (2017). Collapsing to politician$\times$period means and keeping the 133 members who spoke on both sides of 1894, the within-politician contrast is the difference between the average change of Chamber members ($n = 102$) and that of Senate members ($n = 31$). A member is assigned to one house for the whole window—his house of membership, the one in which he speaks most (§3)—because the test permutes the label *across* politicians and a time-varying label is not exchangeable in that way. Under the sharp null that the reform moved no member differentially by house, the chamber label is exchangeable, so permuting it across politicians delivers the exact null distribution of that difference. We studentize, because the two groups have unequal variances and the raw difference would inherit fat tails that reflect the thinner control rather than the design (Janssen, 1997; Chung and Romano, 2013).

The result depends on how members are weighted, and we report both. Weighting each member by his speech, as the regressions do—by the in-vocabulary tokens of the thinner of his two sides—the contrast is $+0.0097$ and the studentized permutation gives a two-sided p of 0.032 on five thousand draws (Table 66). Weighting members equally, it reverses: the average change of Chamber members is $+0.0026$ and that of Senate members $+0.0123$, a few senators moving a long way (their median change is $+0.0037$), and the two-sided p is 0.36 on twenty thousand draws (Table 67). The test that makes no cluster-count assumption therefore certifies a divergence of speech, carried by the members who speak most, not a movement of the median member, and we keep the wild cluster bootstrap ($p = 0.021$; Table 10) beside it. Clustering in two dimensions, arrondissement and legislature, leaves the coefficient unchanged at $+0.0094$ with $p = 0.010$ (Table 67); we report it as a sensitivity rather than a remedy, since the second dimension supplies only ten clusters in Panel A. We do not report one-sided values, because the direction is not pre-registered by anything other than the hypothesis being tested.

The number is what a two-group design yields, and it is worth being precise about what it does and does not say. It asks whether a different set of members would have shown the same divergence—a question about group membership. It makes no use of the fact that the divergence begins at the reform and at no other date, which is the question §6.2 answers with an exact test: the 1894 cutoff produces the largest break of any of the fourteen feasible ones. That test bears directly on the identifying claim; this one bears on a claim the design does not make. Three further facts bound the concern: the reversal of the unweighted comparison is carried by its thinnest members—once the decile of members with the least speech is trimmed, the unweighted contrast is $+0.0016$ ($t = 0.7$), while the contrast that weights each member by his total speech barely moves ($+0.0056$ to $+0.0058$; §A.2, Table 67)—so the weighting decides between a positive and a null contrast, not between a positive and a negative one; the leave-one-arrondissement-out range is $[0.0051, 0.0103]$, with a median of 0.0094 and one district moving it materially—Liège, without which it falls to 0.0051 (without Kortrijk it rises to 0.0103); and the largest break the same specification produces at any cutoff inside the two

reform-free windows is 0.0040, over two fifths of the estimate (Table 66). We accordingly report the coal districts as a change in representation, which the counts establish without inference, and drop the rhetorical reading the coefficients were carrying.

Table 6: Inference on timing. The first row re-estimates the bicameral interaction at every feasible cutoff and reports the rank of the true reform date; the second reports the largest relative-magnitude violation of parallel trends under which the identified set still excludes zero.

	Statistic	Exact p
Rank of the 1894 break among all feasible cutoffs	1 of 14	0.071
smallest attainable with 14 cutoffs		0.071
1894 effect, conventional 95% interval and p	$[0.001, 0.017]$	0.026
HonestDiD breakdown value $\bar{M}$	0.75	
identified set at $\bar{M} = 1.0$	$[-0.001, 0.021]$	
identified set at $\bar{M} = 1.5$	$[-0.003, 0.023]$	

Date randomization inference over every legislature leaving two periods on each side, 1880–1912; politician, arrondissement and party fixed effects, speeches weighted by in-vocabulary token count. Rambachan and Roth (2023) relative-magnitude restrictions on the within-politician event study, six pre-reform leads against three post-reform effects, 1892 omitted; the breakdown value is the largest $\bar{M}$, on a grid of 0.25, at which the lower bound of the identified set is above zero (at $\bar{M} = 0.75$ it is 0.000 to three decimals). Conley–Taber ($p = 0.22$) is not reported in the table: with a single control group and eight control differences its feasible form admits a smallest p of $1/9$ and has no resolution. Produced by `date_randomization.do`.

7.2 OCR quality

Both corpora are OCR-derived and the Senate's scans are older; what protects the contrast is that the control's error process is stable over time, so it differences out, and a fixed-vocabulary degradation exercise in Appendix A shows that corrupting up to 10% of the Chamber's characters leaves the Chamber–Senate gap in Π between 0.0035 and 0.0038, against 0.0036 on the clean text (Table 21).

8 Conclusion

The Belgian reforms of 1894 and 1900 changed parliamentary politics in different ways. The extension of the franchise made Catholic and Liberal speakers in the Chamber more distinctive from one another, relative to the Senate. This was largely a change in the speech of politicians already in office. Catholic incumbents changed their language; among Liberals, the more distinctive voices came mainly from the members elected after the franchise extension. Most of the divergence occurred within policy topics, particularly labour and social policy, rather than because the parties turned to different subjects.

The timing supports an interpretation centred on the reform. The gap opens in 1894, not during the preceding years of labour mobilization, and the same politicians account for much of the change. The estimate is similar under several specifications and measurement approaches, though its size depends in part on how the vocabulary and scoring yardstick are chosen. Inference also requires care: with only two houses, conventional clustering at the arrondissement level does not provide independent treatment-level variation. The cutoff-date and member-permutation tests therefore matter alongside the regression estimates (Sections 6.2 and 7.1). The latter test finds stronger evidence when members are weighted by the amount they speak than when each member receives equal weight.

We cannot make the same causal claim about proportional representation. Both houses were renewed under the new electoral rule in 1900. The Chamber's partisanship measure falls, but the Senate's rises, so

their difference cannot isolate the effect of the reform on parliamentary language. What we can observe is substantial turnover. Catholics regained representation in coal districts where majoritarian elections had largely excluded them. The reform changed which politicians could speak for those districts, even if its effect on the language of a given politician remains uncertain.

The distinction matters beyond Belgium. Research that treats electoral reform as a single event risks combining changes in political incentives with changes in the composition of the legislature. Comparing differently exposed houses can help separate reform effects from national shocks, but only when the comparison house is sufficiently less affected and is not itself reorganized by the reform. The Belgian Senate provides useful variation for the 1894 franchise extension, not a clean control for the 1900 seat-allocation change.

Finally, our measure captures differences in parliamentary language, not necessarily ideological distance, party discipline, or legislative gridlock. We find no robust link between the rise in rhetorical distinctiveness and the adoption of bills. Our companion study documents the related change in attention to labour and social policy (Graziani and Vanschoonbeek, 2026). Together, the findings show that electoral institutions can change both what established politicians say and which politicians have the opportunity to speak. A comparison across other bicameral systems, together with more direct measures of local socialist competition, would help establish how far the Belgian pattern generalizes.

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A Additional measurement, calibration and robustness material

Table 7: Event study on the Catholic–Liberal axis, base legislature 1892. Column (1) is the clean pre-1900 window; column (2) extends to 1912: after 1900 the coefficients return to the 1892 base or slightly below it (-0.0031 to $+0.0009$ through 1910, -0.0069 in 1912), as the Chamber falls back and the Senate's renewal in 1900 moves the control (§6.6). On French speech alone the post-1900 coefficients lie between -0.008 and $+0.000$ (Table 61).

	(1) 1880–1898	(2) 1880–1912
1880	0.000353 (0.00228)	0.00137 (0.00241)
1882	-0.00214 (0.00340)	-0.000766 (0.00340)
1884	-0.00207 (0.00277)	-0.00137 (0.00273)
1886	-0.00235 (0.00173)	-0.00180 (0.00167)
1888	-0.00167 (0.00156)	-0.00118 (0.00159)
1890	-0.000693 (0.00161)	-0.000280 (0.00144)
1894	0.00932** (0.00402)	0.00897** (0.00380)
1896	0.00970*** (0.00317)	0.00983*** (0.00286)
1898	0.00510** (0.00235)	0.00490** (0.00210)
1900		-0.00258* (0.00139)
1902		0.000938 (0.00171)
1904		-0.00309** (0.00131)
1906		0.000556 (0.00246)
1908		-0.00256 (0.00194)
1910		-0.00115 (0.00196)
1912		-0.00694*** (0.00218)
Constant	0.504*** (0.00128)	0.505*** (0.00108)
Observations	206116	392051
R^2	0.153	0.122

Within-politician specification with politician, arrondissement, party, and legislature fixed effects; weighted by in-vocabulary token count; standard errors clustered at arrondissement. Joint test of the six pre-reform leads in column (1): $F(6,40) = 0.79$, $p = 0.58$. A linear Chamber×legislature trend estimated on the pre-reform legislatures (1880–1892): $+0.0003$ per legislature (s.e. 0.0005). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.1 The coal districts in full

The mining results invite a reading in terms of rhetoric. Scoring each Catholic or Liberal Chamber speech for how socialist its phrases sound (the proximity outcome of (15)), Catholic speech from the mining districts would move closer to the socialist register in 1894–99 and further from it after 1900: a courtship of the newly enfranchised worker followed by a retreat once proportional representation removed its reward. The membership data show why that reading fails. Its first leg has nothing to stand on: the corpus records no Catholic deputy for the mining districts between the reforms, so with politician fixed effects the coefficient it needs is not identified at all. The post-1900 leg is estimated from 1894 on, and no Catholic deputy for these districts sat between the reforms, so within a member it moves only for the few members whose district changes across 1900: it is small and imprecise (-0.002, s.e. 0.004; Table 10), and with arrondissement fixed effects it is not identified either (Appendix B.8).

Member counts clarify the problem. Table 8 gives Catholic presence in the four TopMining arrondissements—Charleroi, Liège, Mons and Soignies—by window, assigning each intervention to the arrondissement its speaker represents in that legislature. Between 1894 and 1900 there is none: no Catholic deputy for the four arrondissements speaks in the Chamber. Before 1894 it is two deputies, the Charleroi members Adolphe Drion du Chapois and Ferdinand Noël, who speak in three of the seven legislatures and 188 times in fourteen years; after 1900 it is sixteen deputies, between eight and twelve per legislature, and 18,312 interventions. The window on which the "accommodation" was estimated is empty, and the window on which the "retreat" was estimated holds the members proportional representation brought in. Whatever those two coefficients compare, it is not the same deputies before and after.

Three checks establish that this is a fact about representation rather than about our data. It is not a coverage failure of the constituency crosswalk: the mining flag is non-missing on 102,007 of the 102,221 Chamber speeches in 1894–99, a rate in line with the other two windows. It is not a general silence of the governing parties in the coal belt: Liberals from the same districts speak 2,137 times in the same window. And it is not Catholic silence in general: Catholics speak 58,699 times elsewhere in the Chamber over those years. Catholics are barely returned from the coal districts under the majoritarian rule before the socialist entry, and not at all after it; the career file of §3 lists no Catholic member of the Chamber for the four arrondissements in 1894–1900 either.

This pattern fits the electoral institutions of the period. Under majoritarian rules in multi-member constituencies a party that loses the local majority, at the first round or at the run-off, loses the seats. Before 1894 these were Liberal seats: twenty to twenty-seven Liberal deputies per legislature speak for the four arrondissements, against at most two Catholics. At the 1894 election the Belgian Workers' Party took an outright majority of the vote in Charleroi (50.6%) and in Mons (54.8%), the arrondissement that contains the Borinage, and came second to the Catholics in Soignies (33.8% against 37.1%); the seats passed largely from the Liberals to the socialists—in Liège the two shared them—and the Catholics, a local minority throughout, held none. Proportional representation gives that minority seats: from none between the reforms to as many as twelve Catholic deputies per legislature, and 18,312 interventions over 1900–1912. Whatever the reform did to the distance between the governing parties, it put Catholics on the benches of the coal belt.

We therefore report the coal districts as evidence for the recomposition channel of §6.6 rather than as a within-deputy mechanism, and we drop the accommodation-and-retreat interpretation. The proximity coefficients remain in Appendix B.8 with these counts beside them, because a reader who meets them elsewhere should be able to see why we do not build on them. The substantive point survives the correction and is in fact sharper: the supply-side reform changed who sat, and the coal districts are where that change is largest and most legible.

Table 8: Catholic presence in the TopMining arrondissements, by reform window. The 1894–99 column is the window on which the socialist-proximity "accommodation" was estimated.

	pre-1894	1894–99	post-1900
Catholic speeches, mining districts	188	0	18,312
distinct deputies	2	0	16
Liberal speeches, mining districts	29,715	2,137	14,769
Socialist speeches, mining districts	0	27,616	46,814
Catholic speeches, all other districts	88,978	58,699	115,607

Chamber only; all interventions assigned to a legislature, 1880–1912, with the dataset's speaker links and without the turns of the chair. Each intervention is assigned the arrondissement its speaker represents in the legislature, which the dataset records with the member's mandate. TopMining is the indicator for the top of the 1896 coal-employment distribution used throughout: Charleroi, Liège, Mons and Soignies. The 188 pre-1894 Catholic interventions are by the two Charleroi members Ferdinand Noël (95) and Adolphe Drion du Chapois (93). In the estimation sample (speeches with at least one in-vocabulary phrase) the Catholic rows are 150, 0 and 14,723 speeches by 2, 0 and 16 deputies. The mining flag is non-missing at comparable rates in all three windows, so the empty 1894–99 cell is not an artefact of the constituency crosswalk.

A.2 Setting, data and robustness detail

Restricted suffrage and elite insulation. Before 1893, Catholics and Liberals dominated Belgian national politics under a tax-based franchise: tax and property qualifications confined the vote to a small fraction of adult men (Witte et al., 2009). The duopoly was already organized around a deep confessional cleavage, and our window opens at one of its peaks: the "school war" of 1879–1884, in which a Liberal government's secularization of primary schooling provoked Catholic counter-mobilization, a rupture of diplomatic relations with the Vatican, and, at the 1884 election, a Catholic landslide that opened three decades of uninterrupted Catholic government (Lorwin, 1966; Kossmann, 1978; Witte et al., 2009). Catholic–Liberal distance is thus not an arbitrary pair for a distinctiveness measure: it was *the* axis of Belgian elite conflict on the eve of mass politics, which is what makes its response to the reforms the natural object. Industrialization meanwhile generated mounting social pressure that the restricted franchise barely transmitted into Parliament (Deneckere, 1997). The Senate reinforced this design. Inherited from the Dutch constitutional model and conceived as a conservative chamber, it combined high tax thresholds and a minimum age of forty with a membership drawn almost exclusively from the propertied and professional classes; the pool of eligible senators numbered in the low thousands out of a population above six million (Belgische Senaat, 2019; Luykx and Platel, 1985). Because every law required the assent of both houses, this recruitment made the Senate a structural safeguard for incumbent elites against the policy consequences of mass politics in the lower chamber.

Plural suffrage, 1893–1894. Under sustained socialist mobilization and a general strike, Belgium adopted plural male suffrage in 1893, first applied at the 1894 election. The design reflected the strategic priorities of the incumbent parties rather than a full concession (Deneckere, 1997; Witte et al., 2009). Three features matter. Plural voting preserved unequal weight through supplementary votes for older, propertied, and educated men, capped at three per elector, so the electorate expanded from a censitary base of 136,775 voters (Stengers, 2004) to roughly 1.355 million (Witte et al., 2016) while the relative weight of working-class voters was contained (Gilissen, 1958; Witte et al., 2005). Compulsory voting, introduced simultaneously, was a strategic instrument the Catholic party used to activate its less organized rural base and offset the left's urban turnout machine (Malkopoulou, 2016, 2015). The net effect was to bring mass politics into Parliament while diluting its consequences—"democratization managed from above." This simultaneity matters for interpretation, and we flag it here to return to it in the results. The 1894 treatment is not a

pure franchise expansion but a *bundle*: an enlarged, plurally weighted electorate *and* compulsory voting introduced at the same election. The two need not act through the same channel—the first changes *who* can vote, the second *who turns out*, and it was designed precisely to mobilize the Catholic rural base. Our design identifies the effect of the 1894 reform as a package; it cannot separate the franchise component from the turnout component, a limit we make explicit where we interpret the Catholic-specific response (§6.2). Crucially for our design, the median-voter shift was *national and uniform*: Catholic and Liberal members from both pressured and unpressured districts now faced an enlarged, more labour-sensitive electorate. The Belgian Workers' Party entered the Chamber for the first time in 1894, winning 28 of the 152 seats against 104 for the Catholics and 20 for the Liberals (Belgische Senaat, 2019).

Proportional representation, 1899–1900. The second reform changed not the size of the electorate but the translation of votes into seats. Before 1900, majoritarian rules in multi-member constituencies let locally dominant parties convert local majorities into disproportionate seat shares; as socialist support grew in industrial districts, this squeezed the Liberals between Catholic rural dominance and socialist urban mobilization (Boix, 1999; Emmenegger and Walter, 2019). Proportional representation was, on the canonical reading, a Catholic–Liberal deal: it offered the Liberals an institutional lifeline against extinction under majoritarian rules and, for the Catholics, contained socialist seat growth in industrial strongholds (Boix, 1999; Witte et al., 2009)—though the Catholics lost seats at the first proportional election, and Barzachka (2023), weighing the competing accounts against the historical record, finds electoral threat and extra-institutional threat both operative at different stages of the reform process. Belgium is often treated as the paradigmatic case of incumbents adopting proportional representation to survive a rising flank (Boix, 1999)—a reading contested by Cusack et al. (2007), for whom co-specific investments rather than the left threat drive the adoption of proportional rules, and by Emmenegger and Walter (2019), who observe that the Catholics were dominant, advantaged by the existing rules, and lost twenty-six seats at the first proportional election; nothing in our design adjudicates the origin question, and our estimates concern the reform's consequences, not its causes. The Senate's socialist presence remained minimal and largely indirect well past 1900—the Senate's own record dates the party's first directly elected senator to 1904, and even that seat was lost on a challenge to the eligibility census (Table 30)—so the upper house remained the less exposed chamber throughout (Belgische Senaat, 2019).

Socialist entry and the hardening of the governing duopoly. The Belgian historiography describes the years after 1894 as precisely the kind of transition our measure is built to detect. The party system of the *Belle Époque* was organized along the religious and class cleavages that Lipset and Rokkan (1967) treat as canonical, and Belgium is a textbook case of their freezing into segmented political "worlds": from the 1890s each camp built its own dense network of unions, mutual societies, cooperatives, newspapers, and schools—the pillarization (*verzuiling*) analysed by Huyse (1970) and, for its formative period, by Strikwerda (1997)—so that political competition increasingly ran between organizationally closed blocs rather than across a shared notabiliar milieu (Lorwin, 1966; Kossmann, 1978). The socialist entry of 1894 accelerated the hardening on the Catholic side in particular: the party countered the Workers' Party's mobilization with a social-Catholic apparatus of its own—the organizational response to socialist mobilization that Kalyvas (1996) identifies as the engine of Christian-democratic party formation across Europe—the *Ligue démocratique belge* (founded 1891), in the wake of *Rerum Novarum* (1891)—while the tension between its conservative and christian-democratic wings, dramatized by the Daensist schism of the mid-1890s, made the workers' question an internal battleground as well (Strikwerda, 1997; Witte et al., 2009). The Liberals, squeezed between the clerical–anticlerical and the class cleavage, collapsed to a parliamentary rump at the elections of 1894–1898 before proportional representation restored their representation (Witte et al., 2009; Emmenegger and Walter, 2019; Boix, 1999). The qualitative record thus points to an intensified Catholic–

Liberal differentiation after the socialist entry—each governing party retreating into a more distinctive idiom as competition reorganized around the newcomer—but it is qualitative. Our contribution is to measure that distance at the level of the individual speech and to ask, with a design, whether the reforms caused it to move. The stakes of the answer reach beyond the period: the same segmented blocs would, after the First World War, become the raw material of Belgium's celebrated elite accommodation—the consociational bargaining analysed by Lijphart (1977) and Huyse (1970). Our window is the formative stage of that system, when the pillars were still being built and the distance between their elites was still widening; a measured account of when and why it widened is a boundary condition for the accommodation that followed.

Label and dating audit. We audited politician-level data before estimation, especially because the mining comparisons draw on only a few arrondissements. Using stable politician identifiers, we checked party labels against biographical records, separated speeches by members from interventions by the chair, and corrected mistaken speaker matches. We also assigned each speech to a legislature using its sitting date rather than its calendar year, so that speeches around elections fall on the correct side of the reform. The corrections are applied *upstream* of the text scoring, inside the corpus loader, so that mislabeled speech cannot shape the partisan vocabulary or the leave-out yardstick $\hat{\rho}$ before the regressions ever see it. All estimates in the paper are computed on the dataset with the registry applied; the full registry, entry by entry, is in Appendix B.3.

What underwrites parallel trends. The comparison assumes that, without the reforms, the two houses would have followed parallel trends. We assess that assumption in several ways. First, the Senate placebo (Table 2): the plural-suffrage dummy carries no break on the control house, while the same dummy on the Chamber is positive and significant—the asymmetry the design requires. At 1900, when the control was itself renewed, it does break, which is why Panel B is not read causally (§6.6). Second, the Chamber's own pre-1894 drift on the distinctiveness axis is small: a linear Chamber $\times$ legislature trend on the pre-reform legislatures is $+0.0003$ per legislature (s.e. 0.0005, n.s.), and in the event study of §6.3 the six pre-reform leads are jointly indistinguishable from zero ($F(6,40) = 0.79$, $p = 0.58$; Table 7). The mining triple, negative on both measures but resting on four arrondissements whose Catholic benches were empty between the reforms (§A.7), stays in the appendix as a gradient rather than a second identified effect. The per-legislature trajectories of the two houses are shown for transparency in Figures 3 and 5; given the Senate's smaller corpus its per-legislature intervals are wide, so the inferential weight rests on the pooled difference-in-exposure and on the placebo rather than on the visual gap.

Diagnosticity. Fix a window p (pre-1894, 1894–99, post-1900) and let $C^1_{j,p}$ and $C^0_{j,p}$ be the counts of phrase j by Catholic and Liberal speakers in that window, with token totals M^1_p and M^0_p. The posterior probability that a randomly drawn token of phrase j was spoken by a Catholic is

$$\rho_{j,p} \;=\; \frac{C^1_{j,p}/M^1_p}{C^1_{j,p}/M^1_p + C^0_{j,p}/M^0_p}, \tag{9}$$

the same object as ρ_j of §4.2 but estimated within the window rather than pooled. The *diagnosticity* of phrase j is the probability of guessing the speaker's party correctly from that phrase alone,

$$\zeta_{j,p} \;=\; \max\{\rho_{j,p},\, 1-\rho_{j,p}\} \in [\tfrac{1}{2},\, 1], \tag{10}$$

so $\zeta_{j,p} = \frac{1}{2}$ marks a phrase that carries no partisan information and $\zeta_{j,p} \to 1$ a phrase used by only one side. A phrase is assigned to the Catholic list when $\rho_{j,p} > \frac{1}{2}$ and to the Liberal list otherwise, and within each list phrases are ranked by $\zeta_{j,p}$.

Why diagnosticity alone is not enough. Ranking on ζ favours rare phrases: a collocation used four times, all by Catholics, attains $\zeta = 1$ and tells us nothing. Two restrictions discipline the list. First, the global vocabulary filter of §4.1 already removes phrases below τ_c occurrences and τ_s distinct speakers. Second, and specific to this exercise, we require a phrase to be used by at least five distinct members *within the window*, so that no entry is the idiosyncratic tic of a single talkative deputy, and we report alongside $\zeta_{j,p}$ both the predicted frequency per 100,000 phrase tokens, $f_{j,p} = 10^5 (C^1_{j,p} + C^0_{j,p}) / \sum_k (C^1_{k,p} + C^0_{k,p})$, and the number of distinct speakers. Diagnosticity says how sharply a phrase separates the parties; frequency and speaker counts say how much of the separation it actually carries. Both belong in the table, and a phrase that scores high on the first and low on the other two should be read as decoration rather than evidence.

Period and topic decompositions. Estimating (9)–(10) separately by window makes the lists period-specific: a phrase can be Catholic-coded before 1894 and contested after, and that migration is itself the object of interest. The same construction applies with the window replaced by a topic, or by a topic × legislature cell, which is how we report where the divergence occurred (§A.6). One caution governs the topic version and we state it once. The cell-specific $\rho_{j,p}$ of (9) is appropriate for *describing* a cell's vocabulary, but it is not comparable across cells of different size: in thin cells the leave-out correction pulls the index toward $\frac{1}{2}$, so a small cell looks less partisan whether or not it is. Wherever we compare partisanship *across* cells—over time, or across topics—we therefore score every speech against a single yardstick held fixed across cells—the global $\hat{\rho}$ of §5.4 in the regressions, its counterpart estimated on the Chamber's pool in the descriptive Chamber series—and reserve the cell-specific estimate for the descriptive lists in this subsection.

What the diagnostic phrases are. Table 23 in Appendix B.1 applies (9)–(6) window by window, and the lists are worth reading for themselves. They hold the speakers' own words. The reactions of the floor, which the *Annales* print inside the speech they interrupt (*Très bien! à droite*, *Rires à gauche*, *Protestations à gauche*), would otherwise head them, because a member's own benches applaud him and the opposite benches protest: with the reactions left in the text, *bien_droit* heads two of the three Catholic lists and *bien_gauch* two of the Liberal ones. Partisan by construction and no part of what the speaker said, they are taken out of the text before the phrases are formed (§4.1).

What the lists show is the measure tracking a reallocation of vocabulary rather than noise. Before 1894 the Catholic list is about poor relief—*domicil_secour* (the settlement that fixed which commune owed assistance), *dépôt_mendiqu* (the vagrancy depots), *assist_publiqu*—and about the Flemish provinces (*pay_flamand*), and the Liberal list holds electoral reform (*réform_électoral*), spirits (*eau_vi*), communal autonomy (*autonom_communal*) and the school war (*part_clérical*, *supprim_écol*, *écol_mixt*). On the eve of mass politics, in other words, the governing party of the Catholic bloc had a distinctive language of charity but none of labour.

After plural suffrage that changes. The Catholic list of 1894–99 (Table 23) is the social-Catholic apparatus almost item by item: *union_professionnel* (the professional unions, $\zeta = 0.91$, the most frequent diagnostic phrase of the window), *caiss_épargn* (savings banks, 0.90), *contrat_travail* (the labour contract, 0.76) and *conseil_industr* (the councils of industry and labour, 0.79), beside the vicinal railways (*fer_vicinal*). These are the institutions of the *Ligue démocratique belge* and of the post-*Rerum Novarum* Catholic social programme described in §2, and they enter the Catholic vocabulary exactly when the reform enfranchises the men they address. The Liberals become identifiable instead by the franchise itself (*suffrag_universel*) and by local government (*députpermanent*, *conseil_communal*).

After 1900 the Catholic list keeps part of its social core (*caiss_prévoi*, the provident funds, *chef_entrepris* and *patron_ouvri*), while the Liberal list returns to the school (*écol_communal*, *enseign_public*, *loi_scolair*) and to the audit of the public accounts (*cour_compt*). The party that proportional representation rescued from extinction speaks, distinctively, of the cleavage on which it was founded.

We report this as illustration, not as evidence: the phrase lists are descriptive summaries of the same index the regressions use, so they cannot corroborate it independently. Their value is that they make the aggregate legible. The arrival of the labour lexicon on the Catholic side across 1894 is the movement that §A.6 locates in the labour topic.

What the index registers, and what it does not. A measure of elite division ought to be checked against an episode historians already call divisive, and our window opens on one: the school war of 1879–1884, which broke diplomatic relations with the Vatican and ended in the Catholic landslide of 1884 (§2). Figure 4 shows the index at chance through those years. Restricting to confessional speech does not recover a signal: on a yardstick likewise held fixed across legislatures, Catholic–Liberal partisanship on confessional and education speech is 0.5024 in 1880–1884, against 0.5007 for all speech in the same legislatures (Table 72), and it stays within five thousandths of $\frac{1}{2}$ for the entire window—0.4990 in 1894–99—while the rest of the Chamber's speech rises after 1894. We take this seriously rather than set it aside, because it identifies what the estimator measures. The leave-out index scores how well a phrase betrays a speaker's party—lexical distinguishability, not the intensity of disagreement. The school war was fought by both camps in a shared vocabulary: *instruction*, *ecole*, *liberté d'enseignement*, *neutralité* belong to Catholics and Liberals alike, and opposed positions expressed in common words leave a partisanship index at chance. What changes after 1894 is not that conflict begins but that the vocabularies separate—*union professionnel* and *caisse d'épargne* on the Catholic side, a Liberal list with no labour content on the other—and separation is what the measure is built to detect. The consequence for the paper's claim is a narrowing, and we state it rather than let it be inferred. What we identify is not elite polarization in general but the divergence of party vocabularies, and §A.6 locates that divergence on the distributive axis rather than the confessional one. That the 1894 effect is 78.6% within-topic on the share decomposition (Table 19) says the parties did not start discussing different subjects; that the confessional domain never moves says they did not diverge on the older cleavage either. The reform reorganized the language of the social question, and left the language of the school war where it was. This is a sharper finding than an undifferentiated rise, and the null of 1879–1884 is what licenses it.

Assigning topics. Every speech is assigned to a single policy domain by a CamemBERT classifier fine-tuned on the corpus. Because no hand-labelled corpus of nineteenth-century Belgian parliamentary speech exists, the training labels are generated by a large language model and the classifier is then distilled onto the full corpus: a random sample of speeches, stratified on chamber and reform window and on nothing else, is labelled by GPT-4o against a written codebook, and CamemBERT is trained on those labels and applied to every speech. We state this openly because it governs how the results should be read, and we validate the end product against human coding rather than against the annotator: on a blind subsample the classifier agrees with a human coder at $\kappa = 0.80$ (accuracy 0.86), and the annotator itself at $\kappa = 0.77$ (Appendix B.10, Table 54). The classifier was trained and applied on the interventions of the PLENUM corpus; the dataset's interventions are different units, so its labels are carried over by a second, linear distillation, which reproduces them out of sample on 86% of interventions ($\kappa = 0.77$; Table 74).

Two features of the design matter here. First, sampling for the training set is on the design cells—chamber $\times$ reform window—and never on topic keywords. Keyword-stratified sampling would have anchored the topic partition to the same lexicon that generates s_{distinct}, making a within-topic test partly circular; stratifying on cells that are exogenous to language avoids this. Second, the classifier reads the raw French text while the partisanship measure lives on lemmatized bigrams, so the two objects are estimated on different representations of the same speech.

The taxonomy is six substantive domains plus a procedural residual: labour and social policy; the confessional question; economic and fiscal matters; the state, its rules and its coercion; foreign affairs and defence; and territory and infrastructure. Two boundary choices deserve statement because they are not mechanical.

Education is grouped with religion rather than with governance: in Belgium between 1880 and 1912 the schools question *is* the confessional cleavage, and the window opens on the school war of 1879–1884 (§2), so separating them would break the bucket meant to carry the Catholic–Liberal axis. Mining and industry are grouped with the economy rather than with labour, so that the labour bucket is not defined by coal-sector vocabulary. A finer 22-domain taxonomy is estimated alongside and reported as a robustness.

Table 9: Robustness of the plural-suffrage difference-in-exposure: baseline, bandwidths, sample windows, and donut. All columns absorb politician, arrondissement, party and legislature fixed effects, so column (1) is the Panel A specification with the common time effect saturated (+0.0093, against +0.0094 with Post-1894 alone in Table 3, column 3); (2)–(3) restrict to ±2 and ±3 legislatures around 1894; (4) extends the window to the 1900 legislature and drops every later one; (5) drops 1892 and 1896, the legislatures adjacent to the reform.

	(1) Panel A	(2) BW2	(3) BW3	(4) drop post-1900	(5) donut
Chamber_dk × Post1894	0.0093***	0.0087***	0.0085***	0.0074***	0.0083**
	(0.0028)	(0.0024)	(0.0022)	(0.0024)	(0.0031)
Observations	206116	112233	134404	227489	158567

Standard errors clustered at arrondissement in parentheses. Produced by `polarization_did.do`; `senate_placebo_bounding.do` writes a fragment of the same name without the legislature effects (+0.0077 to +0.0094). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Wild cluster bootstrap (Webb weights, 9,999 replications), arrondissement clusters. The plural-suffrage effect survives at the five per cent level on the politician-fixed-effects specification; the mining-district rows are the socialist-proximity coefficients, on which the paper does not build (§6.7).

Coefficient	Clusters	WCB p
Robust — the plural-suffrage difference-in-exposure		
Chamber × Post-1894 (politician FE)	41	0.021
Chamber × Post-1894 (FE only)	41	0.001
Senate placebo, Post-1894	31	0.881
The socialist-proximity coefficients, mining districts		
Accommodation, TopMining × Post-1894	—	not identified
Accommodation, $\text{mine}_{\text{cont}}$ × Post-1894	38	0.276
Retreat, TopMining × Post-1900	40	0.681
Retreat, $\text{mine}_{\text{cont}}$ × Post-1900	40	0.844

"Clusters" is the number of arrondissements in the estimation sample; the TopMining indicator varies across four of them. Between the reforms no Catholic deputy for the TopMining arrondissements speaks, so the accommodation row has no within-politician variation (Appendix A.1); the mechanism rows are reported for completeness, not as evidence on the mechanism. The retreat coefficient on TopMining, with politician fixed effects, is −0.00245 (s.e. 0.00387).

What the index does not yet buy: legislative outcomes. A standardized effect and a ratio both describe the measure on its own scale. Neither says what a movement of 0.0094 (Table 3, column 3) corresponds to in the working of the legislature, and a reader is entitled to ask. We looked, and report that we did not find it.

The natural external object is legislative output. We assembled the 2,917 bills deposited in the Chamber between 1880 and 1912, each with its deposit date, its adoption outcome, and a dictionary classification of its subject, and asked whether legislatures in which the two governing parties speak more distinguishably adopt a smaller share of what is put before them. Pooled across all bills the association does not even have the sign the gridlock channel predicts (+4.80, s.e. 6.03; Table 70), and disaggregation does not give it one.

Two populations of bills have to be separated. Government bills—*projets de loi*, carrying no named parliamentary author—are 71% of the corpus and are adopted at 74.8% (Table 70). Members' bills—*propositions de loi*—are adopted at 21.8%. The government's share of deposits also swings between 52% and 84% across legislatures (Table 70), so the pooled rate mixes two processes with a moving denominator. Separated, neither shows an association: government bills +3.34 (s.e. 7.39), members' bills +1.73 (s.e. 4.96; Table 70).

Splitting the members' bills by the author's party does not change that. Bills authored by Catholics and Liberals—691 of the 858, and the only ones whose fate could speak to the distance *between* those two parties—show no association (+7.14, s.e. 5.97; +8.25 with a linear trend; Table 70). The 167 bills authored by Socialists and Daensists show a negative one, large and imprecise (−12.89, s.e. 16.35; Table 70): socialists deposit and are voted down, and the legislatures in which this happens are the legislatures in which Π is high, because both follow the same reform. Whatever that coefficient holds is a fact about representation that the paper already documents, not a calibration of Catholic–Liberal distance.

We cut the data several ways before concluding: by legislature, with a linear trend and without the war-truncated legislature of 1912; by government against parliamentary initiative; and by authoring party (Table 70). Every cut agrees. We therefore report no legislative calibration of s_{distinct}: on this corpus there is no association to calibrate against, pooled or disaggregated.

This limits what we can conclude. What the paper establishes is that a dated institutional change moved the rhetorical distance between the governing parties, that the movement is correctly timed, and that it is overwhelmingly within-topic. What it does not establish is what that distance did. Section A.6 shows where the divergence concentrated—on labour and social policy—and Section 6.7 shows the coal districts changing hands, but neither converts 0.0094 (Table 3, column 3) into a legislative consequence. The measure is calibrated against the reform that moved it and against a permutation null, not against what the chamber produced. The comparison that disciplines the magnitude best is the setting closest to ours: in the century-long Finnish series we follow (Simola et al., 2025), estimated with the same family of measures, the average partisanship of a phrase moves within a comparably narrow band around $\frac{1}{2}$ over a hundred years; a single Belgian electoral reform shifts the exposed chamber by an amount on the order of that entire secular movement.[3] Two features of the design push both readings down—the fixed yardstick discards the partisan signal carried by vocabulary that changes sides (§5.4), and the Senate's residual exposure moves the control with the treated house (§5.2)—and two may push them up: a ruler that the treated cell cannot calibrate returns four fifths of the estimate (§A.2), and the leave-out correction weakens in the Chamber and strengthens in the Senate across 1894 (§5.4). The distinctiveness effect should be read as sharp rather than sweeping: reform-timed, within-incumbent, one-signed, and significant at the five per cent level on the wild cluster bootstrap with politician fixed effects (§7.1)—a thin measure moved, rather than a large movement in rhetoric.

Our preferred specification gives +0.0094 (Table 3, column 3): the Chamber×Post-1894 interaction with politician, arrondissement and party fixed effects, on the medium vocabulary, weighted by in-vocabulary token count and clustered at arrondissement. It is the specification that identifies adaptation rather than turnover, which is the paper's substantive claim, and we prefer it although it is the less precise of the two main columns—a standard error of 0.0029 against 0.0021 without the politician fixed effect, and a wild

[3]The much larger raw swings reported for the U.S. *Congressional Record* (Gentzkow et al., 2019) are not a like-for-like benchmark: that corpus is orders of magnitude larger and its index sits well above $\frac{1}{2}$, so its units are not comparable to a thin nineteenth-century parliamentary corpus whose index hugs the no-information benchmark.

cluster bootstrap p of 0.021 against 0.001 (Tables 3 and 10).

Other estimates vary the specification, sample, or measurement yardstick. Dropping the politician fixed effect gives $+0.0095$ (Table 3); the same specification on the sample of the topic decomposition, which is that same sample, gives the same (§6.4); on the strict and loose vocabularies the reference specification gives $+0.0088$ and $+0.0098$ (Table 52); on the split-sample yardstick $+0.0119$, on the leave-cell-out ruler $+0.0073$ and on a vocabulary built on pre-1894 speech $+0.0071$ (§A.2); on French-only speech $+0.0107$ (§A.2); and $+0.0071$ is the Chamber side of the Senate placebo (Table 2), not a headline at all. Each is labelled where it appears. The argument for the reference estimate is the timing. Two exact tests make that argument rather than assert it, and neither requires the forty-one arrondissements to be independent draws on treatment.

The first asks whether the break belongs to the reform date. Re-estimating the bicameral interaction at every feasible cutoff in the window—each legislature that leaves at least two periods on either side—the 1894 reform produces the largest break of the fourteen. Its rank is first, so the exact one-sided p-value is $1/14 = 0.071$ (Table 6), which is the smallest value fourteen cutoffs can attain: no alternative date generates a comparable movement, and the test cannot say more than that with a window of this length. The placebo cutoffs inside the two reform-free windows make the same point in magnitudes rather than ranks: the largest break any of them produces is 0.0040, over two fifths of the estimate (§7.1; Table 66).

The second asks how much the parallel-trends assumption would have to fail before the estimate went away. The joint test of the pre-reform leads does not reject ($F(6,40) = 0.79$, $p = 0.58$), but on a two-group design that is a low-powered test, and not rejecting a pre-trend is not the same as having none: the leads drift mildly across the 1880s, from -0.0021 in 1882 to -0.0007 in 1890 (Table 7). Under the relative-magnitude restrictions of Rambachan and Roth (2023) the identified set for the 1894 effect, whose conventional 95% interval is $[0.001, 0.017]$ ($p = 0.026$), excludes zero for every $\bar{M}$ up to 0.75—barely at 0.75, where its lower bound is 0.000 to three decimals—and admits it at $\bar{M} = 1$, where it is $[-0.001, 0.021]$ (Table 6): the post-reform violation of parallel trends would have to exceed three quarters of the largest movement between consecutive pre-reform legislatures before the 1894 effect could be zero. The pooled within-politician estimate is more precise than any one event-study coefficient, and the timing claim rests on the leads, the rank test and this bound together.

Two qualifications keep the tests honest. The rank test treats the fourteen cutoff dates as exchangeable under the null, an assumption a drifting series would strain. And the joint test of the leads inherits the clustered standard errors of the event study: if those are anti-conservative, as §7.1 argues they can be on this design, the test is too lenient. We flag rather than fix this: the exact tests of §7.1, not the clustered intervals, are what carry the paper's inferential weight.

What the design cannot do is treat the arrondissements as independent units, and §7.1 states what that costs.

Two features of this result deserve emphasis because they recur throughout the paper. The first is the *channel*: pooled across the two parties, the movement is concentrated on members who sat both before and after 1894, so plural suffrage changed the behaviour of the existing elite rather than simply rotating in a different one. This is the same channel our companion study of legislative behaviour finds for the same reform on a different outcome: there, plural suffrage raises the *salience* of labour and social policy through within-politician adaptation among sitting incumbents (Graziani and Vanschoonbeek, 2026). Enfranchisement thus moved both *what* the Chamber debated, toward labour, and *how distinctly* the two governing parties debated it, apart—and, pooled, moved both through sitting members. The second feature is the *asymmetry between parties*, which lies in the channels rather than in the direction: split by party, the Catholic incumbents move while the Liberal incumbents do not, and what Liberal movement there is runs through the members first returned from 1894 (Appendix B.4). A median-voter account in which the larger governing party must reach hardest for the newly enfranchised worker—the social-Catholic mobilization the next section makes concrete—predicts a Catholic movement, and the Catholic side delivers one, through the sitting members

themselves.

One caveat on interpretation follows from the bundled nature of the 1894 reform (§2). We read the Catholic-specific divergence as social-Catholic mobilization to court the newly enfranchised worker—a median-voter response—but an observationally near-equivalent story is available: compulsory voting, introduced at the same election, activated the Catholic rural base, so the Catholic response may be to that changed *active* electorate rather than to enfranchisement as such. Both are consequences of the 1894 package. For the estimate the bundling is harmless, and for a reason worth stating rather than asserting: compulsory voting bound the same electors who returned both houses, so its direct effect is a shock common to the two arms and differences out of β like any other national force. What it complicates is only the reading of the Catholic-specific response: the median-voter reading is one of two compatible interpretations, not the only one. We note the distinction rather than adjudicate it, since nothing in the bicameral contrast can.

A second ambiguity of the same kind runs deeper, and the mining results bear on it directly. Plural suffrage did two things to the Chamber at once: it changed the *electorate* its members faced, and it changed the *chamber* they sat in, by seating twenty-eight socialists in 1894. The Senate received the first too—the same electorate returned both houses—and the second only faintly: its recruitment was loosened only at the margin—a lower *cens* and the new provincial seats—and while provincially elected socialists sat from 1894, the Senate's own record dates the first directly elected one to 1904; the note to Table 30 states the counting rule and the single member the 1900–1902 entries turn on. Socialist speech accordingly makes up three in ten Chamber interventions in 1894–99 (30%) but under a tenth of the Senate's (8%; Table 1), so exposure to the new interlocutor is itself a difference in degree rather than a clean presence–absence contrast—which means the bicameral design attenuates this channel as it does the others, but cannot separate it from the electorate channel. This does not threaten the estimate: socialist entry is a consequence of the franchise reform, not a confounder of it, so an effect that runs through the new entrant is part of the reform's total effect and is correctly attributed to it. What it threatens is the reading of the *mechanism*. Our account is a median-voter one, in which the governing parties differentiate themselves because they are reaching for newly enfranchised workers. An alternative is a chamber-level one, in which the two governing parties become more distinguishable because both are now repositioning relative to a third pole, whatever their districts look like.

The two accounts differ in a testable way, and our own geographic evidence favours neither. If the divergence were driven by the newly enfranchised electorate, it should be stronger where that electorate is denser—the coal districts, where the working-class share of the new voters was highest. The mining triple-difference on the Catholic–Liberal axis provides exactly this test, and it finds the opposite (Appendix B.6): on the continuous measure the reform-timed widening is smaller in the mining constituencies than elsewhere (-0.003 per standard deviation of the mining share within politician, $p < 0.01$; Table 3, column 6), and the TopMining contrast is negative on average and in each of the three post-reform legislatures, although part of that gradient predates the reform (§A.7). Nor is it the uniformity across districts that a chamber-level account predicts, since every deputy—coal-belt or not—sits with the same twenty-eight socialists.

The companion study points the same way on a different outcome. Estimating the identical triple difference on the salience of labour and social policy, it returns a negative Chamber $\times$ Post-1894 $\times$ Mining coefficient, significant at conventional levels in its baseline specification (Graziani and Vanschoonbeek, 2026). Two measures built on different vocabularies—one positional, one salience-based—therefore agree that the response to plural suffrage was weaker in the coal belt.

Read as a puzzle this is awkward for a median-voter account; read as an account of its own it is a prediction. Call it the *contestability* account, and note that it is the same channel §5.2 builds the design on: what moves an incumbent is not the size of the new electorate but the threat of losing the seat to a candidate who claims it. That threat is non-monotone in local socialist strength. Where the socialists are absent there is no one to lose the seat to, and no reason to adapt. Where they are strong but not yet winning, the seat is in play and adaptation should peak. Where they have already won—Charleroi, Mons and the

Borinage—the seat is gone: a Catholic returned from those districts represents the confessional minority of a place whose working-class vote has gone to the socialists and will keep going there; courting it is expensive and unlikely to pay, and his incentive is to consolidate his own constituency instead—the same centrifugal logic that Appendix B.8 discusses for proportional representation, operating here through the entry of a competitor rather than through a change in the seat formula. The mining constituencies are, by construction, the third case: they are where the Belgian Workers' Party *won* in 1894. A weaker response there is therefore not an anomaly for this account but its prediction, and the three accounts separate on exactly this shape. The chamber-level account predicts *uniformity* across districts, since every deputy sits with the same twenty-eight socialists; the median-voter account predicts a response *increasing* in the density of the new electorate; the contestability account predicts a *peak* at intermediate socialist strength and little at either end. The dose that discriminates among them is the local socialist vote share, not the coal share.

Until that shape is estimated we offer the account as a hypothesis, not a test passed: our own mining gradient and the companion's are both negative, which the contestability account predicts and neither of the others does, but a negative coal gradient is one point of the shape rather than the shape, ours partly predates the reform, and 1894 supplies no variation in socialist entry independent of the franchise, so the chamber-level account is not excluded. What the two papers jointly establish is narrower and firmer—that the response to plural suffrage was a movement of the governing parties across the country, weakest rather than strongest where the new electorate was densest, and not a local response to local electorates, consistent with the national character of the organizations that carried it, the *Ligue démocratique belge* and the post-*Rerum Novarum* programme (§2).

We therefore state the mechanism with the qualification it deserves. The evidence that the effect operates through sitting members rather than turnover is strong when the parties are pooled, and geographic in none of the three accounts (§6.2); split by party, the movement runs through the Catholic incumbents and, weakly, the Liberal entrants (Appendix B.4). Among the three accounts, the negative mining gradient on our measure and on the companion's is what the contestability account predicts and neither the median-voter nor the chamber-level account does; the flexible dose in local socialist strength flagged above is the test that would separate them, and until it is run we cannot rule out that much of the aggregate divergence is a response to the new interlocutor in the chamber rather than to the threatened seat in the constituency. All are effects of plural suffrage; separating them fully would require variation in socialist entry that is independent of the franchise, which Belgium in 1894 does not provide.

A final reading of the effect's *content* deserves a test of its own. That enfranchisement moved both *what* the Chamber debated (toward labour) and *how distinctly* the parties debated it (apart) raises the possibility that the second is merely the first: if labour collocations are party-diagnostic, a Chamber that simply debates more labour will register as more distinctive even with each party's position on each topic unchanged. This is not idle—it is precisely the agenda shift our companion study measures as rising labour salience (Graziani and Vanschoonbeek, 2026)—and it would make the two papers two views of one reallocation. We separate genuine within-topic divergence from agenda reallocation with a topic decomposition in §6.4, and find the plural-suffrage effect is overwhelmingly within-topic—78.6% of it on the share decomposition, 94.1% on the regression, on the whole Panel A window (Table 19).

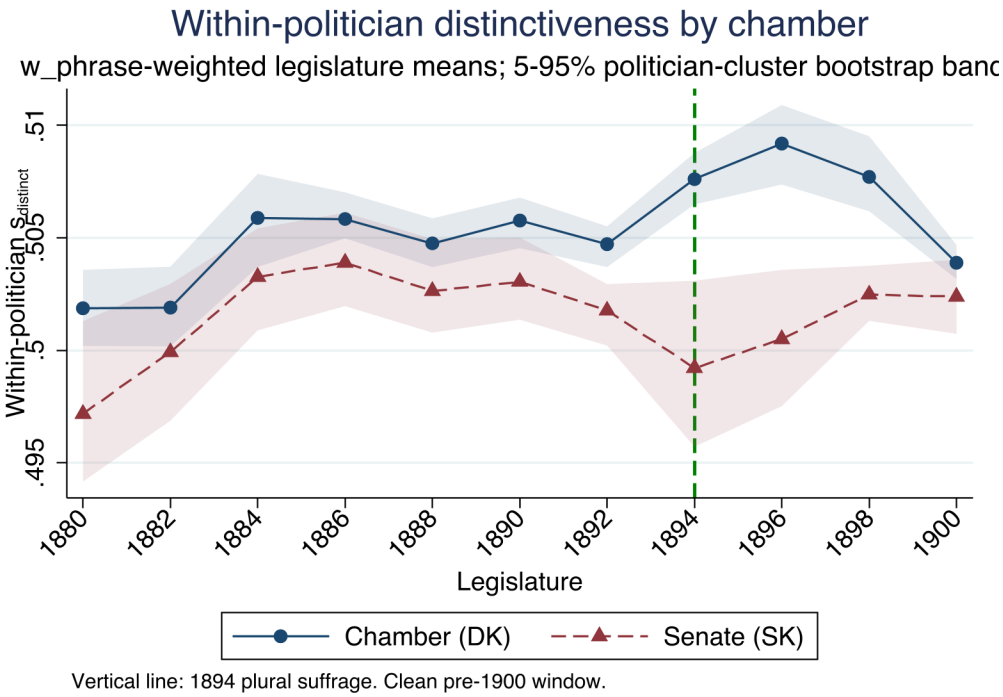


Figure 3: **Within-politician distinctiveness by chamber, plural-suffrage window 1880–1900.** Each series is the $w^{(i,s)}$-weighted legislature mean of the per-speech leave-out outcome $s^{(i,s)}_{\text{distinct}}$ (the single globally fixed yardstick $\hat{\rho}$ of §5.4), computed on the within-politician panel that the politician fixed effect exploits; shaded bands are 5–95% politician-cluster bootstrap intervals. The two houses track together before 1894; at plural suffrage the fully exposed Chamber (treated) turns up, the level-form counterpart of the $+0.0094$ ($p < 0.01$) politician-fixed-effects estimate in Table 3. The Senate (control) does not follow, but its band is wide: the thinner upper house cannot, on its own, reject a flat path per legislature, which is why identification rests on the pooled difference-in-exposure regression and not on the visual gap. We restrict the figure to the clean pre-1900 window; the Senate's own post-1900 drift, and the full 1880–1912 trajectory, are shown in Figure 5 and treated in the Senate placebo of §5.2. The vertical line marks plural suffrage (1894).

How informative is the Panel B contrast? Equivalence bounds and detectable effects. A coefficient near zero can mean "no effect" or "no power", and Table 11 is built to separate the two: it reports two-one-sided equivalence tests (TOST) on the Panel B contrast against two pre-specified bands ($|\beta| < 0.0105$ and $|\beta| < 0.0053$, about the size of the plural-suffrage coefficient and half of it), together with the minimum detectable effect. The contrast is not near zero. On the politician-fixed-effects specification, which here also absorbs legislature fixed effects, it is -0.0100 (s.e. 0.0028) with a 90% confidence interval of $[-0.0148, -0.0052]$; with fixed effects only it is -0.0111 (s.e. 0.0019; Table 11). The interval excludes zero, and the equivalence tests cannot bound the contrast inside either band ($p = 0.434$ and 0.948; Table 11). The minimum detectable effect at 80% power is 0.0054–0.0079 (Table 11), between about three fifths and four fifths of the plural-suffrage coefficient: the design has the power to register a contrast of this size, and it registers one. What it cannot do is attribute it to proportional representation. The Senate placebo puts about two fifths of the movement in the control house, whose directly elected seats were all renewed under the new rule in 1900 (§6.6), so the contrast carries the reform's effect on the Senate as well as on the Chamber, and with both houses treated at once the two cannot be separated.

Table 11: Equivalence bounds and minimum detectable effects for the Panel B contrast. TOST p-values test $H_0: |\beta| \geq \delta$ against equivalence; rejection means the data affirmatively bound the effect inside $(-\delta, \delta)$.

	FE	Politician FE
Chamber × Post-1900	-0.0111	-0.0100
	(0.0019)	(0.0028)
90% CI	[-0.0143, -0.0079]	[-0.0148, -0.0052]
TOST p, $\|\beta\| < 0.0105$ (about the size of PS)	0.622	0.434
TOST p, $\|\beta\| < 0.0053$ (about half of PS)	0.998	0.948
MDE (80% power)	0.0054	0.0079

Both columns absorb arrondissement, party and legislature fixed effects, the second also politician fixed effects. Standard errors clustered at arrondissement; degrees of freedom cluster-adjusted. Bands: 0.0105 and 0.0053, about the size of the plural-suffrage coefficient (+0.0094) and half of it. MDE = 2.80×SE (two-sided $\alpha = 0.05$, power 0.80). Produced by `panelB_equivalence.do`.

Splitting the post-1900 window bears on the same question. Directly elected socialist senators rise from one to ten across it (Table 30), so if the negative contrast reflected a growing contamination of the control it should deepen from 1900–1906 to 1908–1912. It does not do so detectably. With fixed effects the interaction is −0.0101 (s.e. 0.0019) on 1900–1906 and −0.0109 (s.e. 0.0022) on 1908–1912; with politician fixed effects the later half is deeper (−0.0079, s.e. 0.0020, against −0.0173, s.e. 0.0055), but estimated jointly the two are −0.0094 and −0.0114, and their equality is not rejected ($p = 0.21$; Table 69). The contrast arrives in 1900 and stays, which is what a one-time renewal of the control produces; a drift that tracks the socialist senators would have to show on the joint test, and it does not. And the recomposition evidence is *characterization*, not identification: the incumbent/entrant split describes *how* the post-1900 field is organized, not a reform coefficient that survives falsification. The clearest trace of that recomposition is territorial: proportional representation brought Catholic members into the coal districts, from none between the reforms to as many as twelve per legislature (§6.7), which is the same channel read off presence rather than off a coefficient.

Language is not driving the result. Because the corpus is bilingual, French–Dutch usage is partly collinear with party—Flemish members lean Catholic—so part of the measured Catholic–Liberal distinctiveness could in principle be a French-versus-Dutch contrast rather than a partisan one.

The language-composition series is informative because a change in language shares could explain the 1894 estimate only if those shares also changed around 1894. It barely moves at 1894. The Dutch share of interventions is 0.6% in the 1892 legislature and 0.8% in 1894 and in 1896, against 0.4–0.6% from 1880 (Table 12). The larger step is at 1898—to 1.2%, around the language-equality law of that year—after which it stays between 1.1% and 1.3% for the rest of the window. Whatever a French/Dutch contrast contributes to the level of measured distinctiveness, it cannot generate a discontinuity in 1894 out of a composition that moves by a quarter of a percentage point across 1894. That is the identifying window, and Panel A is identified off that break.

Restricting the corpus to French speech is the obvious complement, and on the dataset it passes. Re-estimating the whole pipeline on French-only text—vocabulary rebuilt on that text, 33,955 phrases against the pooled 36,297—gives a plural-suffrage effect of +0.0107 ($p < 0.01$) against +0.0094 on the full sample with politician fixed effects, 114% of it, and +0.0117 against +0.0095 with fixed effects only (Table 60). The restriction is close to neutral at the reform: a difference-in-differences on the probability that a speech is also scored on French-only text returns −0.017 at 1894 (the share is 67% in the Chamber before and after 1894, 66% in the Senate in both windows; Table 60), so the two coefficients estimate nearly the same

quantity, and they agree.

Where language does matter is after 1900. The restriction removes more speech from both houses after 1900, as the Dutch share steps up from 1898, and more from the Senate's (+0.033 on the same probability for the Chamber), and on French text the Panel B contrast is slightly more negative, not less: −0.0108 against −0.0099 with politician fixed effects (−0.0130 against −0.0110 with fixed effects only; Table 60). The level series moves with all speech: on French speech the Chamber's token-weighted mean (by venue) is 0.5028 before 1894, 0.5115 in 1894–99 and 0.5051 after 1900, against 0.5031, 0.5120 and 0.5054 for the paper's all-speech scores (0.5028, 0.5117 and 0.5053 when all speech is re-scored by the same pipeline as the French series; Table 60). The event study of §6.3, re-estimated on the French-only scores, keeps the break at 1894—the Chamber interactions are +0.009, +0.011 and +0.006 in 1894, 1896 and 1898 ($p < 0.05$, $p < 0.01$ and $p < 0.01$)—but its leads are negative in the 1880s, from −0.006 in 1882 ($p < 0.05$) to −0.001 in 1890, and jointly reject zero ($F(6,40) = 2.66$, $p = 0.03$; Table 61): on French speech the Chamber's edge was rising before the reform, by less than the step it took at it, as on the split-sample ruler (§A.2). In the window extended to 1912 the post-1900 interactions lie between −0.004 and +0.000 through 1910 and at −0.008 in 1912, against −0.003 to +0.001 and −0.007 on all speech (Tables 61 and 7): on French speech, as on all speech, the Chamber's edge over the Senate after 1900 is back at its 1892 level or slightly below it. Since Panel B is not identified on this axis (§6.6), this is one more reason not to build on it, and one more argument for reading the proportional-representation result as compositional rather than as an effect on distance.

Table 12: Dutch share of plenary interventions by legislature. The series rises by a quarter of a percentage point at plural suffrage and by nearly twice that at 1898, around the language-equality law of that year. A language confound cannot produce a break where the composition barely moves.

Legislature	1880	1882	1884	1886	1888	1890	1892	1894	1896
Dutch share	.006	.004	.004	.004	.005	.005	.006	.008	.008
Legislature	1898	1900	1902	1904	1906	1908	1910	1912	
Dutch share	.012	.011	.012	.013	.012	.013	.012	.012	

Share of plenary interventions detected as Dutch from their lemmas, both houses, by legislature, on the corpus as estimated: without the turns of the chair and the unresolved speakers (Appendix B.3). Interventions with no marker of either language (between two and three in five) count in the denominator. The 1914 legislature is omitted: it contributes nineteen interventions and a share of 0.11 that reflects that thinness rather than the composition of the record. Produced by `pi_diagnostics.py`.

Does the yardstick score itself? Section 5.4 argues that holding $\hat{\rho}$ fixed attenuates, and lists the channels that push the contrast toward zero. One channel pushes the other way, and because it is the load-bearing objection we test it rather than argue it. The posteriors are estimated on a pool that is 89.1% Chamber by speeches (86.1% by in-vocabulary tokens; Table 71), and the leave-out correction removes only the speaker's own counts—with several hundred members across three decades, subtracting one speaker barely moves C_j. The treated cell therefore largely scores itself, while the Senate and the pre-reform Chamber are scored against a ruler calibrated on text that is not theirs; and because a foreign ruler pulls a cell toward $\frac{1}{2}$, both displacements raise β. The coverage check of §5.4 does not speak to this: it establishes that the *share* of tokens inside the vocabulary is stable across the reform, not that $\hat{\rho}$ is calibrated evenly within it. We therefore re-estimate the headline contrast on two rulers that cannot score their own cell. The *leave-cell-out* ruler removes each chamber×legislature cell from the frequencies used to score it, so the treated cell cannot calibrate itself at all; its cost is that the ruler becomes cell-specific, which both abandons the cancellation argument of §5.4 and adds estimation variance to every score. It returns +0.0073, 78% of the baseline, with a standard error close to the baseline's (0.0030 against 0.0029; $p = 0.035$ on the wild cluster

bootstrap; Tables 13 and 63). The *split-sample* ruler splits the speakers into five folds and scores each fold against $\hat{\rho}$ estimated on the other four, pooling all chambers and periods: speaker self-scoring is gone while the yardstick stays common across the arms of the contrast, so the cancellation argument survives intact. It returns +0.0119 against +0.0094 on the fixed ruler—127% of the baseline—with $p = 0.013$ on the wild cluster bootstrap (Tables 13 and 63). Its event study keeps the break at 1894 (+0.010, +0.013 and +0.006 in 1894, 1896 and 1898; $p < 0.05$, $p < 0.01$ and $p < 0.01$), but its leads are negative throughout the 1880s, from −0.007 and −0.008 in 1880 and 1882 to −0.002 in 1890, the 1882 lead is significant at the five per cent level and those of 1880 and 1888 at the ten, and the six leads jointly reject zero ($F(6,40) = 2.92$, $p = 0.019$; Table 62): on this ruler the Chamber's edge was already rising before the reform, by less than the step it took at it.

The two rulers remove different components of self-scoring—the first the cell's, the second the speaker's—and they move the estimate in opposite directions: removing the speaker's own text raises it by a quarter, removing the cell's own text takes a fifth of it away. A third check, on the phrase set rather than on the ruler, points the same way as the first. Selecting the vocabulary on pre-1894 speech alone, at the same thresholds, keeps 26,883 phrases against the 36,297 of the pooled set; scored on it with the split-sample ruler, the contrast is +0.0071 (s.e. 0.0021) with politician fixed effects and +0.0087 with fixed effects only—three quarters and nine tenths of the respective baselines—and with the leave-cell-out ruler +0.0042 and +0.0063, all four with $p < 0.01$ (Table 64). Two readings fit these results, and the design cannot separate them. On the first, part of the baseline is the treated cells calibrating their own ruler and the treated period its own phrase set, and the estimate overstates the divergence. On the second, the reform brought partisan vocabulary that did not exist before it—the most Catholic phrases of 1894–99 are those of that decade's social legislation, professional unions, savings banks, the labour contract and the councils of industry and labour (Appendix B.1)—and a ruler that cannot see the treated cell, or a phrase set that cannot see the treated period, cannot measure a divergence that happens in new words. We keep the fixed-yardstick estimate as the paper's—it lies between the split-sample ruler, which shares its common yardstick and returns more, and the rulers blind to the treated cell or to the treated period, which return less—and we read it as neither a lower nor an upper bound.

Table 13: The plural-suffrage effect under three yardsticks. "Fixed pooled" is the paper's baseline; "leave-cell-out" removes each chamber×legislature cell from the frequencies used to score it; "split-sample" scores each of five speaker folds against the posteriors estimated on the other four, which removes self-scoring while keeping the yardstick common across chambers and periods.

	Fixed pooled (baseline)	Leave-cell-out	Split-sample (5-fold, by speaker)
Chamber × Post-1894 (politician FE)	0.0094	0.0073	0.0119
as % of baseline	100	78	127
wild cluster bootstrap p	–	0.035	0.013
Ruler self-scores the treated cell?	yes	no	no
Ruler common across chambers/periods?	yes	no	yes

Politician fixed effects, speeches weighted by in-vocabulary token count, standard errors clustered at arrondissement; the bootstrap column uses Webb weights and 9,999 replications. Produced by `gst_leave_cell_out.py` and `leave_cell_out_did.do`.

The two de-biasers agree where it matters. We do not privilege one bias correction: the GST leave-out jackknife and the Simola et al. (2025)-style penalized maximum-likelihood estimator (§A.4) are reported side by side as co-equal routes to the same target. They agree on both panels: with politician, arrondissement, party and legislature fixed effects the penalized outcome reproduces the plural-suffrage divergence

(+0.0094, $p < 0.01$, against +0.0093 for the leave-out outcome with the same fixed effects, Table 9, column 1) and the negative Panel B contrast (−0.0090, $p < 0.01$, against −0.0099 for the leave-out reference specification; Table 17). They agree on the mining triple as well, which is negative on both (−0.004, and −0.003 for the leave-out reference specification, both $p < 0.01$) and stays in the appendix for the reasons given in §A.7—its support and its object—and not because the two de-biasers disagree on it.

Venue or membership? The ministers. The chamber indicator is the member's majority house, the one in which he delivers most of his interventions (§3). Ministers speak in both houses, and members who changed house appear in both, so the indicator and the venue of a speech differ for 21,429 interventions, 11,325 of them in the Panel A window. The concern this raises is specific: a speech assigned to the wrong house places a treated individual in the control cell, or the reverse, which attenuates β; and because these are the corpus's heaviest speakers, token weighting gives the misclassification more weight than head counts would.

Before re-estimating we check that these are ministers or members who changed house (§3) rather than two people merged under one identifier, which is the failure the audit of §A.2 was built to catch. The discriminant is the constituency: a minister carries his own seat into the other house, whereas a merged pair would show two. Of the 120 members who speak in both houses, 94 have the same modal arrondissement in both, 18 have no arrondissement recorded in one of the two—most of them ministers, whose interventions in the other house carry none—and 8 do not; of those eight, one carries three distinct arrondissements across the corpus, the signature of a change of seat or of noisy constituency metadata rather than of two careers, and four speak in their minority house fewer than a dozen times. Forty-seven of the 120 have a clean break in time between the two venues, the pattern of a change of house.

Table 14 then re-estimates the plural-suffrage contrast under four chamber definitions. On the majority house, the reference, it is +0.0094; on the venue of each speech +0.0115; dropping the interventions delivered outside the member's majority house gives +0.0103; and assigning each intervention to the house in which the career file places the member on the day he spoke—a definition that differs from the majority house on 15,928 interventions—gives +0.0085. The other three definitions move the reference by at most 0.0021, under a quarter of the effect, and the largest move, on the venue, is upward (Table 14). The weighting channel is equally quiet: unweighted, the four definitions give between +0.0063 and +0.0111, the reference at +0.0081—a little below the weighted estimate, as §A.2 reports for the reference definition—and all with $p < 0.05$ (Table 14).

A fifth estimate is worth reporting for what it is not. Dropping the members who speak in both houses altogether leaves +0.0083 (s.e. 0.0035, $p < 0.05$; +0.0073 unweighted; Table 14), on 63,888 observations against 206,116: these members carry seven in ten Panel A speeches, so this is a different population—members who spoke in one house only—rather than a corrected estimate. Read as such it is informative, and it is reassuring: among the members who spoke in one house only, the plural-suffrage effect is nearly nine tenths of the reference, so the estimate is not carried by the government and by the members who changed house.

Weighting by speech length. The outcome is weighted throughout by $w^{(i,s)}$, the speech's in-vocabulary token count, for two reasons that are worth separating. The first is definitional: within each party the w-weighted mean of s_{distinct} reproduces exactly the token-weighted aggregation that defines π^R and π^D (§4.3), so the weighted regression estimates the reform effect on *partisanship of speech* with phrase tokens as the units over which the observer's guessing game is played; across the two parties it weights by tokens rather than equally. The second is statistical: s_{distinct} is a generated regressand, the average of $w^{(i,s)}$ noisy token-level posteriors, so its measurement-error variance scales as $1/w^{(i,s)}$ and w-weighting is precision weighting. Dropping the weights therefore changes the *estimand*—from token-weighted partisanship to an

Table 14: The plural-suffrage and proportional-representation contrasts under four definitions of the chamber indicator. "Majority house" is the reference: the house in which the member delivers most of his interventions, constant within politician. "Venue" is the house in which the intervention was delivered. "Drop minority house" removes interventions delivered outside the member's majority house. "Career file" assigns each intervention to the house in which the career file places the member on the day he spoke.

	Majority house (reference)	Venue	Drop minority house	Career file
Panel A: Chamber × Post-1894				
Weighted	0.0094^{***}	0.0115^{***}	0.0103^{***}	0.0085^{**}
	(0.0029)	(0.0026)	(0.0033)	(0.0038)
Unweighted	0.0081^{**}	0.0111^{***}	0.0087^{***}	0.0063^{**}
	(0.0032)	(0.0029)	(0.0031)	(0.0030)
Observations	206,116	206,116	197,230	206,116
Panel B: Chamber × Post-1900				
Weighted	-0.0099^{***}	-0.0094^{***}	-0.0106^{***}	-0.0115^{***}
	(0.0028)	(0.0026)	(0.0030)	(0.0030)
Unweighted	-0.0068^{**}	-0.0094^{**}	-0.0073^{**}	-0.0081^{***}
	(0.0028)	(0.0036)	(0.0030)	(0.0029)
Observations	245,268	245,268	234,389	245,268

Politician, arrondissement and party fixed effects; where the definition varies within politician (venue, career file) the indicator also enters on its own. Speeches weighted by in-vocabulary token count except where stated; standard errors clustered at arrondissement. Dropping the members who speak in both houses gives $+0.0083$ (s.e. 0.0035; unweighted $+0.0073$, s.e. 0.0039) on 63,888 observations in Panel A and -0.0084 (s.e. 0.0033; unweighted -0.0068, s.e. 0.0040) on 102,680 in Panel B, but changes the population rather than the definition and is discussed in the text. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

equal-weighted average over speeches, in which a one-line interjection counts as much as a programmatic address—as well as the efficiency. Table 15 re-estimates the headline contrasts unweighted. The plural-suffrage effect survives unweighted (+0.0092 FE-only, $p < 0.01$, and +0.0081 with politician fixed effects, $p < 0.05$), so Panel A does not depend on the weighting. The Panel B contrast stays negative unweighted with fixed effects only (−0.0123, $p < 0.01$) and shrinks within politician (−0.0068, $p < 0.05$; Table 15): part of the post-1900 signal the weighted within-politician specification picks up lives in long, programmatic speeches—the token-weighted margin the measure is defined on—rather than in the average intervention. Either way Panel B is not read causally (§6.6). We keep the weighted specification as the main one—it is the estimand the measure defines—and report the unweighted run as the robustness column.

One property of the weights is worth stating, because it governs how the weighted estimate should be read. Speech is concentrated: a few members hold the floor far more than the rest—an allocation that, in disciplined parliaments, is the party's choice as much as the member's (Proksch and Slapin, 2012), so who speaks is itself an outcome, and the within-politician specification conditions on the speaker but not on the party's selection of speakers, a channel we flag rather than resolve—and token weighting inherits that. At the politician level, where the adaptation result lives, the weighted comparison has a Kish effective sample size of 28 against 133 politicians observed on both sides of the reform, with the five heaviest carrying 34% of the total weight (Table 67). The comparison is thinner than the politician count suggests, and it leans on a few members: with each member weighted by his total in-vocabulary tokens, the within-politician difference is +0.0056 ($t = 1.7$), against −0.0097 ($t = -1.0$) unweighted; dropping a single politician moves the weighted difference anywhere between 0.0037 and 0.0067 (the unweighted one between −0.0126 and −0.0016); and dropping the thinnest decile of politicians, which leaves the weighted figure unchanged (+0.0058), turns the unweighted one to +0.0016 ($t = 0.7$; Table 67). The sign of the unweighted contrast is therefore set by the members with the least speech, whose means are the noisiest; weighted, the contrast is positive under every single deletion, and its precision is limited by how few members carry it. (Weighting each member by the thinner of his two sides, as the permutation test of §7.1 does, gives +0.0097; Table 66.)

Table 15: Robustness: unweighted regressions. The main specifications of Table 3 re-estimated without the $w^{(i,s)}$ (in-vocabulary token count) analytic weights.

	(1) A: FE	(2) A: Turnover	(3) A: Pol. FE	(4) B: FE	(5) B: Turnover	(6) B: Pol. FE
Chamber × Post-1894	0.0092***	0.0103***	0.0081**			
	(0.0033)	(0.0035)	(0.0032)			
Chamber × Post-1900				-0.0123***	-0.0122***	-0.0068**
				(0.0027)	(0.0027)	(0.0028)
Incumbent (1894)		0.0077**				
		(0.0030)				
Entrant (1894)		0.0000				
		(.)				
Incumbent (1900)					-0.0002	
					(0.0011)	
Entrant (1900)					0.0000	
					(.)	
Observations	206,146	206,146	206,116	245,296	245,296	245,268
R^2	0.019	0.021	0.046	0.009	0.009	0.029

Standard errors clustered at arrondissement in parentheses. Produced by `unweighted_robustness.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The weighting argument above is about the estimand; a separate question is whether either specification depends on the short speeches that are common in the corpus. In the scored corpus the median intervention contributes six in-vocabulary phrases, 18.5% contribute exactly one, and those carry 0.8% of the total weight (Table 16). Short interventions are a property of parliamentary transcripts rather than a defect of the corpus—many interventions are procedural, and a chamber's record is mostly not oratory—but they give the headline a long left tail, and a reader is entitled to ask what happens without it.

Table 16 imposes a rising minimum length and re-estimates both headline contrasts, weighted and unweighted. The plural-suffrage effect stays between $+0.0093$ and $+0.0098$ across the ladder while the estimation sample falls from 206,116 speeches to 65,332, and remains significant at the one per cent level on every rung, its standard error unchanged at 0.003. The Panel B contrast barely moves along the ladder, between -0.0098 and -0.0100, and is significant on every rung. In Panel A weighted and unweighted estimates differ by at most 0.0034 at any threshold (Table 16); in Panel B the unweighted contrast is about two thirds of the weighted one on the first rungs and grows in magnitude as short interventions are removed, staying smaller than the weighted one.

Two things follow. The first is that the reform effect is not an artefact of one-line interjections: it is present, at the same magnitude, among the speeches of twenty phrases or more, under a third of the sample, where the measure is least noisy. The second is that the choice of weights, which the preceding paragraphs defend on definitional grounds, matters little for Panel A—we retain w-weighting because it is the estimand the measure defines, not because the result requires it.

Table 16: Robustness: a minimum-length ladder. The headline bicameral contrasts re-estimated on speeches carrying at least the stated number of in-vocabulary phrases, with and without the $w^{(i,s)}$ analytic weights. Specification as in Table 3, column (3): politician, arrondissement and party fixed effects, errors clustered at arrondissement.

	Panel A (1894)		Panel B (1900)	
Minimum length	Weighted	Unweighted	Weighted	Unweighted
All speeches	0.0094	0.0081	−0.0099	−0.0068
$\geq$ 1 phrases	0.0094	0.0081	−0.0099	−0.0068
$\geq$ 2 phrases	0.0094	0.0087	−0.0099	−0.0077
$\geq$ 3 phrases	0.0094	0.0084	−0.0100	−0.0086
$\geq$ 5 phrases	0.0093	0.0060	−0.0100	−0.0087
$\geq$ 10 phrases	0.0096	0.0077	−0.0098	−0.0074
$\geq$ 20 phrases	0.0098	0.0085	−0.0100	−0.0084

The median speech of the estimation sample carries seven in-vocabulary phrases: ≥ 5 retains 58.4% of its speeches and ≥ 20 retains 31.7%. In the analysis file of Catholic and Liberal speeches, 1880–1912 (399,131), the median speech carries six, 18.5% carry exactly one, and those carry 0.8% of the total weight. Rungs whose estimation sample falls below 500 observations or 20 clusters are not reported. Produced by `short_speech_tail.do`.

A distinct concern is the *width* of the vocabulary rather than its hygiene: the leave-out estimator is defined on a fixed global phrase set, and the count and speaker thresholds that admit a phrase could in principle drive the findings. We therefore re-run the entire pipeline over a three-rung ladder that varies only vocabulary width, holding the hygiene filters fixed: a *strict* set (≥ 50 total occurrences, ≥ 8 distinct speakers), a *loose* set (≥ 25, ≥ 5), and a *medium* set (≥ 35, ≥ 6) chosen a priori as the geometric midpoint, which is the specification reported throughout the main text. The ladder divides cleanly. Everything the causal claim rests on is invariant: the plural-suffrage divergence is $+0.0092$, $+0.0095$, and $+0.0098$ (all $p < 0.01$) from the strictest to the loosest vocabulary, holds under politician fixed effects on every rung ($+0.0088$ to $+0.0098$, all $p < 0.01$), and keeps its concentration on incumbents ($+0.0114$ to $+0.0124$, all $p < 0.01$; Table 52); the negative continuous mining triple and the Panel B contrast are equally stable. The socialist-proximity coefficients do not change the picture on any rung: the Catholic coefficient is not identified on any rung, after either reform (Table 75)—one more reason the paper does not build a mechanism on them (§6.7). The side-by-side ladder is in Appendix B.9.

Table 17: Robustness: penalized Poisson-LASSO partisanship outcome s^{L1}_{distinct}. Columns (1)–(2): Panel A, difference-in-exposure and continuous mining triple; (3)–(4): the same for Panel B. All columns absorb politician, arrondissement, party and legislature fixed effects.

	(1)	(2)	(3)	(4)
Chamber × Post-1894	0.00944***	0.00472***		
	(0.00195)	(0.00109)		
Chamber × Post-1894 × Mining		-0.00390***		
		(0.000822)		
Chamber × Post-1900			-0.00899***	-0.00374*
			(0.00243)	(0.00212)
Chamber × Post-1900 × Mining				0.00113
				(0.00122)
Observations	206116	206116	245268	245268

Standard errors clustered at arrondissement in parentheses. Produced by `polarization_did_extensions.do` (block D3) on the post-audit L_1 scores. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 18: Cluster-bootstrap inference re-estimating ρ on resampled arrondissements

Coefficient	Bootstrap mean	95% CI
Chamber × Post-1894	0.014	[0.006, 0.021]
Chamber × Post-1894 × Mining	-0.009	[-0.017, -0.002]
Chamber × Post-1900 × Mining (entrants)	0.003	[0.001, 0.005]

Note. 200 block-bootstrap replications resampling arrondissement clusters; ρ re-estimated on each resampled Catholic/Liberal pool and the per-speech score recomputed before re-fitting with arrondissement, party, year (and, for the level arm, politician) fixed effects. CIs are wider than the clustered analytic SEs, as expected for a generated regressand.

The decomposition. Speeches carrying no substantive policy content—procedural exchanges, communications, calls to order—are a large share of any parliamentary record, and here they are 61.0% of the corpus (Table 55). We retain them as a seventh bucket rather than dropping them. The reason is identification, not completeness: the rate at which speeches carry a substantive topic is not balanced across the design. The share of classified interventions falls by 2.6 points in the Chamber across 1894 and rises by 0.6 in the Senate, a difference-in-differences of -3.2 percentage points (Table 55), in the direction that would select the estimation sample on treatment at exactly the reform that identifies the headline result if we restricted to classifiable speech. Keeping the residual removes the selection by construction, and it is also the substantively right call: if the Chamber shifts toward procedural speech after 1894, that *is* agenda reallocation, and belongs in the agenda component rather than outside the sample. The decomposition on the restricted sample is reported as a robustness and gives the same answer.

For chamber c and window p, aggregate distinctiveness—the token-weighted mean of s_{distinct} of §4.3—is exactly the topic-token-share-weighted average of within-topic distinctiveness,

$$\Pi_{c,p} = \sum_k w_{c,p,k}\, d_{c,p,k}, \qquad \sum_k w_{c,p,k} = 1, \tag{11}$$

where $w_{c,p,k}$ is the share of chamber c's window-p in-vocabulary tokens in speeches tagged to topic k and $d_{c,p,k}$ is the token-weighted mean of s_{distinct} over those speeches. The change across a reform splits, Oaxaca-

fashion, into a component from shifting *what* is debated and one from shifting *how distinctly* each topic is debated,

$$\Delta\Pi_c = \underbrace{\textstyle\sum_k \Delta w_{c,k}\, \bar{d}_{c,k}}_{\text{agenda reallocation}} + \underbrace{\textstyle\sum_k \bar{w}_{c,k}\, \Delta d_{c,k}}_{\text{within-topic divergence}}, \tag{12}$$

with $\bar{w}, \bar{d}$ the two-window averages. Because the decomposition is linear, the bicameral difference-in-exposure inherits it, $\mathrm{DiD}(\Pi) = \mathrm{DiD}_{\text{within}} + \mathrm{DiD}_{\text{agenda}}$, where $\mathrm{DiD}_{\text{within}}$ is the reform effect *holding the agenda fixed* and $\mathrm{DiD}_{\text{agenda}}$ the part from differential topic reallocation. The regression counterpart augments (7) with topic fixed effects τ_k,

$$s^{(i,s)}_{\text{distinct}} = \beta_W \left(\mathrm{DK}_i \times \mathrm{Post}_t\right) + \tau_k + (\text{two-ways and FE}) + \varepsilon_{i,s}, \quad [aw = w^{(i,s)}], \tag{13}$$

so that the topic fixed effect removes the fact that some topics are intrinsically more party-distinctive—the agenda-composition channel—and β_W identifies the within-topic divergence; comparing β_W to the baseline β on the same sample reads off how much of the effect is agenda-driven. We deliberately do *not* add topic×period interactions: the two houses debate different topics, so a topic×Post term is nearly collinear with Chamber×Post and would mechanically absorb the effect rather than isolate it.

Table 19 reports both objects, and they are two objects rather than one. The decomposition of (12) is arithmetic on the cells: it takes the topic shares $w_{c,p,k}$ and the within-topic means $d_{c,p,k}$ directly and splits each chamber's change, then differences the two components across houses. The regression of (13) is a different estimator of a related quantity, and the difference matters. Topic fixed effects enter in levels: they remove the fact that some topics are intrinsically more party-distinctive, but they do not remove the reallocation of topic *shares* over time, and that reallocation is precisely the agenda channel. Where shares move, β_W is therefore not a clean estimate of within-topic divergence, and a within share read off β_W/β inherits the problem. We report (12) as the decomposition and the regression as a corollary, and we give both rather than the more convenient one.

On the decomposition, the within-topic component is $+0.0091$ against an agenda component of $+0.0025$, a within share of 78.6% (Table 19). On the regression, the baseline effect on the topic-augmented sample is $\beta = +0.0095$ ($p < 0.01$) and adding topic fixed effects leaves it at $\beta_W = +0.0090$ ($p < 0.01$), a ratio of 94.1% (Table 19). The two differ by fifteen points, and the direction is the one the reasoning predicts: the regression overstates the within share, because topic fixed effects in levels leave the reallocation of shares in the residual and that reallocation is the agenda channel. We therefore quote 78.6% in the text and report both in Table 19. The totals differ too, and for a separate reason worth stating: the decomposition is raw cell arithmetic with neither fixed effects nor clustering, so the gap between its total and β is the part of the headline coefficient that comes from the arrondissement and party structure rather than from the topic composition of speech.

Fifteen points is a correction, not a reversal. An agenda component of 21.4% is larger than the 5.9% the regression residual would suggest (Table 19), and the claim that survives is the narrower one: about four fifths of the plural-suffrage effect is divergence *within* topics, and the remainder is the parties debating a differently composed agenda.

The decomposition runs on the whole Panel A window—the 206,146 speeches of the fixed-effects specification of Table 3—so it describes the same object the headline estimate identifies, not a classifiable minority of it. On either measure the reading is the same and it is the one the title claims: the parties did not merely talk more about labour; within the labour domain—the classifier's second best-identified bucket after the procedural residual—they came to speak more distinctively. This is what separates the distinctiveness result from the companion paper's salience result. The two are correlated by construction, but a divergence that survives holding the agenda fixed is not a mechanical re-expression of the agenda shift.

The conclusion does not turn on the two choices the decomposition requires. Restricting to speeches with a substantive topic—the selected sample—gives $\beta = +0.0098$ and a within share of 93.5%; moving

from the six-bucket taxonomy to the 22 fine domains gives 90.0% on the full sample and 88.4% on the restricted one (Appendix Table 56). The finer control, which should absorb more of an agenda channel if there were one to absorb, takes a few points further from the effect and no more. The fine labels are, however, measured with more error than the coarse ones, and a noisier control absorbs less; the same logic applies, more weakly, to the coarse figure itself. The coarse labels agree with human coding at $\kappa = 0.80$ before the distillation and with the classifier at $\kappa = 0.77$ after it (Tables 54 and 74), so the attenuation is modest, and bounding it explicitly leaves the reading intact: even under an assumed reliability of 0.5—below what the validation supports—the regression's agenda residual doubles to 0.0011 and its within ratio is still above 88%. The bound is computed on the regression, which is the estimator the topic labels enter as a control; the share decomposition is not a regression and attenuation enters it differently, through the cell means rather than through an omitted control.

Two features of the completed decomposition deserve to be stated, because both strengthen the reading and neither is visible in the summary share.

The first is the control. The Senate's change decomposes into an agenda component of -0.0006 and a within-topic component of -0.0030 (Table 19): the upper house moves in the opposite direction to the Chamber on *both* margins, not merely staying flat on the total. A shock common to the two houses would have to move them together on at least one; this one moves them apart on each.

The second is the size of the agenda shift itself, which is what makes the small agenda component informative rather than trivial. The Chamber's agenda was transformed across the reform. The confessional share of its in-vocabulary tokens falls from 19.8% before 1894 to 7.8% after, and labour and social policy rises from 5.0% to 13.3% (Table 73)—between them, a fifth of the floor changing hands, which is the agenda shift our companion study measures directly as rising labour salience (Graziani and Vanschoonbeek, 2026). That reallocation contributes little to the distinctiveness effect, and the reason is visible in the cells: before the reform the confessional and labour domains scored close to each other on partisanship (0.5018 and 0.5070; Table 73), so moving weight from one to the other moved it between cells that were nearly equally undiagnostic. What changed was not which cells carried weight but what happened *inside* the labour cell, where distinctiveness rises by $+0.021$ against -0.003 on the confessional one (Table 73). Parliament changed subject dramatically; the divergence did not come from the change of subject.

Table 19: Topic decomposition of the plural-suffrage distinctiveness effect. The upper block computes (12) from the cell shares $w_{c,p,k}$ and the within-topic means $d_{c,p,k}$, splitting each chamber's change and differencing the components across houses; the lower block reports the regression (13). They are different estimators of related quantities and are reported as such: topic fixed effects enter in levels, so β_W leaves the reallocation of topic shares in the residual and the ratio β_W/β overstates the within share.

	Chamber	Senate	Difference
Decomposition (12)*, from cell shares*			
within-topic component	+0.0062	−0.0030	+0.0091
agenda component	+0.0019	−0.0006	+0.0025
total change in Π	+0.0080	−0.0036	+0.0116
within-topic share of the difference			78.6%
Regression (13)*, with fixed effects*			
baseline β (no topic controls)			0.0095***
within-topic β_W (topic FE)			0.0090***
ratio β_W/β			94.1%
Speeches			206,146
DiD on P(in sample)			0.0000

Seven coarse buckets, procedural residual retained, all speeches in the Panel A window carrying a topic and a distinctiveness score. The decomposition is exact cell arithmetic and carries no fixed effects and no clustering, so its total need not equal β; the gap is the part of the headline coefficient that comes from the arrondissement and party structure rather than from topic composition. Standard errors on the regression clustered at arrondissement, speeches weighted by in-vocabulary token count. Topic labels from the CamemBERT classifier trained on GPT-4o annotations of a design-stratified random sample and distilled onto the dataset's interventions (Appendix B.10). Topic fixed effects enter as levels only: the two houses debate different topics, so a topic×Post term is nearly collinear with Chamber×Post and would absorb the effect rather than isolate it. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Produced by `topic_oaxaca.do` and `topic_decomposition.do`.

What a fixed yardstick does to the contrast. Holding $\hat{\rho}$ fixed has a cost that deserves to be stated, because it is not the usual generated-regressand concern. The posteriors are estimated on the pooled sample, which includes speech from *after* the reforms, so the yardstick against which every speech is scored is itself a weighted average of the pre- and post-reform partisan mapping. That average would be innocuous if the mapping were stable. It need not be, and the phrase lists of §4.6 show the party-diagnostic vocabulary changing across 1894—the social-Catholic lexicon of professional unions, savings banks and the labour contract heads the Catholic list only after the reform (Table 23)—which is precisely the substantive movement the paper is about.

The consequence is mechanical and it runs one way. A phrase that is highly diagnostic within each period but points to opposite parties across periods receives a pooled posterior near $\frac{1}{2}$, and therefore contributes almost nothing to s_{distinct} in either period, even though a period-specific estimator would score it as strongly partisan in both. Fixing the yardstick thus *discards* exactly the partisan information carried by vocabulary that changes sides, and understates the level of partisanship in every cell. The same logic applies to the contrast: to the extent that the reform reallocates phrases between the parties rather than merely sharpening existing ones, the fixed-yardstick difference-in-exposure recovers less than the true divergence.

We accept this deliberately, for two reasons. First, comparability: a period-specific $\hat{\rho}$ would make the pre- and post-reform indices quantities on different scales, so that a difference between them would confound a change in speech with a change in the measuring instrument—the same problem that makes cell-specific estimates unsuitable for cross-cell comparison (§4.6). A single yardstick is what makes β interpretable as a

movement of speech rather than of the ruler. Second, direction: this channel biases toward zero.

Because that second claim is the load-bearing one, we checked the channels that would push the contrast the other way. The first is selection of the vocabulary over the pooled window: if speech volume were unbalanced in time, the phrases clearing the frequency thresholds would be disproportionately those of the better-represented period, and the other period would be measured with a yardstick calibrated on the wrong text—mechanically depressing its Π and inflating the contrast. It is not: the share of bigrams falling inside the vocabulary is flat across the reform, 0.269 before 1894 against 0.265 in 1894–99 in the Chamber, and 0.236 against 0.233 in the Senate (Table 71). That figure speaks to the size of the in-vocabulary slice rather than to its composition, so we check the composition directly: re-estimating the vocabulary on pre-1894 speech alone, at identical thresholds, retains 26,883 phrases against the 36,297 of the pooled set—fewer, as the same absolute thresholds applied to about a third of the speech must give, so the count cannot say whether the pooled thresholds favour the treated period. The estimate can, and it does not settle the question: scored with the pre-1894 vocabulary the contrast falls to between three quarters and nine tenths of the baseline on the split-sample ruler and to between less than half and two thirds on the leave-cell-out ruler (§A.2), which is what selection on the treated period would produce, and also what partisan vocabulary born with the reform would produce. This channel cannot be signed from the data. The second is the sample-size dependence of the leave-out correction, which bites harder where speakers are fewer and so pulls thin cells toward $\frac{1}{2}$: were it to weaken in the treated house, Chamber partisanship would rise mechanically. On this corpus it does: the token-weighted mean share of a speaker in his party's pool falls by 0.0020 in the Chamber across 1894 and rises by 0.0017 in the Senate (Table 71), so this channel runs with the contrast rather than against it. The third is the arrival of the socialists, who take floor time from the two governing parties and thin the Catholic–Liberal pool exactly in the treated house: the governing parties' share of Chamber interventions falls from 98% before 1894 to 66% in 1894–99, against a fall from 100% to 92% in the Senate (Table 1). A thinner pool means a sharper leave-out correction, and again in the treated house. The third channel runs with the attenuation already identified, the second against it, and the first cannot be signed. A fourth runs the same way as the second and deserves its own test rather than an argument: because the pooled sample is overwhelmingly Chamber speech and the leave-out correction removes only one speaker at a time, the treated cell is close to scoring itself, while the arms scored against a ruler calibrated elsewhere are pulled toward $\frac{1}{2}$—both of which raise the contrast. We take this up directly in §A.2: a split-sample ruler that removes the speaker's own text returns a quarter more than the estimate, while a leave-cell-out ruler that removes the treated cell's own text returns four fifths of it, so we make no claim that the estimate is a lower bound.

A.3 Proximity to the socialist lexicon

The Catholic–Liberal partisanship of (4) measures distance on the *governing* axis. A mechanism running through the socialist vote (§6.7) needs a second, orthogonal measure: how close a Catholic or Liberal speech sounds to the *socialist* register that entered the Chamber in 1894. We build it by the same logic as s_{distinct}, simply changing the contrast from Catholic-vs-Liberal to Socialist-vs-rest.

On the post-1894 Chamber pool of all three parties, and in the *same* fixed vocabulary of §4.1, let $z_i \in \{0,1\}$ flag a socialist speaker. For phrase j define the socialist phrase share and its complement,

$$S_j^1 = \sum_{i:\, z_i=1} X_{ij}, \quad S_j^0 = \sum_{i:\, z_i=0} X_{ij}, \qquad N^1 = \sum_{i:\, z_i=1} m_i, \quad N^0 = \sum_{i:\, z_i=0} m_i,$$

so that the posterior probability that a token of phrase j was spoken by a socialist is

$$\rho_j^S = \frac{S_j^1/N^1}{S_j^1/N^1 + S_j^0/N^0} = \mathbb{P}(\text{Socialist} \mid \text{phrase } j). \tag{14}$$

A phrase used disproportionately by socialists (*class_ouvri*, *lutt_class*) has ρ_j^S near 1; a generic or confessional phrase has ρ_j^S near 0. Because the speeches we will *score* are Catholic and Liberal—never socialist—the speakers being scored are not in the socialist group, so no leave-out correction is needed for them and (14) is the plain pooled posterior on the three-party pool.[4]

The *socialist proximity* of a Catholic or Liberal speech s is then the token-weighted average socialist-ness of the phrases it contains,

$$\text{prox}_S^{(i,s)} = \frac{\sum_j x_{ij}^{(s)} \rho_j^S}{\sum_j x_{ij}^{(s)}}, \qquad w^{(i,s)} = \sum_j x_{ij}^{(s)}, \tag{15}$$

exactly parallel to (5) but reading each phrase's socialist posterior (14) in place of the self-party posterior. A Catholic deputy who borrows socialist collocations scores high; one who speaks a confessional or paternalist idiom scores low. In the same triple-difference design it asks whether reform moved the governing parties *toward* or *away from* the socialist register; the coefficients are in Appendix B.8.

A.4 The second de-biasing route: penalized maximum likelihood

The leave-out estimator de-biases by jackknife; a second, independent route de-biases by regularization. Gentzkow et al. (2019) obtain the partisanship index from the choice probabilities of an L_1-penalized multinomial logit, and we implement it so that our conclusions rest on two distinct bias corrections rather than one. Estimating the multinomial directly is infeasible at our vocabulary size, so—exactly as in the Finnish application we follow (Simola et al., 2025)—we use the result of Taddy (2015) that a factorial of independent Poisson regressions, conditioned on each speaker's log token total, approximates the conditional multinomial likelihood. For each phrase j we therefore fit a penalized Poisson regression of the count on the party indicator, with $\log m_i$ as an exposure offset and an L_1 penalty on the slope:

$$X_{ij} \sim \text{Poisson}\big(\exp\{a_j + g_j y_i + \log m_i\}\big), \qquad \hat{g}_j = \arg\min_{g_j} \; \ell_j(a_j, g_j) + \alpha\,|g_j|, \tag{16}$$

where ℓ_j is the negative Poisson log-likelihood and α the penalty ($\alpha = 10^{-3}$). The offset $\log m_i$ is what controls for verbosity: a phrase cannot look partisan merely because one party speaks at greater length. From the fitted coefficients we form penalized, within-group–normalized choice probabilities

$$q_j^{\text{cat}} \propto e^{a_j + g_j}, \quad q_j^{\text{lib}} \propto e^{a_j}, \quad \rho_j = \frac{q_j^{\text{cat}}}{q_j^{\text{cat}} + q_j^{\text{lib}}},$$

and aggregate as in (4), $\Pi_{L_1} = \frac{1}{2}\big(\sum_j q_j^{\text{cat}} \rho_j + \sum_j q_j^{\text{lib}} (1 - \rho_j)\big)$. The penalty shrinks the slopes of uninformative phrases to zero, sending their ρ_j to $\frac{1}{2}$, so that only phrases carrying replicable signal move the index. This estimator is a penalized maximum-likelihood object (the distributed-multinomial form of Taddy, 2015), not a separate heuristic: it is the same target as the leave-out index reached by shrinkage rather than by jackknife. We treat the two as co-equal de-biasing routes and report both throughout. One caveat governs how we read its standard errors: the L_1 penalty selects phrases, so the analytic Poisson standard errors are conditional on that selection and understate uncertainty; for inference we therefore lean on the cluster bootstrap of §5.4, which re-estimates the measure on each draw, rather than on the penalized analytic errors.

[4] The posterior ρ_j^S is the Socialist-vs-rest posterior estimated once on the post-1894 Chamber three-party pool in the shared vocabulary, and held fixed across the periods being compared. A Catholic or Liberal speaker's own counts do enter the non-socialist totals S_j^0, N^0, but with a pool of this size any single speaker's contribution to the pooled frequencies is negligible, so the own-inclusion component of (14) is second order.

A.5 Permutation bands

For the descriptive trajectories we attach a falsification band to each cell. We re-compute Π under random relabelling of party, holding group sizes fixed, B times, and report the 5th–95th percentile of the resulting draws. A cell whose true Π falls inside this band carries no partisan signal distinguishable from chance, whether because it is genuinely non-partisan or merely too thin. The band is the natural null for a measure whose chance level is $\frac{1}{2}$ only in expectation, not in finite samples; its width therefore varies with the size and balance of the cell, and is widest exactly where the cell is thinnest.

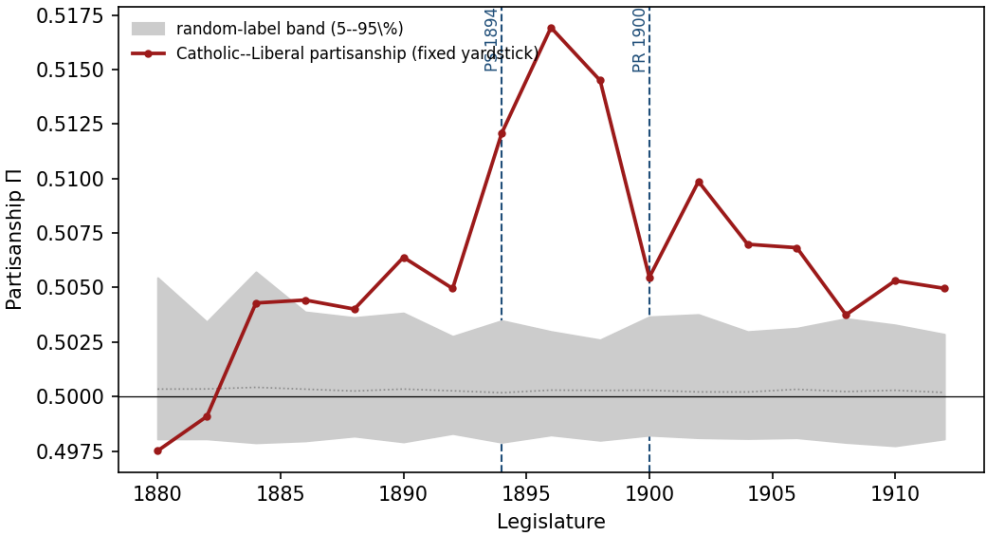


Figure 4: **Catholic–Liberal partisanship and its random-label null.** Partisanship by legislature (red) against the null (grey band: 5th–95th percentile of Π under random permutation of party across members, with group sizes held fixed). Chamber only. Both the series and the band are computed on a single yardstick, estimated on the Chamber's Catholic–Liberal pool (the regressions' yardstick also pools the Senate) and held fixed across legislatures so that legislatures of different size remain comparable, and they weight the two parties by their token shares (§6.1); re-estimating ρ within each cell, as a purely descriptive aggregate would, compresses thin cells toward $\frac{1}{2}$ and is not the quantity of interest here (§4.6). The reading is direct. In 1880–1884 the two governing parties are statistically indistinguishable: the series sits inside its null band, in 1880 marginally below it. From 1886 it lies just above the band (0.504–0.506 against upper bounds of 0.503–0.504), a small pre-reform signal. At the 1894 legislature it steps up (0.512 against an upper bound of 0.503), rises to a peak of 0.517 in 1896, and never returns: from 1894 onward every point lies above its own null, after 1900 by less (0.504–0.510 against upper bounds of 0.503–0.504; in 1908 only just). The break is at plural suffrage; because the series weights the two parties by their token shares, part of the 1894 step reflects the jump in the Catholic share of the governing parties' tokens (§6.1), and from 1898 a small part of the level reflects Dutch speech, which a vocabulary pooling both languages reads as Catholic (§A.2). Vertical lines mark plural suffrage (1894) and proportional representation (1900); the window ends in 1912, the last full legislature before the war. Produced by `permutation_series_fixed.py`.

No covariates in the choice model. One comparison is the century-long study of Finnish parliamentary speech by Simola et al. (2025), who enter government status, gender, and region as controls in the phrase choice probabilities to net out compositional shifts in the parties. None of the three is operative in our setting. *Government status* is collinear with party: the Catholics governed continuously from 1884 to the

end of our window, and the single Liberal government (1878–1884) sits at the very start of the sample, so a government control would be a near-perfect proxy for the Catholic label the measure is built to predict. *Gender* is constant—the Belgian parliament was entirely male throughout 1880–1912. *Region* we do control for, but at the stage where it belongs: the arrondissement enters as a fixed effect in the regression (§6.2), not as a covariate in the generated regressand. Holding it out of the choice model is deliberate. The leave-out yardstick ρ must be common across chambers and periods so that it differences out of the bicameral contrast (§5.4); a region-specific choice model would make ρ vary by arrondissement and break that cancellation. Composition is therefore handled by the permutation band above and by fixed effects downstream, not inside the measure.

A.6 Where the parties diverged: polarization by topic

The decomposition of §6.4 establishes that the divergence is within-topic rather than agenda-driven, but it treats the topics as a block. The natural next question is which subjects the parties pulled apart on. We answer it by computing, for every topic $\times$ legislature cell, the same partisanship index on a single yardstick held fixed across cells, estimated on the Chamber's Catholic–Liberal pool—the comparability requirement of §4.6, without which thin cells would appear spuriously less partisan—and then aggregating to reform windows. Table 20 reports the result, ordered by the change at plural suffrage.

The ranking is unambiguous. *Labour and social policy* moves most, and by a wide margin: Catholic–Liberal partisanship on labour speech rises from 0.503 before 1894 to 0.530 in 1894–99, a change of $+0.027$ (Table 20)—more than twice the next-largest topic. Agriculture follows at $+0.013$, then budget ($+0.011$), infrastructure ($+0.011$) and justice ($+0.010$). At the bottom sit electoral and constitutional policy ($+0.008$), governance and administration ($+0.001$) and education and science (-0.003; Table 20). Industry and economic policy moves the other way (-0.026; Table 20), but on four speeches in 1894–99, and governance on a few hundred, so neither cell can carry a reading.

That contrast is the substantive point. The reform that enfranchised working-class men moved the governing parties apart on the rules of representation—the very subject the reform concerned, and the one on which a purely institutional account of elite conflict would predict the sharpest division—by under a third of what it did on labour. It drove them apart on *what to give the newly enfranchised*. The parties did not chiefly come to disagree about how democracy should be organized; they came to disagree about the social question that the new electorate brought into the Chamber: labour emerges as the single subject on which the two governing parties became most distinguishable.

Two features of the table temper the reading. First, the pre-1894 column is nearly flat—every topic but agriculture, and the thin industry cell, sits within half a point of the no-information benchmark—so the post-reform ordering is a ranking of *changes* rather than a re-expression of pre-existing topic differences. Second, after 1900 every topic that rose gives back about half of its rise or more—labour about half, budget and infrastructure four fifths—and labour remains the most partisan topic (0.517; Table 20); we do not read that movement against the Panel B contrast, which is not identified (§6.6). We report this descriptively—the cells are Chamber-only and the classifier assigns a single dominant topic per speech—so the exercise characterizes where the measured divergence sits rather than identifying a topic-specific reform effect.

Table 20: Catholic–Liberal partisanship by topic and reform window, Chamber, on a single yardstick fixed across cells. Topics ordered by the change at plural suffrage.

Topic	pre-1894	1894–99	post-1900	Δ at PS
Labor_and_Worker_Social_Policy	0.5027	0.5296	0.5165	+0.0269
Agriculture_and_Rural_Policy	0.5109	0.5236	0.5140	+0.0126
Budget_and_Fiscal_Policy	0.5021	0.5134	0.5037	+0.0113
Infrastructure__Transportation	0.5029	0.5136	0.5047	+0.0107
Justice_and_Legal_System	0.5036	0.5137	0.5062	+0.0101
Electoral_and_Constitutional_P	0.5019	0.5097	0.5031	+0.0078
Governance_and_Administration	0.5034	0.5044	0.4997	+0.0010
Education_and_Science	0.5006	0.4977	0.5013	-0.0029
Industry_and_Economic_Policy	0.5078	0.4818	0.4957	-0.0260

Token-weighted means of the per-speech leave-out outcome within each topic × window cell, scored against a single $\hat{\rho}$ estimated once on the Chamber's Catholic–Liberal speeches that carry a topic label, so that cells of different size remain comparable. Topics from the 22-domain speech classifier, distilled onto the dataset's interventions (Appendix B.10); speeches with no substantive policy topic are excluded. Produced by `gst_topwords_and_topics.py`.

A.7 Mining: a gradient, not a second reform effect

The mining triple-difference points the same way on both measures, and it is the opposite of the way an account driven by the new electorate would point. On the continuous measure it is negative for plural suffrage (Chamber × Post-1894 × Mining $= -0.006$ with fixed effects and -0.003 with politician fixed effects, both $p < 0.01$; Table 3), and it keeps its sign under the bandwidths and the donut (-0.003 to -0.006, all $p < 0.01$), with an explicit pre-trend interaction of zero (Table 32); the two placebo cutoffs, at 1886 and 1888, lean the same way (-0.003 at both, one at the ten per cent level), so part of the gradient may predate the reform. On the TopMining indicator it is -0.004 ($p < 0.05$; Table 31). The event study (Table 33), whose omitted legislature is 1900, puts the gap at -0.007 in 1894, -0.014 ($p < 0.01$) in 1896 and -0.007 ($p < 0.01$) in 1898, but its leads, only the 1892 one significant and at the ten per cent level, drift down toward the reform: measured against 1892 they fall from $+0.012$ in 1884 to zero, and the three post-reform terms are -0.003, -0.010 and -0.003. A placebo reform at 1896 gives -0.005 (n.s.; Table 35), as large as the estimate. The widening of the bicameral contrast was smaller where the new electorate was densest, but a gap that was already narrowing before the reform accounts for part of that. The proportional-representation mining triple is small and unstable across specifications—negative but insignificant on TopMining, positive at the five per cent level with fixed effects and null with politician fixed effects on the continuous measure (Tables 31 and 3)—and its event study shows no sustained shift at 1900 (Table 34).

We nonetheless keep the mining triples and their event studies in the appendix, and it is worth being precise about why, because two reasons are easily conflated. The first is the support: the TopMining indicator rests on four arrondissements whose Catholic benches were empty between the reforms (Appendix A.1), so that in the plural-suffrage window the gradient compares the Liberal deputies of the coal districts with the Catholic and Liberal deputies elsewhere; its leads drift the same way; and the continuous measure, which spreads the variation over all arrondissements, leans the same way at the placebo dates. The second is the object: the triple says where the widening was weaker, not why, and the three accounts set out earlier in this appendix read the same sign differently. Neither reason touches the headline plural-suffrage result, which is identified off the *bicameral* contrast and contains no mining interaction, and whose own falsification tests—placebo dates, bandwidths, donut, and the wild cluster bootstrap—pass. Both are also why §6.7 reports the coal districts as a change in presence rather than as a within-deputy trajectory: a movement estimated off a

window that holds no Catholic mining deputy (Appendix A.1) cannot be read as a deputy's response, and the coal-district evidence in this paper is accordingly a count of who sits, with the gradient beside it as a fact about where the reform bit least.

A.8 Differential OCR quality across the two houses

Both corpora are OCR-derived (§3), so measured distinctiveness could in principle reflect text quality rather than partisan language. The logic is mechanical: OCR noise inserts non-discriminating garbage tokens, which push the leave-out posterior toward the no-information floor of $\frac{1}{2}$, so a noisier house looks *less* distinctive by construction. Two facts bound the concern before any test. First, a *time-invariant* quality difference is a level: it loads on the chamber and politician fixed effects and differences out of β, so only Senate OCR quality that *changes over time around the reform dates* could manufacture a spurious differential trend. Second, because noise can only pull a posterior *toward* $\frac{1}{2}$, it can erode a genuine distinctiveness gap but never fabricate one; of the two directions the concern might take, only the erosion of real signal, not the invention of spurious signal, is mechanically available. We probe both the temporal-stability requirement and the magnitude of any contamination directly.

(i) Temporal noise diagnostic. For each chamber×legislature cell we compute three model-free proxies that rise with OCR garbling—the hapax (once-occurring) token share, the share of tokens shorter than three characters, and the type–token ratio—and report all three for *both* houses across all three reform windows (Table 21, Panel A). We do not read the cross-chamber *levels* as OCR quality: these proxies also rise with corpus size and lexical variety, and the Chamber—larger and more heterogeneous—scores higher on all three, which reflects its breadth rather than the fidelity of its transcription. A time-invariant difference of that kind is a level: it loads on the chamber and politician fixed effects and differences out of β. What could manufacture the break is quality that *changes* around a reform, in either house, and that is what the by-window cells are for.

Nothing in the proxies at 1894 could manufacture the break. In the Chamber all three rise a little across the reform—the hapax share from 0.4297 to 0.4437, the short-token share from 0.0066 to 0.0072 and the type–token ratio from 0.0120 to 0.0122 (Table 21)—movements in the direction of more garbling, which would pull the Chamber's measured distinctiveness toward $\frac{1}{2}$, not away from it. In the Senate the hapax share is flat (0.3997 to 0.4001; Table 21) and the short-token share moves by 0.0005; its type–token ratio is lower in 1894–99 (0.0268 against 0.0323), but that reflects a decline running through the whole pre-reform period, from 0.043–0.044 in 1880–82 to 0.027 in 1892 and 0.024 in 1894 (Table 76), rather than a step at the reform. The largest movements in either house arrive after 1900—the Chamber's hapax share rises to 0.4781, the Senate's to 0.4389 (Table 21)—which is neither the date that identifies the clean test nor a degradation signature: it moves toward *more* lexical diversity, not the repetition a failing scan produces, and it coincides with the growth of the Dutch-language component discussed above. Because Panel A is identified off the 1894 break, and across it the Chamber's proxies move in the direction that would erode its distinctiveness while the Senate's barely move or continue a pre-reform decline, a time-varying OCR story cannot produce the result.

(ii) Placebo degradation of the Chamber text. As a direct magnitude check we run the feared artefact in reverse: we inject OCR-like character-level noise (visually-confusable substitutions, deletions, and duplications) into the *Chamber* corpus at increasing rates and re-score both houses, asking how much degradation collapses the Chamber's distinctiveness onto the Senate's (Table 21, Panel B). One design choice governs what the test can show. The vocabulary must be held *fixed* at the one estimated on the clean corpus. Rebuilding it on each degraded corpus lets noise change both the scores and which phrases are eligible to be scored, and the second channel dominates: the rebuilt vocabulary concentrates on robust, high-frequency and disproportionately partisan phrases, so measured partisanship can *rise* with noise—an artefact of reselection, not a property of degradation.

With the vocabulary held fixed the test says something simple. Corrupting 2% of the Chamber's characters leaves the gap at 0.0035, against 0.0036 on the clean text; 5% gives 0.0036; 10% gives 0.0038; and 15%—far beyond any plausible error for this material—gives 0.0047 (Table 21, Panel B). Chamber Π moves within four ten-thousandths of its clean value across the whole grid. The gap is not eroded by noise at any rate we can plausibly entertain, and the residual movement is in the fourth decimal of an index that lives in the third: sampling variation, not a trend. One detail is worth stating because a reader recomputing this will find it: the Senate's Π drifts down from 0.5016 to 0.5009 (Table 21, Panel B) although its text is never touched, because the posteriors are estimated on the joint pool and corrupting one house perturbs $\hat{\rho}$ for both. The effect is small and it works *against* the gap closing, so it does not soften the conclusion. The differential-OCR concern is not driving the result: the static component differences out, the time-varying component runs at 1894, in the Chamber, in the direction that would close the gap, and deliberately degrading the exposed house does not move its edge.

Table 21: OCR robustness. Panel A: noise proxies for both chambers by reform window—what would threaten the design is a proxy stepping at 1894 in the direction that sharpens the Chamber's text; the Senate's type–token ratio declines throughout the pre-reform period, without a break at 1894. Panel B: placebo degradation of the Chamber text, with the vocabulary held fixed at the clean corpus so that injected noise can only move tokens within an unchanged phrase set; Π is the token-weighted mean leave-out partisanship.

Panel A. Temporal noise proxies

House	Proxy	pre-1894	1894–99	post-1900	Range
Chamber	hapax share	0.4297	0.4437	0.4781	0.0483
Chamber	short share	0.0066	0.0072	0.0083	0.0017
Chamber	type token	0.0120	0.0122	0.0173	0.0053
Senate	hapax share	0.3997	0.4001	0.4389	0.0391
Senate	short share	0.0047	0.0052	0.0056	0.0009
Senate	type token	0.0323	0.0268	0.0337	0.0069

Panel B. Placebo degradation of the Chamber

OCR noise rate	Chamber Π	Senate Π	Gap
0.00	0.5053	0.5016	0.0036
0.02	0.5051	0.5016	0.0035
0.05	0.5050	0.5015	0.0036
0.10	0.5050	0.5012	0.0038
0.15	0.5056	0.5009	0.0047

Panel A: higher values indicate more lexical garbling; proxies are token-weighted within window. The Chamber's short-token share and type–token ratio rise across 1894, in the direction that would pull its measured distinctiveness toward $\frac{1}{2}$; the Senate's type–token ratio declines through the pre-reform period; the post-1900 rise in both is toward greater lexical diversity and coincides with the growth of Dutch-language speech. Panel B: character-level OCR-like noise injected into the Chamber text at the stated rate, vocabulary fixed at the clean corpus. The gap stays between 0.0035 and 0.0047 at every rate, against 0.0036 on the clean text. The Senate's Π drifts down slightly although its text is untouched, because $\hat{\rho}$ is estimated on the joint pool; that drift widens rather than narrows the gap. Produced by `ocr_robustness.py`.

A.9 Pairwise polarization, 1894–1912

Scored on the paper's vocabulary with dyad posteriors estimated on post-1894 speech (Table 22), Chamber leave-out partisanship sits at 0.5157–0.5211 for Catholic–Socialist across the two post-1894 windows and at 0.5037 and 0.5136 for Liberal–Socialist, against 0.5015 and 0.5051 for the Catholic–Liberal reference. On a vocabulary built on the three-party pool itself (Figure 6), pooled over 1894–1912, both socialist dyads are well above their permutation null bands (0.519 and 0.516, against upper bounds of 0.504 and 0.503; Table 22) and the Catholic–Liberal dyad narrowly so (0.508 against 0.503). Legislature by legislature, where thinner cells widen the band, the Catholic–Socialist dyad lies above its band in every legislature, the Liberal–Socialist one from 1900 on, and the Catholic–Liberal one, narrowly, in 1898, 1902 and 1906 (Figure 6 in Appendix B.6). The socialist entrant does separate from the governing parties in the aggregate vocabulary; what the design identifies is the reform-timed movement between the governing parties, which neither dyadic exercise is designed to register. Within-party dispersion is lower in the Chamber than in the Senate, for both parties (Table 39: Catholic 0.798 vs. 0.875; Liberal 0.760 vs. 0.832); because the distance of a member's phrase distribution from his party's centroid falls mechanically with the volume of his speech, and Chamber members speak far more, we do not read the level either way.

Table 22: Pairwise leave-out distinctiveness of the three party dyads, by chamber and reform window, 1894–1912.

Dyad	Chamber	1894–99	post-1900
Cath.–Lib.	DK	0.5015	0.5051
Cath.–Lib.	SK	0.4934	0.4988
Cath.–Soc.	DK	0.5211	0.5157
Cath.–Soc.	SK	0.4555	0.5058
Lib.–Soc.	DK	0.5037	0.5136
Lib.–Soc.	SK	0.4553	0.5127

Token-weighted window means of the pairwise leave-out outcome, scored on the paper's vocabulary with dyad posteriors estimated on post-1894 speech. Produced by `dyad_polarization_1900.py`. The by-legislature trajectories with permutation bands, on a vocabulary built on the three-party pool (`dyad_polarization.py`), are in Figure 6; pooled over 1894–1912 on that vocabulary the three dyads score 0.5186 (Catholic–Socialist), 0.5156 (Liberal–Socialist) and 0.5077 (Catholic–Liberal), against upper bounds of the permutation band of 0.5039, 0.5030 and 0.5027.

B Additional tables

B.1 Event-study coefficients and the partisan lexicon

This appendix holds two tables that the main text reads but does not display, because both are long and neither adds to the argument beyond what the accompanying discussion already states.

Table 7 reports the event study of §6.3 coefficient by coefficient, in the clean pre-1900 window and over the full 1880–1912 span. The pattern discussed in the main text is visible here in detail: the six pre-reform leads are individually small and jointly indistinguishable from zero, the 1894, 1896, and 1898 interactions rise, and the post-1900 coefficients in column (2) return to the 1892 base or slightly below it. On French speech alone the pre-reform leads are negative and the post-1900 coefficients slightly lower (§A.2). Figure 2 plots column (1).

Table 23 lists the most diagnostic phrases by party and reform window, ranked by the contribution $w_{j,p}\zeta_{j,p}$ of §4.6. The substantive reading—the arrival of the labour lexicon on the Catholic side across 1894, and the Liberal return to the school after 1900—is given in §4.6 and not repeated here.

Table 23: Most diagnostic phrases by party and reform window, ranked by contribution $w_{j,p}\zeta_{j,p}$. Phrases are restricted to those used by at least eight distinct members with $\zeta_{j,p} \geq 0.70$ and a frequency of at least five per 100,000 tokens. The reactions of the floor printed in the record (*Très bien! à droite*, *Rires à gauche*, *Protestations à gauche*) are taken out of the text before the phrases are formed (§4.1) and do not appear in the lists.

Window	Phrase	ζ_j	per 100k	Speakers
pre-1894 — most Catholic phrases				
	fond_commun	0.79	30	62
	domicil_secour	0.74	28	50
	dépôt_mendiqu	0.90	22	48
	pay_flamand	0.74	19	83
	assist_publiqu	0.77	15	52
	droit_auteur	0.77	15	38
	disposit_gouvern	0.71	15	105
	droit_recour	0.83	12	45
pre-1894 — most Liberal phrases				
	réform_électoral	0.76	14	69
	eau_vi	0.70	15	51
	autonom_communal	0.71	13	76
	caiss_prévoi	0.74	12	46
	supprim_écol	0.71	10	49
	part_clérical	0.91	7	46
	compagn_nord	0.82	8	47
	écol_mixt	0.77	8	28
1894-1899 — most Catholic phrases				
	union_professionnel	0.91	121	58
	caiss_épargn	0.90	46	56
	contrat_travail	0.76	49	47
	cahi_charg	0.89	37	51
	fer_vicinal	0.89	36	63
	conseil_industr	0.79	38	42
	grand_central	0.85	34	40
	travail_est	0.75	37	81
1894-1899 — most Liberal phrases				
	députpermanent	0.77	80	74
	conseil_communal	0.71	77	88
	suffrag_universel	0.81	66	61
	une_major	0.71	57	87
	affair_étranger	0.72	43	63
	per_famill	0.76	40	66
	etat_congo	0.94	29	14
	gouvern_belg	0.79	30	58
post-1900 — most Catholic phrases				
	caiss_prévoi	0.85	36	56
	chef_entrepris	0.79	18	56
	patron_ouvri	0.70	19	85
	recherch_patern	0.84	13	26
	central_examin	0.81	13	95
	roi_honneur	0.99	10	24
	conseil_min	0.75	13	44
	gouverneur_général	0.73	13	29
post-1900 — most Liberal phrases				
	écol_communal	0.70	58	96
	chef_cabinet	0.72	36	117
	cour_compt	0.78	32	69
	enseign_public	0.79	27	69
	écol_moyen	0.73	25	86
	écol_publiqu	0.74	23	65
	abstenu_motif	0.72	19	147
	loi_scolair	0.73	18	79

$\zeta_{j,p}$ is the probability of identifying the speaker's party from phrase j alone in window p (10); "per 100k" is the phrase's predicted frequency per 100,000 phrase tokens; "Speakers" is the number of distinct members using it. Produced by `gst_topwords_and_topics.py`.

B.2 What the vocabulary filter retains

The hard filter of §4.1 keeps 36,297 phrases, and in the Catholic and Liberal speech of the window they account for 26% of all bigram tokens (Table 71) (adjacent pairs of lemmas, function words included). The rest is removed by construction: procedural boilerplate, surnames, party self-labels, lemmatization fragments, pairs of function words, and phrases too rare to estimate are all removed precisely because they either predict party mechanically, carry no content, or cannot be estimated without finite-sample bias. It is the same logic that leads Gentzkow et al. (2019) to discard the great majority of their own candidate phrases, and neither they nor Simola et al. (2025) report a coverage share; we report ours because the question is a natural one to ask of a nineteenth-century corpus.

Two facts bound the concern. First, coverage is stable across the reform: the share of bigram tokens falling inside the vocabulary is 0.269 before 1894 against 0.265 in 1894–99 in the Chamber, and 0.236 against 0.233 in the Senate (Table 71), so the pre- and post-reform periods are measured with equally representative slices of their own speech (§5.4).A filter that captured the two periods unequally would bias the contrast; this one does not. Second, the vocabulary ladder varies the thresholds by roughly $\pm 40\%$ in each direction and moves none of the paper's conclusions (Appendix B.9), a wider stress test than the 10% tightening Gentzkow et al. (2019) report. What the filter costs is statistical power rather than validity: a thinner vocabulary is a noisier index, and noise pushes estimated effects toward zero rather than away from it.

B.3 The correction registry

This appendix lists the pre-estimation corrections summarized in §A.2. They form a verified registry applied inside the corpus loader, upstream of the text scoring, so that mislabeled speech cannot shape the partisan vocabulary or the leave-out yardstick $\hat{\rho}$; each entry is checked against the biographical record (the Maitron dictionary and the constituency rolls). The registry has six components. In the Chamber, taken from the Belgian Chamber Debates dataset, much of what it repairs arrives repaired: the dataset links each speech to a member or minister in office on the day and takes his party from the member and government records. The registry runs over both houses, and we report what it does in each.

Party fixes. The decisive case for the design is Léon Furnémont, POB deputy for Charleroi from 1894: labelled "Liberal", as a purely modal consolidation of inconsistent raw labels can make him, he would place socialist speech in the Liberal $\times$ Chamber $\times$ mining cell exactly after 1894—the cell, and the direction, that most flatter the headline estimate. The dataset labels him Socialist on every intervention that carries a label (6,416 of 6,731), and the fix confirms it. The same layer assigns Pieter Daens to his own Daensist label, which the dataset already gives on 2,505 of his interventions, and sets the party of the ministers it lists to the one the biographical record gives—Jules de Burlet is Catholic throughout. The consolidation on each member's career mode fills 17,560 blank labels—nearly half of them in the 1912 legislature, for which the dataset gives no party to 8,114 interventions of members it labels in the others—and overrides 274 stray ones.

Speaker linking. In the Chamber the link is the dataset's (§3): of the 594,396 interventions of the 1880–1912 legislatures, 551,702 are matched exactly to a member who sat on that day, 26,671 on a near match and 16,023 to a minister. The digitization of the Senate attaches each intervention to a member page by fuzzy matching on the speaker field, and the match is not reliable enough to carry fixed effects and entrant status. We therefore re-link by rule every one of its 62,690 interventions of the 1880–1914 legislatures that remain once the 8,525 rows delivered by *le Président*, the identity repairs and the party-entry rule described below have been applied: the speaker must be a member who sat in that house on that day, or a minister in office, whose surname matches the speaker field, with ministerial and other roles resolved by date. The rule confirms the digitization's link for 48,360 interventions (and for 904 more on a weaker match, and 253

delivered outside the member's recorded mandate); it moves 12,364 to a different member (385 of them on a weaker match), and with the new links the party label changes on 1,350 Senate interventions; and it leaves 661 unresolved and 148 ambiguous between two members, all of which are excluded.

Chair speech. Interventions from the chair are procedural rather than member speech and are dropped. In the Chamber the dataset attributes the turns of the chair to the presiding officer, and we leave them out before anything else (97,959 blocks from June 1880 to August 1914); in the Senate the loader drops the 8,525 rows delivered by *le Président* but fuzzy-matched to a member page.

Identity repairs. Two pages covering two real politicians each (father–son merges) are split so that fixed effects and entrant status are computed per person, and anachronistic fuzzy-match identities—members who cannot have sat in this parliament—are excluded outright (477 interventions).

Whitelisted switchers. Genuine party switchers are exempted from modal consolidation so that the corrections cannot overwrite real movement: Auguste Houzeau de Lehaie, a Liberal who joined the POB as socialist senator for Mons (for Charleroi–Thuin from 1900), keeps his labels as recorded, subject to the party-entry rule below.

Party-entry rule. A party label cannot precede the party's entry into parliament: 27 speeches labelled Socialist before the 1894 legislature, across five politicians, are dropped—22 of them by Léon Defuisseaux.

A residual set of identity flags—members with speeches recorded outside their term of office, or under a page shared with a namesake—is documented in the replication code but left uncorrected, because the affected members are Catholic or Liberal in non-mining districts and so do not enter any cell that identifies a reported coefficient.

B.4 Who moves: adaptation versus recomposition

This block decomposes each reform response into its behavioural source—whether a measured change reflects *the same* members speaking differently or a *different* set of members being elected. The distinction is central to the paper's reading and is the same decomposition the companion study uses (Graziani and Vanschoonbeek, 2026). Table 24 separates the overall effect into an incumbent and an entrant component for each reform: for plural suffrage the Chamber $\times$ Post interaction loads on incumbents ($+0.0103$, $p < 0.01$) with an entrant differential indistinguishable from zero (-0.0043), the quantitative signature of adaptation rather than turnover. Had the rise instead come from a new cohort of more distinctive entrants, the entrant term would carry it; pooled, it does not. Table 25 splits the same decomposition by party, and the two parties turn out to move through different channels. Among sitting members the 1894 rise is carried by the Catholic incumbents ($+0.0067$, $p < 0.01$), while the Liberal incumbents do not move (-0.0033, n.s.). The Liberal side is not inert, but what movement it shows runs through *turnover*: Liberal entrants after 1894 are more distinctive than the Liberals they join ($+0.0136$, $p < 0.10$; Table 25), while the Catholic entrants add nothing distinguishable from the incumbents ($+0.0063$, n.s.). Each party contributes to the widening through a different channel—the governing party by adapting in place, the Liberal rump through the few men it returned after 1894. After 1900 both incumbent terms are negative, significantly so for the Catholics (-0.0095, $p < 0.05$; Table 25): this is the Panel B contrast, which §6.6 does not read causally. Table 26 re-runs the post-1900 mining triple separately for incumbents and entrants: the incumbents carry a small positive mining gradient ($+0.004$, $p < 0.01$), and for the entrants, observed only after 1900, the triple is not identified. Of the two specifications that follow, the first discards the Senate contrast entirely and identifies only off geographic variation inside the lower house. It runs the plural-suffrage mining interaction on the Chamber alone: mining districts move less than the rest after 1894 (-0.0068, $p < 0.01$; Table 27), and the bicameral triple is negative too (Appendix B.6), so the gradient is not only geography common to the two houses: the Chamber's coal districts moved less than its other districts by more than the Senate's did—the gradient §A.7 reads, with its cautions. The second, estimated on both houses, asks the same of proportional representation on the incumbent margin: the negative Panel B contrast is carried by incumbents (-0.0142,

$p < 0.01$; Table 28), with entrants less negative by $+0.0067$ ($p < 0.01$; Table 24). Since the contrast is the Chamber's change net of the Senate's, and the Senate is the house that moved in 1900 (§6.6), this is a statement about the control as much as about the Chamber, and we do not read it as an effect of proportional representation on sitting deputies. Neither specification manufactures the headline result, and neither rescues the objects the main text quarantines: the bicameral contrast is doing the identifying work.

Table 24: Adaptation versus recomposition: incumbent and entrant interactions

	(1) PS 1894	(2) PR 1900
Entrant × Chamber × Post-1894	-0.00426	
	(0.00730)	
Entrant × Post-1894	0	
	(.)	
Chamber × Post-1894 (incumbents)	0.0103***	
	(0.00294)	
Entrant × Chamber × Post-1900		0.00665***
		(0.00178)
Entrant × Post-1900		0
		(.)
Chamber × Post-1900 (incumbents)		-0.0142***
		(0.00178)
Observations	206146	245296

Standard errors clustered at arrondissement in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 25: Adaptation versus recomposition, by party

	(1) Catholic 1894	(2) Liberal 1894	(3) Catholic 1900	(4) Liberal 1900
Entrant × Chamber × Post-1894	0.00628	0.0136*		
	(0.00502)	(0.00726)		
Chamber × Post-1894 (incumbents)	0.00667***	-0.00334		
	(0.00189)	(0.00295)		
Entrant × Chamber × Post-1900			0.00263	0.00225
			(0.00463)	(0.00234)
Chamber × Post-1900 (incumbents)			-0.00950**	-0.00297
			(0.00382)	(0.00418)
Observations	128137	78008	166140	79155

Standard errors clustered at arrondissement in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 26: Recomposition: post-1900 mining DDD by incumbency status

	(1) Incumbents	(2) Entrants
Chamber × Post-1900 × Mining	0.004***	0.000
	(0.001)	
Chamber × Post-1900	-0.010***	0.000
	(0.002)	
Post-1900	0.005***	
	(0.002)	
Observations	167317	77979
R^2	0.064	0.057
Party FE	Yes	Yes
Arrondissement FE	Yes	Yes
Politician FE	No	No

Standard errors clustered at arrondissement in parentheses; speeches weighted by in-vocabulary token count. Produced by `polarization_did.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 27: Plural suffrage within the Chamber: the mining interaction without the Senate contrast

	(1)
Post-1894 × Mining (within Chamber)	-0.00682***
	(0.00113)
Post1894	0
	(.)
Observations	194946

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Chamber speeches, legislatures 1880–1898; politician, arrondissement, party and legislature fixed effects; speeches weighted by in-vocabulary token count; standard errors clustered at arrondissement in parentheses. Produced by `polarization_did_extensions.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 28: Proportional representation on the incumbent margin

	(1)
incumbent1900 × Chamber_dk × Post1900	-0.0142***
	(0.00178)
incumbent1900 × Post1900	0
	(.)
Chamber_dk × Post1900	0
	(.)
Observations	245296

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Both houses, legislatures 1894–1912; chamber, arrondissement, party and legislature fixed effects; speeches weighted by in-vocabulary token count; standard errors clustered at arrondissement in parentheses. The regression also includes Incumbent and Incumbent × Chamber; Chamber × Post-1900 and Incumbent × Post-1900 are collinear with the fixed effects, because every member who sat before 1900 is an incumbent. Produced by `polarization_did_extensions.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Is the Senate's drift real or compositional? The bicameral design treats the Senate as a stable control, but the upper house too renews its membership across the window, so its own trajectory in s_{distinct} could in principle reflect senators entering and leaving rather than sitting senators changing how they speak. Because the same decomposition that identifies adaptation in the Chamber applies to the Senate, we can read the two apart directly. Table 29 reports, for each chamber and reform window, the raw window mean of s_{distinct} and its within-politician counterpart; the gap between them (comp. = raw − within) is the between-member, compositional component, while co-movement of the two is within-person change. The Chamber's plural-suffrage rise is visible in both columns—within politician (+0.0036) as well as between members (+0.0044)—and after 1900 both fall back, the raw mean further than the within-politician one. The Senate's movement is small and is not a decline across the window: its raw mean dips in 1894–99 and recovers after 1900, and the dip is a little more than half compositional—the within-politician mean falls from 0.5051 to 0.5036 while the compositional component falls from −0.0035 to −0.0056—and both halves recover after 1900. With politician fixed effects the compositional part is absorbed and only the within-senator movement enters the contrast.

Table 29: Senate drift: adaptation or composition? Raw versus within-politician mean s_{distinct} by chamber and reform window; comp. = raw − within is the between-member (turnover) component.

	Chamber			Senate		
Window	raw	within	comp.	raw	within	comp.
pre-1894	0.5030	0.5042	−0.0012	0.5016	0.5051	−0.0035
1894–1899	0.5110	0.5078	0.0032	0.4980	0.5036	−0.0056
post-1900	0.5051	0.5043	0.0007	0.5034	0.5052	−0.0017

Window means of the per-speech leave-out outcome on the single globally fixed yardstick $\hat{\rho}$, $w^{(i,s)}$-weighted, on the estimation sample, each member counted in his house of membership. Within-politician columns subtract each member's weighted mean and add back the grand mean. Produced by `senate_decomposition.do`.

B.5 Socialist representation in the Chamber and Senate

This appendix documents the descriptive evolution of socialist representation in the Belgian Chamber of Representatives and Senate over the analysis window. Its purpose is twofold. First, it records when the Belgian Workers' Party (BWP) entered the national legislature and how quickly its parliamentary presence expanded after the extension of the franchise. Second, it makes concrete the institutional distinction between the two houses that motivates the difference-in-exposure design. The Chamber was directly exposed to the post-1893 expansion of the electorate; the Senate was *less* exposed, not sealed off. Before the constitutional reform of 1893 senators were directly elected but subject to restrictive eligibility conditions; after it, the Senate combined directly elected senators with provincial senators chosen by the provincial councils. This matters because socialist entry into the Senate came first through the provincial route, while *directly* elected socialist representation there was limited and delayed relative to the Chamber—and then grew, so that the control chamber becomes progressively more exposed to the same entry that treats the Chamber, most visibly after 1908 (Table 30).

Table 30: Socialist representation by election year, Chamber and Senate, 1880–1912. For the Senate, "directly elected seats" excludes provincial senators and senators by right; the last column—directly elected BWP senators—is the exposure of the control chamber to socialist entry that identifies the contamination discussed in §5.2.

Year	Chamber		Senate			
	BWP seats	Total seats	BWP seats	Total seats	Directly elected seats	Directly elected BWP seats
1880	0	132	0	66	66	0
1882	0	138	0	69	69	0
1884	0	138	0	69	69	0
1886	0	138	0	69	69	0
1888	0	138	0	69	69	0
1890	0	138	0	69	69	0
1892	0	152	0	76	76	0
1894	28	152	2	102	76	0
1896	29	152	2	102	76	0
1898	28	152	1	102	76	0
1900	32	152	6	102	76	1[†]
1902	35	166	7	110	83	1[†]
1904	29	166	8	110	83	4[‡]
1906	31	166	9	111	83	4
1908	35	166	11	111	83	7
1910	35	166	11	110	83	7
1912	39	186	15	120	93	10

Notes: Values report the full standing composition after the year indicated. For the Senate, "directly elected seats" excludes provincial senators and senators by right. In 1896, 1906, and 1910 there was no regular Senate renewal corresponding to a full new Senate composition; those rows report the composition in force after that year, from the composition lists and recorded within-legislature changes. Senate totals include provincial senators from 1894 onward and the senator by right in 1906 and 1908. The last column counts senators sitting among the directly elected and carried with a socialist label on the composition list for that legislature. [†] On that rule the 1900 and 1902 entries are a single member, Auguste Houzeau de Lehaie (arr. Charleroi–Thuin), a Liberal elected to the Senate in 1897 who moved to the Belgian Workers' Party and is carried with a socialist label from the 1900–1902 legislature onward. Because the composition lists give a party label for some members and not others, the two entries are documented minima rather than exhaustive counts. They are consistent with the Senate's own dating of the party's first directly elected senator to 1904 (Belgische Senaat, 2019), since Houzeau was not returned on a party list. [‡] The 1904 figure is itself qualified: Ferdinand Elbers was elected for Brussels on 24 May 1904, but his election was annulled when he declined to pay the eligibility census and the seat passed to a Catholic, and he entered the Senate only as a provincial senator, filling the seat of a Liberal who had died. Entries for 1904–1912 have not been re-checked against the composition lists under the rule stated here.

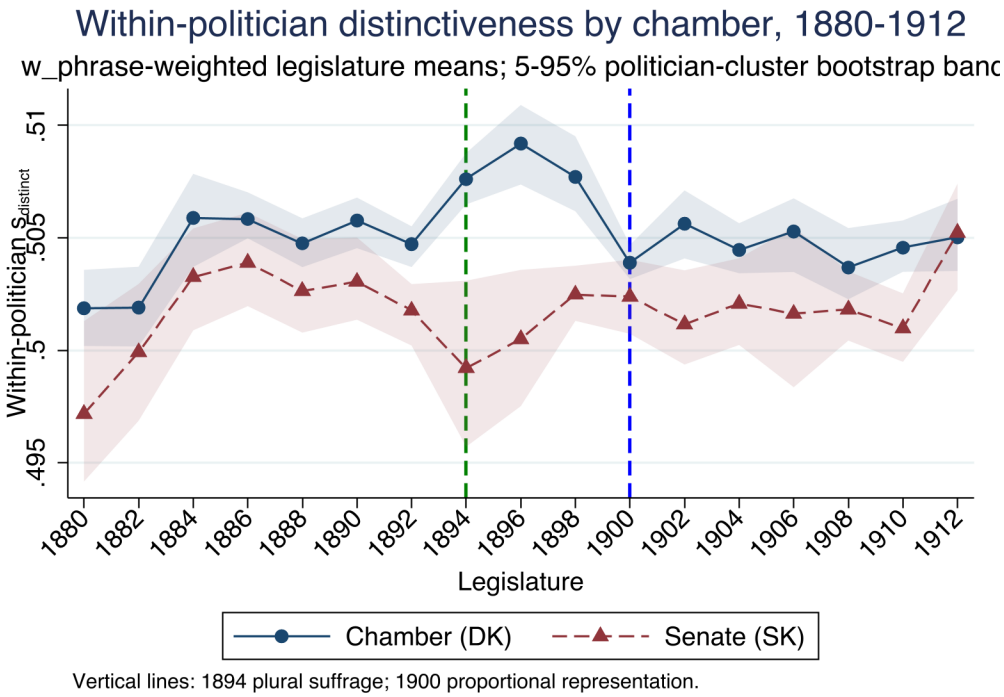


Figure 5: **Within-politician distinctiveness by chamber, full window 1880–1912.** (the main-text Figure 3 restricts to the clean plural-suffrage window). $w^{(i,s)}$-weighted legislature means of $s^{(i,s)}_{\text{distinct}}$ on the within-politician panel, with 5–95% politician-cluster bootstrap bands. The Chamber's plural-suffrage rise is the identifying movement; the Senate's post-1900 path is the compositional/adaptation question addressed in Table 29; the Senate placebo (Table 2) finds no Senate break at 1894 and, at 1900, a positive but insignificant one while the Chamber falls back, when every directly elected Senate seat was renewed under the new rule. Vertical lines mark plural suffrage (1894) and proportional representation (1900).

B.6 The mining triple-difference and its falsification

This block documents, table by table, what the mining triple on the Catholic–Liberal axis shows and why it is read as a gradient rather than as a second reform effect (§A.7). Table 31 reports the TopMining triples themselves: negative for plural suffrage (-0.004, $p < 0.05$), and small and insignificant for proportional representation (-0.001) and for the post-1900 entrants ($+0.001$); on the continuous measure the plural-suffrage triple is negative too (-0.006 with fixed effects and -0.003 with politician fixed effects, both $p < 0.01$; Table 3). Tables 33 and 34 trace the Chamber $\times$ TopMining gap legislature by legislature. Measured against its omitted legislature, 1900, the gap is $+0.008$ in 1884 and drifts down to -0.004 by 1892, the only lead that is significant, and at the ten per cent level; it is -0.007 in 1894, -0.014 ($p < 0.01$) in 1896 and -0.007 ($p < 0.01$) in 1898—-0.003, -0.010 and -0.003 below its 1892 level; and after 1900, measured against 1912, its coefficients range from -0.013 to $+0.006$ with no break at the reform. Table 32 stress-tests the continuous triple under ± 2 and ± 3 legislature bandwidths, a donut that drops the legislatures adjacent to the reform, the explicit Chamber $\times$ trend $\times$ mining interaction and two placebo cutoffs: it is negative under the bandwidths and the donut (-0.003 to -0.006, all $p < 0.01$), the trend interaction is zero, and the placebo cutoffs lean the same way (-0.003 at both, one at $p < 0.10$). Tables 35 and 36 place a *placebo* reform at 1896: the plural-suffrage TopMining triple is -0.005 there (n.s., but as large as the estimate), and the proportional-representation one, on the continuous measure with politician fixed effects,

is flat (−0.001, n.s.). Table 37 repeats the bandwidth and donut checks for the entrant-restricted post-1900 mining triple, insignificant in all three windows, and Table 38 reports the within-party mining triple, near zero in both periods (−0.001 and −0.000)—so the mining gap is between parties, not a within-party reform response. Finally, Table 39 reports within-party dispersion by chamber—the mean leave-one-out distance of each member's phrase distribution from his own party's centroid. It is lower in the Chamber than in the Senate for both parties (Catholic 0.798 vs. 0.875; Liberal 0.760 vs. 0.832). The comparison is a level, and it is confounded by speech volume: a member with more text has a phrase distribution closer to his party's centroid, and Chamber members speak far more than senators. We therefore read it neither as the heterogeneity an exposure-driven account predicts nor as discipline; the reform-timed test of the second moment is the one of §7.

Table 31: Mining triple-difference (TopMining): plural suffrage and proportional representation

	(1) PS (Panel A)	(2) PR (Panel B)	(3) PR entrants
Chamber × Post-1894 × TopMining	-0.00429**		
	(0.00175)		
Chamber × Post-1894	0.00426***		
	(0.00149)		
Chamber × Post-1900 × TopMining		-0.00149	0.00107
		(0.00268)	(0.00243)
Chamber × Post-1900		-0.00446*	0
		(0.00260)	(.)
Observations	206116	245268	77979

Standard errors clustered at arrondissement in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 32: Robustness: mining DDD — pre-trend, bandwidths, donut and placebo cutoffs

	(1) Pre-trend	(2) BW2	(3) BW3	(4) Donut	(5) Placebo 1	(6) Placebo 2
Panel A: plural suffrage, pre-1894 vs. 1894–1899						
Chamber × trend × Mining	-0.000					
	(0.000)					
Chamber × Post/Cutoff × Mining		-0.004***	-0.003***	-0.006***	-0.003	-0.003*
		(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
Chamber × Post/Cutoff		0.004**	0.003*	0.005***	0.007*	0.007**
		(0.002)	(0.002)	(0.002)	(0.004)	(0.003)
Observations	146791	94830	134427	149608	206146	206146
R^2	0.064	0.074	0.072	0.076	0.064	0.063
Panel B: proportional representation, 1894–1899 vs. 1900+						
Chamber × trend × Mining	-0.000					
	(0.000)					
Chamber × Post/Cutoff × Mining		0.006***	0.005***	0.005**	0.003	-0.000
		(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
Chamber × Post/Cutoff		-0.006***	-0.008***	-0.008***	-0.007*	-0.006***
		(0.002)	(0.002)	(0.003)	(0.003)	(0.001)
Observations	206146	77695	129014	206496	245296	245296
R^2	0.073	0.087	0.077	0.053	0.047	0.051
Party FE	Yes	Yes	Yes	Yes	Yes	Yes
Arrondissement FE	Yes	Yes	Yes	Yes	Yes	Yes
Politician FE	No	No	No	No	No	No

Standard errors clustered at arrondissement in parentheses; speeches weighted by in-vocabulary token count. Placebo cutoffs are 1886/1888 (Panel A) and 1896/1904 (Panel B). Produced by `polarization_did.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 33: Event study: Chamber × mining interactions around plural suffrage

	(1) s_distinct
1884.Legislature#c.Ch_mine_top	0.00816
	(0.00525)
1886.Legislature#c.Ch_mine_top	0.0000768
	(0.00244)
1888.Legislature#c.Ch_mine_top	-0.00295
	(0.00390)
1890.Legislature#c.Ch_mine_top	-0.000998
	(0.00229)
1892.Legislature#c.Ch_mine_top	-0.00430*
	(0.00253)
1894.Legislature#c.Ch_mine_top	-0.00715
	(0.00455)
1896.Legislature#c.Ch_mine_top	-0.0139***
	(0.00364)
1898.Legislature#c.Ch_mine_top	-0.00721***
	(0.00252)
1900.Legislature#c.Ch_mine_top	0
	(.)
N	196030

The omitted legislature is 1900, so every coefficient, 1892 included, is measured against it. Standard errors clustered at arrondissement in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 34: Event study: Chamber × mining interactions around proportional representation

	(1) s_distinct
1888.Legislature#c.Ch_mine_top	-0.00207 (0.00773)
1890.Legislature#c.Ch_mine_top	-0.000323 (0.00585)
1892.Legislature#c.Ch_mine_top	-0.00369 (0.00668)
1894.Legislature#c.Ch_mine_top	-0.00547 (0.00727)
1896.Legislature#c.Ch_mine_top	-0.0122** (0.00585)
1898.Legislature#c.Ch_mine_top	-0.00564 (0.00408)
1900.Legislature#c.Ch_mine_top	-0.00115 (0.00611)
1902.Legislature#c.Ch_mine_top	0.00302 (0.00640)
1904.Legislature#c.Ch_mine_top	0.00131 (0.00425)
1906.Legislature#c.Ch_mine_top	0.00634 (0.00479)
1908.Legislature#c.Ch_mine_top	-0.00476 (0.00356)
1910.Legislature#c.Ch_mine_top	-0.0126*** (0.00394)
1912.Legislature#c.Ch_mine_top	0 (.)
N	320368

The omitted legislature is 1912. Standard errors clustered at arrondissement in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 35: Placebo reform at 1896: plural-suffrage mining triple

	(1)
Chamber × Placebo-1896 × TopMining	-0.00544
	(0.00352)
Chamber × Placebo-1896	-0.00103
	(0.00160)
N	59336

Standard errors clustered at arrondissement in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 36: Placebo reform at 1896: proportional-representation mining triple

	(1)
Chamber × Placebo-1896 × Mining	-0.00117
	(0.00167)
Ch_Postpl	-0.00253
	(0.00165)
Observations	59336

Standard errors clustered at arrondissement in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 37: Entrant mining DDD: bandwidth and donut robustness

	(1) BW ±2	(2) BW ±3	(3) Donut
Chamber × Post-1900 × Mining	-0.000422	0.000226	0.00110
	(0.00128)	(0.000942)	(0.00105)
Observations	21166	29830	73068

Standard errors clustered at arrondissement in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 38: Within-party dispersion: mining triple-difference

	(1) PS 1894	(2) PR 1900
Chamber × Post-1894 × Mining	-0.000961	
	(0.00212)	
Chamber × Post-1894	-0.00196	
	(0.00126)	
Chamber × Post-1900 × Mining		-0.000103
		(0.00189)
Chamber × Post-1900		-0.00194
		(0.00332)
Observations	206115	245265

Standard errors clustered at arrondissement in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 39: Within-party dispersion, mean over post-1894 legislatures

Party	Chamber	Dispersion
Catholic	Chamber	0.798
Catholic	Senate	0.875
Liberal	Chamber	0.760
Liberal	Senate	0.832

Note. Mean leave-one-out distance ($1 - \cos$) of each member's phrase distribution from the party centroid, averaged over the post-1894 legislatures up to 1912.

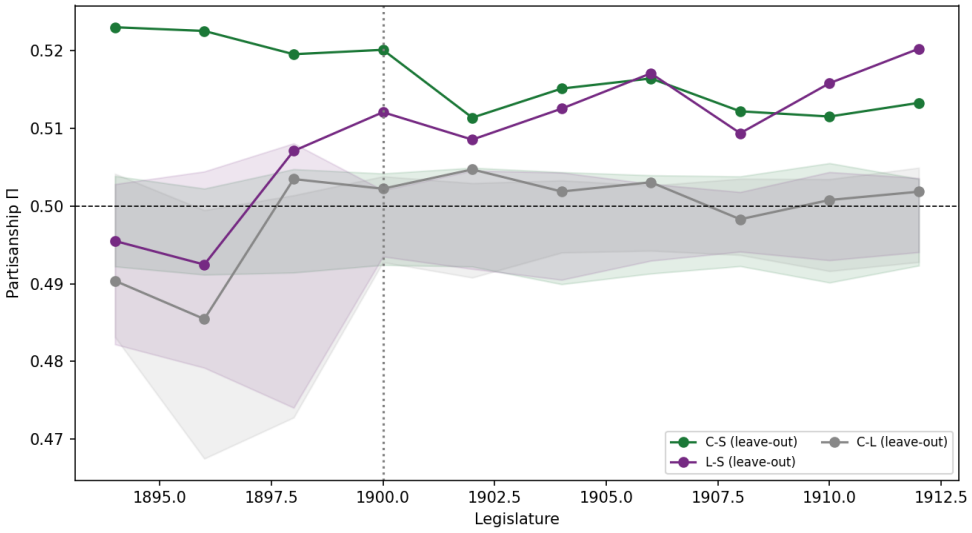


Figure 6: Pairwise elite polarization, 1894–1912. Leave-out Π for the Catholic–Socialist, Liberal–Socialist, and Catholic–Liberal dyads by legislature, Chamber, on a vocabulary built on the post-1894 three-party pool, against permutation null bands (shaded). The Catholic–Socialist dyad lies above its band in every legislature and the Liberal–Socialist one from 1900 on; the Catholic–Liberal dyad lies inside its band except, narrowly, in 1898, 1902 and 1906. The socialist entrant separates from both governing parties; the movement the design identifies is the one between them. The vertical line marks proportional representation (1900).

B.7 The territorial axis: a specificity check

The tables here support the specificity test of §7. They are produced by re-running the leave-out estimator with the speaker partition redefined from party (Catholic vs. Liberal) to geography (mining district vs. rest), holding the pooled yardstick, the vocabulary, and the bicameral difference-in-differences fixed. The outcome, s^{mine}, measures how distinctive a member's mining-versus-rest position makes their speech *within their own party*, so that party composition of the coal belt is differenced out and the object is territorial rather than partisan. This is not the Catholic–Liberal mining triple of Appendix B.6, which interacts mining intensity with the *partisan* outcome; it is the same estimator pointed at a different cleavage, and it serves as a placebo for the headline result. Table 40 reports the bicameral Chamber $\times$ Post interaction, pooled and within-party, for each reform: on plural suffrage the within-party coefficients are small, negative and insignificant (-0.007, and -0.003 with politician fixed effects), while the pooled ones are positive ($+0.006$, $p < 0.10$; $+0.008$, $p < 0.05$, with politician fixed effects)—the pooled partition picks up the party composition of the coal benches, which held no Catholic between the reforms (Appendix A.1); on proportional representation the pooled coefficients are insignificant (-0.001, and -0.009 with politician fixed effects), and so is the within-party one with fixed effects only ($+0.008$), while the within-party coefficient with politician fixed effects is negative (-0.012, $p < 0.01$). Table 41 traces the within-party Chamber–rest gap legislature by legislature (base 1892): the leads are small and insignificant ($+0.006$ and $+0.004$ in 1880 and 1882, near zero from 1884); the 1894 coefficient is $+0.002$ and those of 1896 and 1898 are close to zero; from 1900 the coefficients are negative, between -0.011 and -0.015 from 1902 and significant in all but 1906. The territorial axis therefore does not break at 1894, where the partisan axis does, and after 1900 the mining members of each party become less distinguishable from the rest of their party—the territorial counterpart of the intra-party cohesion of Appendix B.8. It does not move with the reform that sharply moves the partisan axis, which is precisely why the plural-suffrage result cannot be a generic increase in corpus

separability.

Table 40: Territorial axis (mining vs. rest): bicameral difference-in-differences. Outcome is mining-versus-rest distinctiveness s^{mine}; columns pool Catholic and Liberal speakers or restrict to the within-party object, with and without politician fixed effects. Columns (1)–(4) are plural suffrage (Leg. $\leq$ 1899); columns (5)–(8) proportional representation (Leg. $\geq$ 1894).

	(1) Pooled	(2) Pooled, pol. FE	(3) Within-party	(4) W-p, pol. FE	(5) Pooled	(6) Pooled, pol. FE	(7) Within-party	(8) W-p, pol. FE
Chamber $\times$ Post-1894	0.0056*	0.0078**	-0.0067	-0.0033				
	(0.0029)	(0.0032)	(0.0047)	(0.0038)				
Chamber $\times$ Post-1900					-0.0013	-0.0093	0.0076	-0.0117***
					(0.0027)	(0.0064)	(0.0065)	(0.0039)
Observations	206,146	206,116	206,146	206,116	245,296	245,268	245,296	245,268
R^2	0.072	0.137	0.318	0.371	0.056	0.120	0.144	0.231

Standard errors clustered at arrondissement in parentheses. Weighted by in-vocabulary token count on the mining axis. Produced by `mining_polarization_did.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 41: Territorial axis (mining vs. rest): event study around the reforms. Chamber × legislature interactions on within-party mining-versus-rest distinctiveness, base legislature 1892.

	(1) mining-vs-rest distinctiveness, within own party
ChL1880	0.0060
	(0.0079)
ChL1882	0.0039
	(0.0059)
ChL1884	-0.0006
	(0.0031)
ChL1886	-0.0006
	(0.0023)
ChL1888	0.0006
	(0.0038)
ChL1890	0.0014
	(0.0031)
ChL1894	0.0016
	(0.0040)
ChL1896	-0.0032
	(0.0040)
ChL1898	-0.0021
	(0.0034)
ChL1900	-0.0050
	(0.0035)
ChL1902	-0.0119**
	(0.0055)
ChL1904	-0.0126***
	(0.0044)
ChL1906	-0.0112
	(0.0076)
ChL1908	-0.0120***
	(0.0044)
ChL1910	-0.0119*
	(0.0060)
ChL1912	-0.0153**
	(0.0061)
Observations	392,051
R^2	0.312

Standard errors clustered at arrondissement in parentheses. Politician, arrondissement, party, and legislature fixed effects absorbed. The 1892 interaction is the omitted base and 1914 is dropped for collinearity. Produced by `mining_polarization_did.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The pooled axis and recomposition. The runs above net out *who* represents a district (the politician fixed effect), so they cannot see the channel proportional representation runs through—recomposition, the replacement of the sitting bloc. Pooling Catholic and Liberal speakers on the mining-versus-rest axis keeps composition in, and the composition can be seen moving, though not in a form the bicameral contrast certifies: the bicameral Panel B contrast on the pooled axis is -0.0013 (s.e. 0.0027) with fixed effects and -0.0093 (s.e. 0.0064) within politician (Table 42), while the Chamber event study (Table 43) climbs from well below its 1898 base in the 1880s (-0.004 to -0.009, all significant) to the base by 1894, and falls below it again after 1900 (-0.007 in 1902, -0.005 in 1904 and -0.007 in 1912, all $p < 0.01$). The climb and the fall bracket the years in which the coal districts returned no Catholic deputy (Appendix A.1), so the pooled partition tracks who sat for the coal belt; but the climb begins before the reform and there is no bicameral break, so the axis is description rather than a test, and the recomposition reading of §6.6 rests on the turnover of 1900 and on the coal-district counts of §6.7, not on this axis.

Table 42: Pooled territorial axis (mining vs. rest, Catholic+Liberal): bicameral Panel B around proportional representation.

	(1) FE	(2) FE + Turn.	(3) Pol. FE
Chamber × Post-1900	-0.0013	-0.0015	-0.0093
	(0.0027)	(0.0027)	(0.0064)
Incumbent (1900)		0.0012	
		(0.0010)	
Entrant (1900)		0.0000	
		(.)	
Observations	245,296	245,296	245,268
R^2	0.056	0.057	0.120

Standard errors clustered at arrondissement in parentheses. Outcome is the pooled mining-versus-rest distinctiveness $s_{\text{all}}^{\text{mine}}$, weighted by in-vocabulary token count; both houses, legislatures 1894–1912. Columns (1) and (2) absorb arrondissement and party fixed effects and include a post-1900 indicator; column (3) adds politician fixed effects. Column (2) adds indicators for members first seated before 1900 (incumbents) and from 1900 (entrants); the second is the complement of the first and is omitted. Produced by `mining_pooled_1900.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 43: Pooled territorial axis: Chamber event study, base legislature 1898. The coefficients climb to the base through the 1880s and fall below it again after 1900; there is no break at the reform.

	(1) s_mine_all
EL1880	-0.0044**
	(0.0017)
EL1882	-0.0072***
	(0.0015)
EL1884	-0.0079***
	(0.0013)
EL1886	-0.0076***
	(0.0019)
EL1888	-0.0091***
	(0.0021)
EL1890	-0.0065***
	(0.0012)
EL1892	-0.0021*
	(0.0012)
EL1894	-0.0008
	(0.0006)
EL1896	-0.0005
	(0.0007)
EL1900	-0.0017
	(0.0011)
EL1902	-0.0070***
	(0.0011)
EL1904	-0.0051***
	(0.0010)
EL1906	-0.0021
	(0.0018)
EL1908	-0.0009
	(0.0015)
EL1910	-0.0034*
	(0.0017)
EL1912	-0.0066***
	(0.0016)
Observations	369,024
R^2	0.125

Chamber only. Politician, arrondissement, and party fixed effects absorbed; 1898 is the omitted base; legislatures 1880–1912. Standard errors clustered at arrondissement. Produced by `mining_pooled_1900.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 44: Selection at the individual level: post-1900 entrants. The tenure slope is the within-entrant change in distinctiveness per legislature since debut; the debut premium is the entrant–incumbent gap at the entrant's first sitting. A flat slope together with a debut premium of zero indicates that the post-1900 entrants neither entered more distinctive than the incumbents nor became more distinctive with tenure.

Post-1900 entrants, s_{distinct}	Estimate
Tenure slope (per legislature since debut)	0.0003 (p=0.25)
Entrant premium at debut (vs incumbents)	0.0003 (p=0.84)

Catholic and Liberal Chamber speakers only, weighted by in-vocabulary token count; standard errors clustered at arrondissement (29 clusters for the slope, 35 for the premium, so the inference is indicative rather than sharp). Produced by `entrant_tenure_final.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.8 The socialist-proximity mechanism, in full

This appendix holds the apparatus for a lexical reading of the coal districts that the paper does not adopt (§6.7): the socialist-proximity differences-in-differences by party, their per-rung detail, and the phrase-level register tables. It is descriptive throughout, for the reasons stated in the main text (four mining clusters; the imprecision of both legs, §7.1; the non-corroborating three-party distance of Appendix B.7) and, above all, for the counts of Appendix A.1: between the reforms the mining districts return no Catholic deputy, so no coefficient below compares the same deputies across the reforms. On that reading, the reform's informative signal would lie not on the Catholic–Liberal axis but on the *socialist* one. Scoring each Catholic or Liberal Chamber speech for how socialist its phrases sound (the socialist-proximity outcome of (15)) in a within-Chamber difference-in-differences, Catholic speech from mining districts would move *toward* the socialist register after plural suffrage and back *away* from it after proportional representation, and the sign reversal in the same districts would reconcile the divided Chamber of 1894 with the inert Catholic–Liberal axis of 1900: the second reform would undo not the first reform's polarization but its *accommodation*.

None of this survives, and the reason is the counts rather than the estimator. Tables 45 and 46 report the specification with arrondissement, politician and legislature fixed effects. In both windows the Catholic column is not identified: no Catholic deputy for the four TopMining arrondissements speaks in the Chamber between the reforms (Appendix A.1), and with the arrondissement the dataset records with each mandate, the Catholic deputies who speak for the four after 1900 do so for the four throughout. With politician and legislature fixed effects but no arrondissement fixed effects, as in the bootstrap of §7.1, the post-1900 leg is -0.002 (s.e. 0.004; Table 10), identified only by a single deputy whose recorded district changes after 1900; the 1894–99 leg is not identified even there. The Liberal columns point in opposite directions in the two windows ($+0.007$, s.e. 0.007, after plural suffrage; -0.007, s.e. 0.002, after proportional representation; Tables 45 and 46). The pooled columns are positive in both windows, significantly so only after plural suffrage ($+0.015$, $p < 0.01$, against $+0.004$, n.s.; Tables 45 and 46), while no party column carries them; the plural-suffrage one is stable across the vocabulary ladder ($+0.014$ and $+0.015$ on the strict and loose rungs; Table 75), the proportional-representation one stays small ($+0.005$, $p < 0.10$, and $+0.003$, n.s.; Table 75); the Catholic column is not identified on any rung. The reading leans on the reversal rather than on either level, because a constant geographic difference cannot change sign at a date; with no Catholic deputy in the middle window there is nothing for the sign to reverse from. Its economic content is a median-voter response: franchise expansion forces Catholics to court newly enfranchised workers through social-Catholic language, and proportional representation, by decoupling seats from local pluralities, removes the incentive and prompts a retreat to a distinctively confessional paternalism—the quantitative counterpart of the retreat into segmented, pillarized competition that the historiography describes (§2).

Why proportionality might reduce the courtship. The direction of the second leg is not self-evident, and the opposite prediction is easy to construct: under proportional representation every vote adds to a party's list total, so a Catholic deputy in a coal district might be expected to keep courting the socialist-leaning voter rather than abandon him. Three features of the Belgian institutional detail point to a reduction instead; we record them because the alternative is the natural objection.

First, what changes at 1900 is not whether votes count but *whose* vote is pivotal for a given seat. Under the pre-1900 majoritarian rule in multi-member constituencies, a locally dominant party converted a local majority into a disproportionate share of the district's seats (§2), so the marginal working-class voter in Charleroi or Mons could decide whether the Catholic list carried the arrondissement or lost it entirely. Proportional allocation removes precisely that discontinuity: the socialist-leaning vote now accrues to the Belgian Workers' Party's own list, and the Catholic seat count in the district follows the size of the Catholic vote rather than the outcome of a local race against the left. The voter still counts; he no longer counts *for the Catholic seat*.

Second, once the seat return to a marginal vote is proportional rather than discontinuous, the cheapest marginal votes are one's own. Courting the socialist-leaning worker is expensive for a confessional party—it requires borrowing an idiom that its clerical and conservative wing contested, as the Daensist schism of the mid-1890s made plain (§2)—while mobilizing the existing Catholic constituency through the party's own organizational network was, by the late 1890s, comparatively cheap and, under compulsory voting, highly effective. The centrifugal logic is the standard one for proportional systems (Cox, 1997): where seats accrue in proportion to a party's own vote, the return to cultivating one's segment rises relative to the return to reaching across.

Third, the timing and the origin of the reform point the same way. Proportional representation was adopted by the incumbents themselves, and on the canonical reading as a device for containing socialist seat growth without having to outbid the socialists in policy (Boix, 1999): a rule that guarantees you seats is a rule that relieves you of the need to compete for the other side's voters. That the pillarized, organizationally segmented competition of the twentieth century consolidates in exactly these years (§2) is the historiographical counterpart of the same movement.

None of this is testable with our design, and we do not claim it is. The two accounts predict opposite signs for the post-1900 leg, and the Catholic leg is not identified once arrondissement fixed effects are added—the Catholic deputies for the four arrondissements sit for them only after 1900—while with politician fixed effects alone it rests on a single deputy whose recorded district changes after 1900 and is small and imprecise (-0.002, s.e. 0.004; §7.1 and Table 10)—one more reason the mechanism is not part of this paper's claims.

Table 45: Socialist-lexicon proximity, plural suffrage (TopMining × Post-1894), by party

	(1) PS: all	(2) PS: Catholic	(3) PS: Liberal
TopMining × Post-1894	0.0150***	0	0.00722
	(0.00211)	(.)	(0.00679)
N	187251	120094	67157

Members whose majority house is the Chamber; arrondissement, politician and legislature fixed effects (column (1) also party); standard errors clustered at arrondissement in parentheses. Column (2) is not identified (absorbed by the fixed effects): between the reforms no Catholic deputy for the TopMining arrondissements speaks (Appendix A.1). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 46: Socialist-lexicon proximity, proportional representation (TopMining × Post-1900), by party

	(1) PR: all	(2) PR: Catholic	(3) PR: Liberal
TopMining × Post-1900	0.00359	0	-0.00706***
	(0.00299)	(.)	(0.00220)
N	220232	155621	64611

Members whose majority house is the Chamber; arrondissement, politician and legislature fixed effects (column (1) also party); standard errors clustered at arrondissement in parentheses. Column (2) is not identified: the Catholic deputies for the TopMining arrondissements sit for them only after 1900. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The partisan lexicon. The scalar measures so far compress thousands of phrases into a single number. To show *what* divides the parties, we decompose the measure to the phrase level. Following the phrase-level decompositions of Gentzkow et al. (2019) and Simola et al. (2025)—with the simplification noted in §4.6—we rank phrases by ζ_j, the posterior probability of correctly guessing the speaker's party from phrase j alone; the token-weighted aggregate of §4.3 is the count-weighted average of the leave-out version of this quantity. Table 47 lists, for each reform window and for both the Catholic–Liberal and Catholic–Socialist splits, the most party-diagnostic phrases together with their predicted frequency per 100,000 phrases and an a-priori register tag that separates a *socialist* register (class, labour, conflict) from a *confessional-paternalist* one (patronage, mutual aid, charity, family).[5]

Table 48 reports, for Catholic deputies in mining districts, the share of socialist- versus paternalist-register tokens by window. It has no 1894–99 row: no tagged token falls in that window, which holds no Catholic mining deputy (Appendix A.1). Between the outside windows the socialist share is the same, 67% before 1894 (162 of 241 tagged tokens) and 67% after 1900 (9,802 of 14,664; Table 48): the table does not show a post-1900 retreat from the language of *class* and *labour*.

[5]The register tag is an a-priori dictionary of accent-insensitive stems (e.g. *class*, *lutt*, *ouvri* for the socialist register; *patron*, *epargn*, *ateli*, *charit* for the paternalist one), applied to the lemmatised-truncated phrase forms; paternalist cues win ties, so *patron_ouvri* reads as paternalist and *class_ouvri* as socialist. A transparent dictionary is the main specification because it is deterministic and auditable.

Table 47: The register lexicon: the most party-diagnostic phrases within the socialist and paternalist registers, by window and split (highest ζ_j), Catholic–Liberal and Catholic–Socialist splits, with a-priori register tag. Format follows the phrase tables of Simola et al. (2025).

Period	Split	Phrase	ζ_j	Loads on	per 100k	Speakers	Register
pre-1894	CathLib	écol_bienfais	0.93	Cath.	7	21	paternalist
pre-1894	CathLib	assist_publiqu	0.74	Cath.	15	60	paternalist
pre-1894	CathLib	caiss_prévoi	0.74	Lib	12	60	paternalist
pre-1894	CathLib	domicil_secour	0.70	Cath.	28	62	paternalist
pre-1894	CathLib	enseign_religion	0.68	Lib	7	54	paternalist
pre-1894	CathLib	bienfais_publiqu	0.67	Cath.	7	64	paternalist
pre-1894	CathLib	ouvri_agricol	0.63	Cath.	7	88	socialist
pre-1894	CathLib	enseign_religi	0.63	Lib	22	74	paternalist
1894-1899	CathLib	ateli_chômag	0.89	Cath.	12	18	paternalist
1894-1899	CathLib	societ_exploit	0.85	Cath.	5	30	socialist
1894-1899	CathLib	syndicat_élevag	0.84	Cath.	9	24	socialist
1894-1899	CathLib	caiss_retrait	0.84	Cath.	19	54	paternalist
1894-1899	CathLib	capital_est	0.84	Cath.	7	48	socialist
1894-1899	CathLib	comit_patronag	0.83	Cath.	6	33	paternalist
1894-1899	CathLib	affili_caiss	0.82	Cath.	8	40	paternalist
1894-1899	CathLib	assist_publiqu	0.81	Cath.	5	23	paternalist
1894-1899	CathSoc	syndicat_élevag	0.98	Cath.	8	22	socialist
1894-1899	CathSoc	vieux_ouvri	0.91	Soc	6	41	socialist
1894-1899	CathSoc	syndicat_professionnel	0.89	Soc	10	32	socialist
1894-1899	CathSoc	une_grev	0.87	Soc	11	46	socialist
1894-1899	CathSoc	ouvri_social	0.86	Soc	7	37	socialist
1894-1899	CathSoc	part_ouvri	0.86	Soc	31	73	socialist
1894-1899	CathSoc	quand_ouvri	0.86	Soc	6	47	socialist
1894-1899	CathSoc	grev_est	0.85	Soc	6	39	socialist
post-1900	CathLib	ouvri_houilleur	0.79	Cath.	7	42	socialist
post-1900	CathLib	caiss_prévoi	0.79	Cath.	35	80	paternalist
post-1900	CathLib	caiss_commun	0.78	Cath.	9	59	paternalist
post-1900	CathLib	regl_ateli	0.77	Cath.	9	49	paternalist
post-1900	CathLib	caiss_remplac	0.75	Lib	5	31	paternalist
post-1900	CathLib	écol_confessionnel	0.72	Lib	9	68	paternalist
post-1900	CathLib	cet_ouvri	0.71	Cath.	6	52	socialist
post-1900	CathLib	bon_secour	0.71	Lib	5	16	paternalist
post-1900	CathSoc	démocrat_chrétien	0.92	Soc	12	56	paternalist
post-1900	CathSoc	vi_ouvri	0.88	Soc	5	56	socialist
post-1900	CathSoc	inspecteur_ouvri	0.87	Soc	7	26	socialist
post-1900	CathSoc	patron_charbonni	0.87	Soc	7	32	paternalist
post-1900	CathSoc	lois_ouvri	0.86	Soc	5	50	socialist
post-1900	CathSoc	class_ouvri	0.86	Soc	122	168	socialist
post-1900	CathSoc	quand_ouvri	0.84	Soc	6	56	socialist
post-1900	CathSoc	ouvri_avaient	0.82	Soc	6	53	socialist

ζ_j: posterior probability of the speaker's party given phrase j (pooled choice probabilities; > 0.5 loads on the first party of the split). "per 100k": predicted count per 100,000 phrases. Register tag from the a-priori dictionary in `partisan_phrases.py`.

Table 48: Register share among Catholic mining-district deputies, by reform window: socialist versus confessional-paternalist lexicon.

Window	Socialist	Paternalist	n_{soc}	n_{pat}	Ratio S/P
pre-1894	67.2%	32.8%	162	79	2.05
post-1900	66.8%	33.2%	9802	4862	2.02

Share of socialist- versus paternalist-register tokens in the speech of Catholic deputies from mining districts. There is no 1894–99 row: no tagged token falls in that window.

The intra-party face: cohesion, not schism. A natural worry about the between-party divergence is that it conceals a *within*-party reshuffling—that what widened was not the Catholic–Liberal gap but the gap between each party's mining and non-mining wings. The same estimator answers this directly: we score, inside each governing party separately, how well an observer could tell a coal-district member from the rest ($s^{\text{mine}}_{\text{cat}}$, $s^{\text{mine}}_{\text{lib}}$), i.e. the party's *internal* rhetorical division along the mining cleavage. Table 49 reports the levels and Table 50 the Chamber event study (base 1892). The answer is the opposite of a schism, and it completes the adaptation story. The Catholic party entered the 1880s internally *divided* on this line ($s^{\text{mine}}_{\text{cat}} \approx 0.56$, well above chance)—the social-Catholic/conservative tension the historiography documents (§2)—and then *closed* the divide: 0.562 before 1894, 0.550 in 1894–99, 0.528 after 1900 (Table 49), with the event-study terms small in 1894 (-0.004) and negative and significant from 1896 on (-0.010 in 1896 and 1898, -0.024 in 1902 and 1904; Table 50). The Liberals show little internal mining structure (close to chance throughout, 0.504–0.513; Table 49). Two readings follow. First, the between-party divergence of §6.2 is *not* a within-party reshuffling in disguise: as the Catholic party pulled away from the Liberals it became more internally homogeneous, the whole party—both wings—converging on a common register, which is what "within-incumbent adaptation of the party as a bloc" should look like at the intra-party level. Second, the convergence begins before 1894—the leads decline toward the 1892 base through the 1880s, from $+0.006$ in 1880—and then deepens after the reforms (-0.010 in 1896 and 1898, -0.024 in 1902; Table 50), the intra-party counterpart of the adaptation and of the post-1900 recomposition; because the trend predates the first reform we read it descriptively, as corroboration of the adaptation and recomposition readings, not as a further causal estimate.

Table 49: Within-party internal division along the mining cleavage: token-weighted mean of the internal distinctiveness score, Chamber, by window. Chance is 0.5; higher values mean the party's coal-district members are more rhetorically distinguishable from its other members.

Party (internal mining split)	pre-1894	1894–99	post-1900
Catholic	0.5618	0.5497	0.5281
Liberal	0.5037	0.5064	0.5126

Produced by `within_party_internal.do` on the within-party mining scores of `mining_polarization.py`. Descriptive; Chamber only.

Table 50: Within-party internal division: Chamber event study, base legislature 1892. Negative post-1894 terms indicate internal *convergence*—the party closing its mining/non-mining divide.

	(1) Catholic internal	(2) Liberal internal
EL1880	0.0064**	-0.0029
	(0.0026)	(0.0022)
EL1882	0.0022	-0.0064***
	(0.0021)	(0.0011)
EL1884	0.0042**	-0.0051**
	(0.0018)	(0.0019)
EL1886	0.0010	0.0003
	(0.0022)	(0.0057)
EL1888	0.0023	-0.0012
	(0.0027)	(0.0056)
EL1890	0.0008	-0.0013
	(0.0015)	(0.0041)
EL1894	-0.0039	-0.0042**
	(0.0026)	(0.0019)
EL1896	-0.0097***	-0.0027
	(0.0033)	(0.0019)
EL1898	-0.0096***	-0.0023
	(0.0024)	(0.0025)
EL1900	-0.0123***	-0.0012
	(0.0034)	(0.0032)
EL1902	-0.0238***	-0.0046**
	(0.0050)	(0.0022)
EL1904	-0.0244***	-0.0028
	(0.0024)	(0.0022)
EL1906	-0.0181***	0.0000
	(0.0039)	(0.0026)
EL1908	-0.0176***	0.0000
	(0.0036)	(0.0037)
EL1910	-0.0213***	0.0002
	(0.0055)	(0.0028)
EL1912	-0.0267***	-0.0002
	(0.0043)	(0.0036)
Observations	239,292	129,732
R^2	0.202	0.131

Politician and arrondissement fixed effects absorbed; standard errors clustered at arrondissement. The pre-1892 leads are not flat (the convergence predates the reform), so the table is read descriptively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.9 Vocabulary-filter robustness: the three-rung ladder

The estimates in the main text are built on the *medium* global vocabulary (≥ 35 total occurrences and ≥ 6 distinct speakers per phrase). Because that width is a researcher choice, we re-estimate the whole pipeline—vocabulary, leave-out posteriors $\hat{\rho}$, every difference-in-exposure and triple-difference, and the full falsification battery—under a *strict* vocabulary (≥ 50, ≥ 8) and a *loose* one (≥ 25, ≥ 5), with the medium set fixed a priori as the geometric midpoint of the two so that the rung reported in the text is not chosen on results. Only vocabulary width varies; the hygiene filters (minimum within-period count, minimum token length) and the drop-token cleaning are held fixed across rungs. Table 52 places the headline coefficients side by side across the three vocabularies.

Everything the causal claim rests on is invariant, and the ladder is orderly. The plural-suffrage divergence is $+0.0092$, $+0.0095$ and $+0.0098$ ($p < 0.01$ throughout) as the vocabulary widens from $50/8$ to $35/6$ to $25/5$, and $+0.0088$, $+0.0094$ and $+0.0098$ ($p < 0.01$ throughout) under politician fixed effects (Table 52). Moving from the strict to the loose thresholds, which widens the phrase set from 34,338 to 37,529 phrases, moves the headline coefficient by about a tenth, monotonically, in the direction a wider vocabulary should move it: more phrases admitted means more partisan signal retained. Its concentration on incumbents behaves identically ($+0.0114$, $+0.0120$, $+0.0124$, all $p < 0.01$), and so does the mining triple, negative in its continuous form (-0.0051, -0.0056, -0.0057, all $p < 0.01$) and in its TopMining form (-0.0029, -0.0038, -0.0039; Table 52), which Appendix B.6 reads as a gradient rather than as a second reform effect. The Panel B contrast is negative on every rung: -0.0106, -0.0110, -0.0114 with fixed effects, and -0.0092, -0.0099, -0.0105 once the within-politician panel is imposed (Table 52)—the 1900 renewal of both houses (§6.6) read across the ladder.

Table 52 is written directly by the same script that estimates the rungs, one column per run, so the table and the estimates cannot come apart; every ladder figure quoted in the text has the run as its single source.

The reactions of the floor. A fourth choice is not a matter of width. The record prints the reactions of the floor and the stage directions of the sitting inside the speech they interrupt—*(Rires à gauche.)*, *(Très bien! à droite.)*, *(Interruption de M. Woeste.)*, *(M. le président agite la sonnette.)*—and they are not the speaker's words, so we take them out of the text before any phrase is formed. A parenthesized segment is removed when it holds a reaction word (laughter, noise, interruption, protest, exclamation, approval, applause, murmurs, the bell, and their Dutch counterparts), a reaction pair (*très bien*, *aux voix*, *à l'ordre*) or a repeated cry, and its other words belong to the language of stage directions—the sides of the house, intensity and duration, the president, the speaker, the ministers, names. Segments without such a head—numbers, places, editorial notes—and the speaker's own parentheses stay. In the Chamber the rule removes 142,037 segments from 108,822 blocks; in the Senate, whose text arrives lemmatized, the same segments are found in the raw text and their tokens removed from the lemmas of 12,119 interventions (35,505 tokens). The vocabulary built on the cleaned text has 36,297 phrases, against 36,452 with the reactions in. Left in the text, the reactions would enter the vocabulary as party-diagnostic phrases by construction, since a member's own benches applaud him and the opposite benches protest; but the estimates barely depend on them. With the reactions in, the plural-suffrage divergence is $+0.0096$ with fixed effects and $+0.0096$ with politician fixed effects (s.e. 0.0022 and 0.0029, both $p < 0.01$), and the Panel B contrast -0.0107 and -0.0093 (both $p < 0.01$), against $+0.0095$ and $+0.0094$, and -0.0110 and -0.0099, on the cleaned text (Table 59).

One check disciplines the ladder itself. Because a narrower vocabulary leaves more speeches with no in-vocabulary phrase, the three rungs rest on slightly different samples, and one may ask whether that attrition is differential—if the narrow rung dropped disproportionately many Chamber speeches after 1894, the ladder would compare selected samples rather than vocabularies. It does not. Table 51 reports the share of speeches present under the loose vocabulary but absent under the strict one, by chamber and reform window. The rate is at most 0.2% (Chamber 0.0017, 0.0012, 0.0020; Senate 0.0010, 0.0010, 0.0013). The Chamber loses

marginally more speeches than the Senate, as the larger, more lexically varied house with a longer tail of rare collocations should, and a constant difference of that kind differences out of the bicameral contrast exactly as any other time-invariant chamber characteristic does. What would threaten the comparison is a change in attrition at a reform date that differs between the houses, and the differences are under a tenth of a percentage point at both dates.

Table 51: Vocabulary attrition across the ladder: share of speeches present under the loose vocabulary but absent under the strict one, by chamber and reform window. Small in both houses and similar across windows, so the rungs compare comparable samples.

	pre-1894	1894–99	post-1900
Senate	0.0010	0.0010	0.0013
Chamber	0.0017	0.0012	0.0020

Share of the speeches scored under the loose vocabulary that are not scored under the strict one, on the current score files of the loose and strict rungs.

Table 52: Headline coefficients across the vocabulary-filter ladder. Only vocabulary width varies across columns; hygiene filters and drop-token cleaning are held fixed. The medium vocabulary is the main-text specification.

	Strict ($\geq 50, \geq 8$)	Medium ($\geq 35, \geq 6$) (main text)	Loose ($\geq 25, \geq 5$)
Plural suffrage, Catholic–Liberal axis (Panel A)			
Chamber × Post-1894 (FE)	0.0092***	0.0095***	0.0098***
Chamber × Post-1894 (politician FE)	0.0088***	0.0094***	0.0098***
× Mining (DDD, FE)	-0.0051***	-0.0056***	-0.0057***
TopMining triple (PS)	-0.0029*	-0.0038**	-0.0039**
Incumbents only (PS)	0.0114***	0.0120***	0.0124***
Proportional representation, Catholic–Liberal axis (Panel B)			
Chamber × Post-1900 (FE)	-0.0106***	-0.0110***	-0.0114***
Chamber × Post-1900 (politician FE)	-0.0092***	-0.0099***	-0.0105***

Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Every coefficient is estimated on the rung it is printed under and written directly from that run by `ladder_reconcile.do`; the table has no second source to diverge from. Standard errors clustered at arrondissement. Sample sizes differ mildly across rungs because a narrower vocabulary leaves more speeches with no in-vocabulary phrase; Table 51 shows this attrition is negligible and, crucially, flat across chambers and reform windows. The socialist-proximity coefficients on the three rungs are in Table 75.

B.10 Topic classification and validation

This appendix documents the classifier behind the decomposition of §6.4: the taxonomy, how the training labels were produced, the sampling design, held-out performance, validation against human coding, and coverage.

Taxonomy. Speeches are assigned to one of 22 substantive policy domains or to a procedural residual, and the 22 are grouped into the six buckets used in the main text. The grouping is complete by construction—every domain is mapped, there is no leftover category—and is reported in Table 53. Two boundary decisions

are substantive rather than technical and are stated in §6.4: education is grouped with religion, and mining and industry with the economy.

Supervision. No hand-labelled corpus of nineteenth-century Belgian parliamentary speech exists, so the training labels are generated by GPT-4o (snapshot `gpt-4o-2024-08-06`, temperature 0, fixed seed; every raw response retained) against a written codebook that defines each domain and resolves the boundaries most likely to be contested in this corpus—labour against general welfare, labour against industry, education against religion, defence against colonial affairs, police against army, and a budget debate against the policy area it concerns. The prompt returns one primary domain and up to three secondary ones; only the primary enters the classifier's target. The full prompt is reproduced in the replication materials.

Sampling. The labelled frame is a probability sample of 7,000 speeches stratified on chamber $\times$ reform window, with a floor per cell so that the thinner Senate cells are represented, plus a further 1,500 drawn from thin cells for training only. Selection uses no topic keywords: a design that oversampled speeches rich in hand-written topic keywords would have anchored the topic partition to the same lexicon that generates s_{distinct}, and would in addition have made the validation split unrepresentative of the corpus. Test and validation splits are drawn only from the probability sample; the booster enters training alone.

Model. A CamemBERT encoder with two softmax heads: a seven-way head over the six buckets plus the residual, which is the paper's object and carries full weight in the loss, and an auxiliary 23-way head over the fine domains at weight 0.3. Both heads use class-weighted cross-entropy. Long speeches are chunked with overlap and chunk probabilities averaged. Training the coarse target as a single-label softmax rather than as independent binary heads is what makes the rare domains learnable: under a multi-label design the smallest classes sit below 3% prevalence and the model converges on predicting the marginal frequencies. Decision thresholds do not arise, because the coarse assignment is an argmax.

Distillation onto the dataset. The classifier was trained on, and applied to, the interventions of the PLENUM corpus. The dataset's interventions are different units—a member's turn rather than a block of PLENUM's structuring—so the labels are carried over by a second distillation: a linear classifier (tf-idf on the unigrams and bigrams of the lemmatized text, logistic regression fitted by stochastic gradient descent) learns the classifier's probabilities on PLENUM's interventions in two folds split by sitting, each PLENUM intervention receives the prediction of the fold that did not see it, and each of the dataset's interventions the average of the two fold models; the Senate, for consistency, also carries the distilled out-of-fold labels rather than the original ones. Out of fold, the distilled coarse label agrees with the classifier's on 86% of interventions ($\kappa = 0.77$, macro $F_1 = 0.79$), and whether an intervention carries a substantive topic at all on 89% ($\kappa = 0.78$), with no drift across the Chamber's reform windows (Table 74). The 22 domains of the descriptive by-topic table, which come from a second classifier of the same family, are distilled in the same way and less faithfully ($\kappa = 0.56$ on the domain, 0.77 on whether one is assigned; Table 74).

Validation against human coding. Held-out performance measures fidelity to the annotator, not accuracy. To measure accuracy we hand-coded a blind subsample of 250 speeches drawn from the probability sample: the sheet was shuffled, carried no model output, and the chamber and period columns were hidden during coding. Against that standard the classifier reaches an accuracy of 0.86 and $\kappa = 0.80$, and the GPT-4o annotation itself 0.84 and $\kappa = 0.77$ (Table 54). By the conventional reading (Landis and Koch, 1977) this is substantial agreement, at the threshold content analysis asks of a primary measure and comfortably above what is asked of a control variable. That the classifier agrees with the coder at least as closely as its own annotator does indicates that distillation does not compound the annotator's error.

Coverage and the procedural residual. The classifier scores every speech. 61.0% carry no substantive policy content—procedural exchanges, communications, votes and fragments—which is unremarkable for a plenary record. Retaining them as a seventh bucket rather than dropping them is what keeps the estimation sample free of selection on treatment; Table 55 reports the classification rate by chamber and reform window, which is what makes the selection visible in the restricted sample and absent in the full one.

What the classifier is used for, and what it is not. The topic labels enter the analysis only as fixed effects in the decomposition of §6.4 and in the descriptive by-topic table of §A.6. They do not enter the partisanship measure, the vocabulary, the leave-out yardstick, or any specification that identifies the headline effect: the plural-suffrage estimate of §6.2 is identified off the bicameral contrast and contains no thematic interaction. Measurement error in the topic labels therefore attenuates how much the topic fixed effects can absorb, which biases the reported within-topic share *upward*; it cannot generate the effect being decomposed.

Table 53: Topic taxonomy: 22 substantive domains and the six buckets used in the main text. The grouping is complete—every domain is assigned—so the coarse partition has no residual category beyond the procedural one.

Bucket	Fine domains
Labour and social	Labour and worker social policy; general welfare and public assistance; public health and healthcare
Confessional	Religion, secularism and church–state; education and science
Economic and fiscal	Budget and fiscal policy; taxation, customs and property; finance, banking and monetary policy; commerce and trade; industry and economic policy; agriculture and rural policy
State and governance	Governance and administration; electoral and constitutional policy; justice and legal system; public order and policing; citizenship, immigration and minority rights
Foreign and defence	Colonial, foreign and international relations; defence and military
Territory and infrastructure	Infrastructure, transport and public works; land use, housing and urban policy; environment and natural resources; communication, media and intellectual property
Procedural residual	Procedural, ceremonial or fragmentary material with no policy content

Table 54: Classifier validation. Two hundred and fifty randomly drawn speeches, stratified on the design's cells, hand-coded against the seven-bucket taxonomy; agreement is reported against that coding for the fine-tuned CamemBERT model used throughout and for the GPT-4o annotations it was trained on.

	CamemBERT	GPT-4o
Cohen's κ against hand coding	0.80	0.774
Accuracy	0.86	0.84
Macro F_1	0.823	
Hand-coded speeches	250	

Agreement is computed on the coarse seven-bucket assignment, the object the decomposition of §6.4 uses; the fine twenty-three-domain head is auxiliary and is not validated separately. A single coder produced the reference labels, so κ has no human ceiling to be read against—a second independent coder on a subsample would supply one, and we note its absence rather than let the figure stand as if it had.

Table 55: Classification coverage by chamber (house of delivery) and reform window. The share of speeches carrying a substantive policy topic falls in the exposed Chamber at plural suffrage and does not fall in the Senate. Retaining the procedural residual as a seventh bucket removes this selection by construction, which is why the main specification does so.

Chamber	Window	Speeches	Classified	Share
Chamber (DK)	pre-1894	169,472	72,002	0.425
	1894–1899	102,221	40,745	0.399
	post-1900	322,266	110,797	0.344
Senate (SK)	pre-1894	16,346	9,231	0.565
	1894–1899	11,090	6,327	0.571
	post-1900	34,445	16,990	0.493
All	1880–1912	655,840	256,092	0.390
DiD, plural suffrage (raw shares)				−0.032

Table 56: Robustness of the topic decomposition to taxonomy granularity and to the treatment of the procedural residual. *Full* retains every speech, with the non-policy residual as a seventh bucket; *classified* keeps only speeches carrying a substantive topic. The within-topic share is the object reported in Section 6.4.

Taxonomy	Sample	β	β_W	Within share	Speeches
Six buckets	Full	0.0095	0.0090	94.1%	206,146
Six buckets	Classified	0.0098	0.0091	93.5%	106,672
22 topics	Full	0.0095	0.0086	90.0%	206,146
22 topics	Classified	0.0098	0.0086	88.4%	106,672
Headline DiD, all speeches		0.0095	—	—	206,146

Standard errors clustered at arrondissement; speeches weighted by in-vocabulary token count. β is the plural-suffrage difference-in-exposure on the stated sample and β_W the same coefficient with topic fixed effects; the within share is β_W/β. The last line reports the headline estimate on the whole corpus, for comparison with the samples the decomposition can use. Produced by `topic_variants.do`.

B.11 Estimates quoted in the text: supplementary tables

The tables in this section hold the estimates that the text quotes and that no other table or figure reports. Each is produced from the same runs as the rest of the paper; the producing script is named in its note.

Table 57: Partisanship of parliamentary speech by legislature and chamber: the values plotted in Figure 1, the series with the two parties weighted equally, and the Catholic share of the two parties' tokens.

	Chamber				Senate			
Legislature	Π	$\frac{1}{2}(C+L)$	Catholic share	Speeches	Π	$\frac{1}{2}(C+L)$	Catholic share	Speeches
1880	0.4984	0.4974	0.4261	13,829	0.5000	0.5001	0.5113	756
1882	0.4998	0.4982	0.3814	16,737	0.5032	0.5027	0.4554	813
1884	0.5033	0.5039	0.5398	20,938	0.5035	0.5026	0.3929	920
1886	0.5040	0.5041	0.6554	17,569	0.5034	0.5038	0.3835	1,015
1888	0.5045	0.5038	0.6354	20,921	0.4997	0.5007	0.4045	1,253
1890	0.5053	0.5044	0.6696	20,512	0.5003	0.5019	0.3050	1,331
1892	0.5046	0.5046	0.6365	28,277	0.5019	0.5013	0.1596	2,885
1894	0.5099	0.5057	0.8342	25,295	0.4961	0.4970	0.3975	921
1896	0.5125	0.5064	0.8701	16,242	0.4978	0.4983	0.4595	876
1898	0.5109	0.5063	0.8484	17,255	0.5006	0.5006	0.5025	652
1900	0.5043	0.5039	0.6393	20,937	0.5039	0.5053	0.2533	1,521
1902	0.5079	0.5051	0.7076	21,768	0.5025	0.5057	0.1979	1,515
1904	0.5060	0.5043	0.6753	24,754	0.5039	0.5024	0.2414	2,243
1906	0.5063	0.5045	0.6576	20,712	0.5031	0.5083	0.1620	1,670
1908	0.5033	0.5022	0.6572	29,920	0.5025	0.5031	0.1452	1,959
1910	0.5042	0.5034	0.6516	28,158	0.5017	0.5054	0.1641	1,577
1912	0.5040	0.5043	0.5936	31,517	0.5064	0.5038	0.1993	1,883
Window means (excess over $\frac{1}{2}$)								
pre-1894	0.5030 (+0.0030)	0.5032			0.5016 (+0.0016)	0.5017		
1894–1899	0.5110 (+0.0110)	0.5061			0.4980 (−0.0020)	0.4985		
post-1900	0.5051 (+0.0051)	0.5041			0.5034 (+0.0034)	0.5045		

Token-weighted means of the per-speech leave-out outcome s_{distinct} on the globally fixed yardstick $\hat{\rho}$, Catholic and Liberal speeches, each member counted in his house of membership; $\frac{1}{2}(C+L)$ weights the two parties equally; "Catholic share" is the Catholic share of the two parties' in-vocabulary tokens; "Speeches" counts Catholic and Liberal speeches with a score. In the Panel A estimation sample the standard deviation of s_{distinct} is 0.0234 across speeches (weighted; 0.0459 unweighted) and 0.0234 across the weighted means of the 470 politicians. Produced by `08_descrittive.py` (blocks B and D) from the scores of `06_punteggi.py`.

Table 58: The Chamber corpus: the Belgian Chamber Debates dataset against the PLENUM structuring of the same OCR text, and the linking of speakers.

Panel A: by legislature, sittings in both corpora

Legislature	Word ratio	Turn ratio	Correlation
1880	0.81	0.91	0.92
1882	0.82	0.98	0.97
1884	0.77	0.98	0.91
1886	0.71	1.01	0.71
1888	0.65	1.03	0.59
1890	0.60	1.04	0.62
1892	0.67	0.98	0.56
1894	0.46	1.58	0.33
1896	0.53	1.02	0.52
1898	0.60	0.97	0.62
1900	0.50	1.30	0.32
1902	0.45	1.66	0.35
1904	0.50	1.63	0.27
1906	0.49	2.12	0.28
1908	0.49	2.51	0.13
1910	0.40	3.48	0.28
1912	0.48	2.10	0.35

Panel B: speaker linking, Chamber turns of the 1880–1912 legislatures

	Turns	Share
Exact match to a member sitting that day	551,702	92.8%
Near match	26,671	4.5%
Minister	16,023	2.7%
All	594,396	100%

Panel A: the word ratio is the words of PLENUM's interventions over the dataset's in the same sittings; the turn ratio is the median over sittings of the dataset's turns over PLENUM's interventions; the correlation is across sittings, of the two counts. Over the window the word ratio is 0.715 before 1894 and 0.491 from 1894; the dataset has 3,835 sittings and PLENUM 3,881, 3,831 of them in both. On a held-out sample, the lemmatized rendering of the dataset's Chamber text reproduces 98.0% of PLENUM's tokens (precision 93.4%) and 90.6% of its candidate phrases (precision 86.9%). Panel B: the dataset's link of each turn; ministers include those linked by role. Produced by `11_derivati.py` (block A).

Table 59: The headline contrasts to four decimals, with the reactions of the floor taken out of the text (the paper's estimates) and left in.

	Reactions removed (paper)		Reactions in the text	
	FE	Politician FE	FE	Politician FE
Chamber × Post-1894	0.0095^{***}	0.0094^{***}	0.0096^{***}	0.0096^{***}
	(0.0021)	(0.0029)	(0.0022)	(0.0029)
Observations	206,146	206,116	207,663	207,633
Chamber × Post-1900	-0.0110^{***}	-0.0099^{***}	-0.0107^{***}	-0.0093^{***}
	(0.0020)	(0.0028)	(0.0020)	(0.0028)
Observations	245,296	245,268	248,048	248,020
Phrases in the vocabulary	36,297		36,452	

Specifications of Table 3, columns (1) and (3): arrondissement and party fixed effects, and politician fixed effects in the second column of each pair; weighted by in-vocabulary token count; standard errors clustered at arrondissement. The reactions are removed by the rule of Appendix B.9 before the phrases are formed; the right-hand columns re-run the whole pipeline with them left in the text. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 60: French-only speech. The headline contrasts re-estimated with the whole pipeline (vocabulary, posteriors, scores) rebuilt on French-language speech, against the same specifications on all speech.

	(1) All, FE	(2) All, Pol. FE	(3) French, FE	(4) French, Pol. FE	(5) All, FE	(6) All, Pol. FE	(7) French, FE	(8) French, Pol. FE
Chamber × Post-1894	0.0095^{***}	0.0094^{***}	0.0117^{***}	0.0107^{***}				
	(0.0021)	(0.0029)	(0.0019)	(0.0028)				
Chamber × Post-1900					-0.0110^{***}	-0.0099^{***}	-0.0130^{***}	-0.0108^{***}
					(0.0020)	(0.0028)	(0.0020)	(0.0028)
Observations	206146	206116	137640	137606	245296	245268	156222	156192
R^2	0.059	0.146	0.064	0.161	0.053	0.117	0.052	0.130

Columns (1)–(4): Panel A (Chamber × Post-1894), all speech and French only, with fixed effects and with politician fixed effects; columns (5)–(8): the same for Panel B. Standard errors clustered at arrondissement. Selection into the French-only sample: a difference-in-differences on the probability that a speech is also scored on French-only text gives -0.017 at 1894 and $+0.033$ at 1900; the share so scored, pre-1894 / 1894–99 / post-1900, is 0.667 / 0.671 / 0.629 in the Chamber and 0.658 / 0.661 / 0.601 in the Senate. The French-only vocabulary keeps 33,955 phrases, against 36,297 on all speech. Chamber window means (by venue), pre-1894 / 1894–99 / post-1900: French speech 0.5028 / 0.5115 / 0.5051; all speech re-scored by the same pipeline 0.5028 / 0.5117 / 0.5053; the paper's scores 0.5031 / 0.5120 / 0.5054. Produced by `french_only_did.do` and `pi_diagnostics.py`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 61: Event study on the Catholic–Liberal axis on French-only speech, base legislature 1892: the specification of Table 7 on the scores of Table 60.

	(1) 1880–1898	(2) 1880–1912
1880	-0.00357 (0.00329)	-0.00271 (0.00340)
1882	-0.00639** (0.00283)	-0.00514* (0.00284)
1884	-0.00188 (0.00292)	-0.00136 (0.00293)
1886	-0.00352* (0.00179)	-0.00309* (0.00167)
1888	-0.00248 (0.00161)	-0.00212 (0.00163)
1890	-0.00115 (0.00156)	-0.000903 (0.00146)
1894	0.00874** (0.00403)	0.00820** (0.00386)
1896	0.0113*** (0.00298)	0.0112*** (0.00275)
1898	0.00626*** (0.00201)	0.00595*** (0.00194)
1900		-0.00401** (0.00162)
1902		0.0000332 (0.00198)
1904		-0.00240 (0.00150)
1906		-0.000113 (0.00268)
1908		-0.00342* (0.00190)
1910		-0.00230 (0.00204)
1912		-0.00814*** (0.00255)
Constant	0.504*** (0.00105)	0.505*** (0.00101)
Observations	137606	254067
R^2	0.169	0.137

Within-politician specification with politician, arrondissement, party and legislature fixed effects; weighted by in-vocabulary token count; standard errors clustered at arrondissement. Joint test of the six pre-reform leads in column (1): $F(6,40) = 2.66$, $p = 0.029$. Produced by `event_study_main.do` on the French-only scores (`esegui_paper_c.do`, block C9b). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 62: Event study on the split-sample ruler, base legislature 1892, 1880–1898: the specification of Table 7, column (1), on the scores of the five-fold split-sample ruler of Table 13.

	Chamber × legislature
1880	−0.0069*
	(0.0039)
1882	−0.0076**
	(0.0031)
1884	−0.0017
	(0.0035)
1886	−0.0031
	(0.0026)
1888	−0.0027*
	(0.0016)
1890	−0.0016
	(0.0013)
1894	0.0098**
	(0.0038)
1896	0.0128***
	(0.0041)
1898	0.0060***
	(0.0022)

Politician, arrondissement and party fixed effects and legislature indicators; weighted by in-vocabulary token count; standard errors clustered at arrondissement (41 clusters); significance from the t distribution with 40 degrees of freedom. Joint test of the six pre-reform leads: $F(6,40) = 2.92$, $p = 0.019$. Produced by `leave_cell_out_did.do` (`esegui_paper_c.do`, block C8). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 63: The three yardsticks in full: both panels, with fixed effects and with politician fixed effects (Table 13 reports the politician-fixed-effects Panel A coefficients and their bootstrap p-values).

	(1) Fixed, FE	(2) Fixed, Pol. FE	(3) LCO, FE	(4) LCO, Pol. FE	(5) Split, FE	(6) Split, Pol. FE
Panel A: pre-1894 vs. 1894–1899						
Chamber × Post-1894	0.0095***	0.0094***	0.0089***	0.0073**	0.0125***	0.0119***
	(0.0021)	(0.0029)	(0.0023)	(0.0030)	(0.0021)	(0.0034)
Observations	206146	206116	206146	206116	206146	206116
R^2	0.059	0.146	0.038	0.104	0.055	0.134
Panel B: 1894–1899 vs. 1900–1912						
Chamber × Post-1900	-0.0110***	-0.0099***	-0.0085***	-0.0066**	-0.0131***	-0.0105***
	(0.0020)	(0.0028)	(0.0023)	(0.0030)	(0.0021)	(0.0017)
Observations	245296	245268	245296	245268	245296	245268
R^2	0.053	0.117	0.025	0.083	0.040	0.100

"Fixed" is the paper's pooled yardstick, "LCO" the leave-cell-out ruler, "Split" the five-fold split-sample ruler. Weighted by in-vocabulary token count; standard errors clustered at arrondissement. Produced by `leave_cell_out_did.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 64: A vocabulary selected on pre-1894 speech alone, at the same thresholds, scored on the split-sample and leave-cell-out rulers (Panel A).

	(1) Pooled, FE	(2) Pooled, Pol. FE	(3) Pre-1894 split, FE	(4) Pre-1894 split, Pol. FE	(5) Pre-1894 LCO, FE	(6) Pre-1894 LCO, Pol. FE
Chamber × Post-1894	0.0095***	0.0094***	0.0087***	0.0071***	0.0063***	0.0042***
	(0.0021)	(0.0029)	(0.0016)	(0.0021)	(0.0014)	(0.0015)
Observations	206146	206116	203266	203234	203266	203234
R^2	0.059	0.146	0.043	0.112	0.033	0.093

Columns (1)–(2): the paper's pooled vocabulary and yardstick; (3)–(6): the vocabulary selected on pre-1894 speech, 26,883 phrases against the pooled 36,297. Weighted by in-vocabulary token count; standard errors clustered at arrondissement. Produced by `vocab_prereform.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 65: The second moment: does the spread *within* each party narrow in the Chamber after 1894? Bicameral difference-in-differences on five measures of within-party dispersion.

	(1) |deviation|, politician	(2) MAD	(3) SD, adjusted	(4) ln SD, adjusted	(5) SD, raw
Chamber × Post-1894	−0.00734	0.00016	−0.01365	−0.37206	−0.00811
	(0.00579)	(0.00169)	(0.01103)	(0.49759)	(0.00849)
Observations	1,627	40	40	32	40

Panel A window. Column (1): each member's absolute deviation from his own party's contemporaneous centroid in his own house, computed leaving him out, on politician×legislature means, with politician, party and legislature fixed effects and standard errors clustered at the member's modal arrondissement. Columns (2)–(5): the dispersion of politician means within each party×house×legislature cell—median absolute deviation; standard deviation net of the sampling variance of the means, and its logarithm, undefined in the eight cells where the net variance is not positive; raw standard deviation—with chamber, party and legislature fixed effects and heteroskedasticity-robust standard errors. The 95% interval of column (1) is $[-0.019, +0.004]$. Correlation across the seventeen legislatures between the between-party distance and the within-party spread: -0.028 in the Chamber, $+0.145$ in the Senate. Produced by `within_party_dispersion.do` (`esegui_paper_a.do`).

Table 66: Inference that does not rely on the number of clusters: the collapsed design, randomization inference over the chamber label, and leverage.

	Estimate	p
The identifying variation		
Collapsed chamber×legislature 2×2	0.0116	0.020
Within-politician difference (pre/post means)	0.0097	
Inference that does not assume independent clusters		
Randomization inference, studentized (two-sided)		0.0324
Randomization inference, studentized (one-sided)		0.0146
memo: unstudentized, two-sided		0.0050
Leverage		
Leave-one-arrondissement-out range	[0.0051, 0.0103]	
Largest placebo break, reform-free windows	0.0040	

Panel A window. Collapsed 2×2: the difference-in-differences on chamber×legislature cell means, weighted by the cells' tokens. Within-politician difference: the 133 members who spoke on both sides of 1894, each weighted by the in-vocabulary tokens of the thinner of his two sides. Randomization inference: the chamber label permuted across those members, studentized, 4,999 draws. Leverage: the reference coefficient (Table 3, Panel A, column 3) dropping one arrondissement at a time, and the largest coefficient the same specification produces at any cutoff inside the reform-free windows 1880–1892 and 1902–1912. Produced by `few_group_inference.do`.

Table 67: Politician-level inference: two-way clustering, the members observed on both sides of the reform, and the permutation of the chamber label with equal member weights.

	(1) Pol. FE
Chamber × Post-1894	0.0094^{***} (0.0029)
Observations	206116
Two-way cluster	arrondissement, legislature
Bilateral members (chamber = house of first speech)	
Chamber: N / mean / median / share>0	102 / 0.0026 / 0.0018 / 0.578
Senate: N / mean / median / share>0	31 / 0.0123 / 0.0037 / 0.581
Leave-one-politician-out on T	
Unweighted: T / range	-0.0097 / $[-0.0126, -0.0016]$
Token-weighted: T / range	0.0056 / $[0.0037, 0.0067]$
Studentized permutation of the chamber label	
All bilateral members ($n = 133$)	$p = 0.364$
Excluding genuine movers ($n = 133$)	$p = 0.363$

Top panel: the reference specification (Table 3, Panel A, column 3) with standard errors clustered by arrondissement and by legislature (41 and 10 clusters; $p = 0.010$). T is the Chamber–Senate difference in the mean within-politician change of the members observed on both sides of the reform, members weighted equally ("unweighted") or by the volume of their speech ("token-weighted", with the weights of `politician_level_inference.do`); the ranges drop one politician at a time. Token-weighted, T has $t = 1.66$ and a Kish effective sample size of 27.8, the five heaviest members carrying 33.5% of the weight; unweighted, $t = -1.01$. Without the decile of members with the least speech, T is $+0.0016$ unweighted ($t = 0.66$) and $+0.0058$ token-weighted. Permutation: twenty thousand draws, Welch studentization; no member changes house under the rule of first speech, so excluding the genuine movers leaves the same 133. Produced by `politician_level_inference.do` (`esegui_paper_a.do`); Kish size, t statistics and the trimmed decile from `esegui_paper_c.do`, block C11.

Table 68: Debut cohorts: mean distinctiveness of each cohort of members in its first legislature, by chamber, and the difference-in-differences across the reforms.

Cohort	Chamber	Members	Senate	Members
1882	0.5023	23	0.5068	15
1884	0.5022	35	0.5001	15
1886	0.5055	30	0.5000	4
1888	0.4992	14	0.4986	11
1890	0.5100	17	0.5022	8
1892	0.5046	32	0.5044	16
1894	0.5067	40	0.4957	13
1896	0.5142	11	0.5143	3
1898	0.5100	19	0.5044	5
1900	0.5030	40	0.5070	22
1902	0.5258	6	0.5016	6
1904	0.4998	18	0.5051	2
1906	0.5013	10	0.4920	4
1908	0.5086	18	0.4883	7
1910	0.5154	10	0.5000	4
1912	0.5020	34	0.5046	20

	(1) Cohorts 1882–1898	(2) Cohorts 1894–1912
Chamber × cohort of 1894 or later	0.0081	
	(0.0058)	
Chamber × cohort of 1900 or later		-0.0094
		(0.0064)
Observations	23448	22569

Upper panel: token-weighted mean of s_{distinct} over each cohort's speeches in its debut legislature, by the house of debut, and the number of debuting members; the 1880 cohort, left-censored, is dropped. Lower panel: debut-legislature speeches only, column (1) the cohorts up to 1898 and column (2) the cohorts from 1894 on, each regressed on the Chamber indicator interacted with a post-reform cohort indicator, with arrondissement and party fixed effects (no legislature effects, which are collinear with the cohort at debut); weighted by in-vocabulary token count; standard errors clustered at arrondissement. Descriptive: debut cohorts are selected. Produced by `debut_cohort_event_study.do`.

Table 69: The Panel B contrast in the two halves of the post-1900 window, estimated separately and jointly.

	(1) FE	(2) FE	(3) Pol. FE	(4) Pol. FE	(5) Joint
Chamber × Post-1900	-0.0101***	-0.0109***	-0.0079***	-0.0173***	
	(0.0019)	(0.0022)	(0.0020)	(0.0055)	
Chamber × Post-1900, 1900–1906					-0.0094***
					(0.0026)
Chamber × Post-1900, 1908–1912					-0.0114***
					(0.0036)
Observations	151011	153640	150987	153614	245268
Sub-window	1900–1906	1908–1912	1900–1906	1908–1912	joint
Politician FE	No	No	Yes	Yes	Yes
Equality, p					0.2066

Columns (1)–(4): each half of the post-1900 window against the 1894–1899 legislatures; column (5): both interactions in one regression on the whole Panel B sample, with the p-value of their equality. Weighted by in-vocabulary token count; standard errors clustered at arrondissement. Produced by `panelB_split_1908.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 70: Partisanship and legislative outcomes: bills deposited in the Chamber by legislature, and legislature-level regressions of bill adoption on Chamber partisanship Π.

Panel A: bills by legislature

		Government bills		Members' bills		Government share
Legislature	Π	Deposited	Adopted	Deposited	Adopted	of deposits
1882	0.4998	94	67.0%	50	28.0%	65.3%
1884	0.5033	144	82.6%	40	17.5%	78.3%
1886	0.5040	150	81.3%	28	25.0%	84.3%
1888	0.5045	120	71.7%	23	26.1%	83.9%
1890	0.5053	145	69.0%	38	26.3%	79.2%
1892	0.5046	153	52.9%	59	27.1%	72.2%
1894	0.5099	198	75.3%	77	23.4%	72.0%
1896	0.5125	122	77.9%	32	31.3%	79.2%
1898	0.5109	94	72.3%	46	19.6%	67.1%
1900	0.5043	104	82.7%	96	21.9%	52.0%
1902	0.5079	87	79.3%	49	24.5%	64.0%
1904	0.5060	97	87.6%	40	25.0%	70.8%
1906	0.5063	99	81.8%	61	11.5%	61.9%
1908	0.5033	115	83.5%	79	17.7%	59.3%
1910	0.5042	153	68.6%	40	7.5%	79.3%
1912	0.5040	184	73.4%	100	23.0%	64.8%
All		2,059	74.8%	858	21.8%	70.6%

Panel B: adoption regressed on Π

Outcome	Coefficient	s.e.	Legislatures
All bills, adoption rate	+4.80	6.03	16
Government bills, adoption rate	+3.34	7.39	16
Members' bills, adoption rate	+1.73	4.96	16
Government share of deposits	+2.41	7.75	16
Government bills, adoption rate, linear trend	+2.93	7.63	16
Members' bills, adoption rate, linear trend	+2.32	4.47	16
Government share of deposits, linear trend	+3.98	6.95	16
Government bills, adoption rate, without 1912	+3.18	7.78	15
Members' bills, adoption rate, without 1912	+2.07	5.21	15
Government share of deposits, without 1912	+1.23	7.93	15
Members' bills by Catholics and Liberals	+7.14	5.97	16
with a linear trend	+8.25	5.85	16
Members' bills by Socialists and Daensists	−12.89	16.35	13

Bills deposited in the Chamber: government bills (*projets de loi*, no named parliamentary author) and members' bills (*propositions de loi*); adoption is the bill's recorded outcome. Panel B: one observation per legislature; conventional standard errors. Weights: the bills deposited—government bills for the government-bill rows and for the government share of deposits, members' bills for the members'-bill rows, all bills for the first row, and for the rows by the author's party the members' bills of those authors (443 Catholic and 248 Liberal, 159 Socialist and 8 Daensist). The 1912 legislature is truncated by the war. The first row of Panel B pools the two kinds of bill and is computed from Panel A with the same weighted regression, which reproduces the two separate rows below it. Produced by `downstream_laws.do` (`esegui_paper_c.do`, block C14).

Table 71: Vocabulary coverage and the composition of the leave-out pool, by chamber and reform window, Catholic and Liberal speech.

		pre-1894	1894–1899	post-1900
Chamber	share of bigram tokens in the vocabulary	0.269	0.265	0.255
Senate	share of bigram tokens in the vocabulary	0.236	0.233	0.222
Chamber	mean share of a speaker in his party's pool	0.0224	0.0204	0.0153
Senate	mean share of a speaker in his party's pool	0.0148	0.0165	0.0077

Coverage: in-vocabulary phrase tokens over all bigram tokens (adjacent pairs of lemmas, function words included), medium vocabulary; over the whole window the share is 0.257. The pool from which $\hat{\rho}$ is estimated is 89.1% Chamber by speeches and 86.1% by in-vocabulary tokens. The speaker's share is his token-weighted share of his party's phrase counts, the quantity on which the leave-out correction bites. Produced by `11_derivati.py` (blocks C and D) and `08_descrittive.py` (block E).

Table 72: Catholic–Liberal partisanship of confessional speech by legislature, Chamber, on the paper's fixed yardstick.

Legislature	Confessional	Other speech	Legislature	Confessional	Other speech
1880	0.5002	0.4978	1898	0.4987	0.5117
1882	0.5009	0.4995	1900	0.5004	0.5047
1884	0.5048	0.5026	1902	0.4993	0.5088
1886	0.5000	0.5048	1904	0.5020	0.5063
1888	0.5036	0.5047	1906	0.5006	0.5070
1890	0.4980	0.5063	1908	0.5010	0.5036
1892	0.5003	0.5050	1910	0.4981	0.5050
1894	0.4990	0.5111	1912	0.5040	0.5041
1896	0.4994	0.5135			

Token-weighted means of s_{distinct} over the Chamber speeches the topic classifier assigns to the confessional bucket (confessional and education policy) and over all other speeches. 1880–1884: confessional 0.5024, other 0.5001, all speech 0.5007; 1894–1899: confessional 0.4990. Produced by `downstream_validation.do` (`esegui_paper_d.do`, block D4).

Table 73: The cells behind the topic decomposition: each topic's share of the chamber's in-vocabulary tokens (w) and its within-topic partisanship (d), before 1894 and in 1894–1899.

	Chamber				Senate			
	w pre	w post	d pre	d post	w pre	w post	d pre	d post
Confessional	19.8%	7.8%	0.5018	0.4990	13.9%	8.5%	0.5022	0.4916
Economic and fiscal	16.2%	23.5%	0.5047	0.5144	20.0%	24.9%	0.5011	0.4995
Foreign and defence	8.0%	8.3%	0.4994	0.5005	11.4%	8.3%	0.5048	0.5027
Labour and social	5.0%	13.3%	0.5070	0.5280	5.3%	9.1%	0.4938	0.4839
Other	6.5%	7.6%	0.5033	0.5077	4.5%	4.1%	0.5078	0.5059
State and governance	28.5%	26.3%	0.5029	0.5082	24.9%	30.9%	0.5007	0.4977
Territory and infrastructure	16.0%	13.3%	0.5033	0.5090	20.0%	14.2%	0.5017	0.5040

Six policy buckets and the seventh, residual bucket of speeches with no substantive policy content ("Other"); w is the bucket's share of the chamber-window's in-vocabulary tokens, summing to one over the seven, and d the token-weighted mean of s_{distinct} within it. Panel A window, Catholic and Liberal speech. These are the $w_{c,p,k}$ and $d_{c,p,k}$ of (11): they reproduce the window means of Table 57, and Table 19 reports the decomposition they add up to. Produced by `topic_oaxaca.do` (`esegui_paper_d.do`, block D1).

Table 74: Distillation of the topic labels onto the dataset's interventions: out-of-fold agreement of the distilled label with the classifier's, by target and reform window.

Target	Interventions	n	Accuracy	κ	Macro F_1
Coarse bucket (seven)	Chamber, pre-1894	158,456	0.8581	0.7736	0.7865
	Chamber, 1894–1899	74,008	0.8667	0.7711	0.7941
	Chamber, post-1900	161,702	0.8608	0.7468	0.7752
	Chamber, outside 1880–1912	63,426	0.8586	0.7827	0.7701
	Chamber, all	457,592	0.8605	0.7675	0.7942
Substantive topic or procedural	Chamber, pre-1894	158,456	0.8926	0.7802	0.8895
	Chamber, 1894–1899	74,008	0.8996	0.7862	0.8925
	Chamber, post-1900	161,702	0.8882	0.7529	0.8755
	Chamber, outside 1880–1912	63,426	0.8985	0.7952	0.8971
	Chamber, all	457,592	0.893	0.7756	0.8871
Descriptive domain (22)	Chamber, pre-1894	158,456	0.6664	0.5869	0.5812
	Chamber, 1894–1899	74,008	0.6673	0.5718	0.571
	Chamber, post-1900	161,702	0.6609	0.5532	0.5616
	Chamber, outside 1880–1912	63,426	0.6133	0.5031	0.4276
	Chamber, all	457,592	0.6572	0.5638	0.4659
Domain assigned or not	Chamber, pre-1894	158,456	0.8881	0.773	0.8864
	Chamber, 1894–1899	74,008	0.8929	0.77	0.8849
	Chamber, post-1900	161,702	0.8874	0.7432	0.8714
	Chamber, outside 1880–1912	63,426	0.8904	0.7806	0.8903
	Chamber, all	457,592	0.8889	0.7674	0.8836

Each intervention of the PLENUM corpus is labelled by the fold model that did not see it; agreement is with the label the classifier assigned to it. "Substantive topic or procedural" is whether the intervention carries any of the six policy buckets; the twenty-two descriptive domains come from a second classifier of the same family. Produced by `10_argomenti.py`.

Table 75: The socialist-proximity coefficients across the vocabulary ladder: TopMining × Post, within the Chamber, by party.

	Vocabulary	All	Catholic	Liberal
TopMining × Post-1894	Strict (50/8)	0.0144^{***}	n.i.	0.00649
		(0.00208)		(0.00589)
	Medium (35/6)	0.0150^{***}	n.i.	0.00722
		(0.00211)		(0.00679)
	Loose (25/5)	0.0151^{***}	n.i.	0.00776
		(0.00212)		(0.00742)
TopMining × Post-1900	Strict (50/8)	0.00521^{*}	n.i.	-0.00460^{**}
		(0.00286)		(0.00225)
	Medium (35/6)	0.00359	n.i.	-0.00706^{***}
		(0.00299)		(0.00220)
	Loose (25/5)	0.00311	n.i.	-0.00825^{***}
		(0.00306)		(0.00240)

The specification of Tables 45 and 46 (the medium rows) on the three vocabularies of Appendix B.9: socialist-proximity outcome, Catholic and Liberal Chamber speech, arrondissement, politician and legislature fixed effects; standard errors clustered at arrondissement. "n.i.": not identified, since no Catholic deputy for the TopMining arrondissements speaks in the Chamber between the reforms (Appendix A.1). Produced by `socialist_proximity_did.do`. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 76: OCR noise proxies by legislature and chamber: the series behind the window means of Table 21, Panel A.

	Chamber			Senate		
Legislature	Hapax share	Short share	Type–token	Hapax share	Short share	Type–token
1880	0.4745	0.0061	0.0155	0.4277	0.0042	0.0426
1882	0.4558	0.0063	0.0133	0.4198	0.0044	0.0444
1884	0.4247	0.0065	0.0118	0.3940	0.0042	0.0322
1886	0.4178	0.0065	0.0118	0.3931	0.0045	0.0307
1888	0.4143	0.0071	0.0112	0.4000	0.0050	0.0310
1890	0.4100	0.0069	0.0111	0.3915	0.0049	0.0280
1892	0.4196	0.0066	0.0102	0.3901	0.0050	0.0267
1894	0.4477	0.0080	0.0124	0.3869	0.0050	0.0241
1896	0.4263	0.0070	0.0118	0.4052	0.0054	0.0247
1898	0.4555	0.0065	0.0124	0.4107	0.0052	0.0327
1900	0.5309	0.0090	0.0186	0.4488	0.0053	0.0377
1902	0.4505	0.0080	0.0152	0.4348	0.0052	0.0369
1904	0.4561	0.0075	0.0157	0.4393	0.0051	0.0335
1906	0.4801	0.0088	0.0202	0.4309	0.0055	0.0369
1908	0.4808	0.0081	0.0174	0.4409	0.0057	0.0316
1910	0.4633	0.0085	0.0171	0.4155	0.0060	0.0315
1912	0.4835	0.0084	0.0171	0.4551	0.0059	0.0307

Hapax share: the share of once-occurring tokens; short share: the share of tokens shorter than three characters; type–token ratio; each computed on the chamber×legislature cell's text, the 1914 legislature omitted. Produced by `ocr_robustness.py`.